

Unit 1 AP Topics 16

1. Let f be a twice-differentiable function such that $f(2)=5$ and $f(5)=2$. Let g be the function given by $g(x)=f(f(x))$.
Let $h(x)=f(x)-x$. Explain why there must be a value r for $2<r<5$ such that $h(r)=0$.

Part D

1 point is earned for $h(2)$ and $h(5)$

1 point is earned for conclusion, using IVT

Let $h(x)=f(x)-x$. $h(2)=f(2)-2=5-2=3$ $h(5)=f(5)-5=2-5=-3$ Since $h(2)>0>h(5)$, the Intermediate Value Theorem guarantees that there is a value r , with $2<r<5$, such that $h(r)=0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $h(2)$ and $h(5)$

1 point is earned for conclusion, using IVT

Let $h(x)=f(x)-x$.

$$h(2)=f(2)-2=5-2=3$$

$$h(5)=f(5)-5=2-5=-3$$

Since $h(2)>0>h(5)$, the Intermediate Value Theorem guarantees that there is a value r , with $2<r<5$, such that $h(r)=0$.

Unit 1 AP Topics 16

2.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	0	3	6
$f(x)$	-5	4	1	7
$f'(x)$	-1	2	-2	4

The table above gives values of a twice-differentiable function f and its first derivative f' for selected values of x . Let g be the function defined by $g(x) = f(x^2 - x)$.

- (a) What is the value of $g'(3)$?
- (b) It is known that $g''(0) = -1$. What is the value of $f''(0)$?
- (c) Is there a value c , for $0 < c < 3$, such that $g(c) = 5$? Justify your answer.
- (d) Let h be the function with derivative given by $h'(x) = 4e^{\cos x}$. At what value of x in the interval $-3 \leq x \leq 0$ does the instantaneous rate of change of h equal the average rate of change of f over the interval $-3 \leq x \leq 0$?

Part A

Neither point may be earned for a response that does not apply the chain rule. The second point may be earned with a maximum of one error in $g'(x)$, unless the error is failing to apply the chain rule. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes one of the criteria below.

☐ $g'(x)$

☐ $g'(3)$

Solution:

$$g'(x) = f'(x^2 - x)(2x - 1)$$

$$g'(3) = f'(6)(5) = 4(5) = 20$$

Part B

Points may be earned in this part for consistent and correct use of the $g'(x)$ from part (a).

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

☐ $g''(x)$ (2 points for full credit)

☐ answer

Solution:

$$g''(x) = f''(x^2 - x)(2x - 1)^2 + f'(x^2 - x)(2)$$

$$g''(0) = f''(0)(-1)^2 + f'(0)(2)$$

$$-1 = f''(0) + 2(2) \Rightarrow f''(0) = -5$$

Part C

Unit 1 AP Topics 16

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ g is continuous, with reason
- ☐ justification using Intermediate Value Theorem

Solution:

f is twice differentiable. $\Rightarrow f$ is continuous.

g is continuous as the composition of continuous functions.

$$g(0) = f(0) = 4 \text{ and}$$

$$g(3) = f(6) = 7.$$

$$\Rightarrow g(0) < 5 < g(3)$$

By the Intermediate Value Theorem, there is a value of c , for $0 < c < 3$, such that $g(c) = 5$.

Part D

At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

Unit 1 AP Topics 16

The average rate of change of f over the interval $-3 \leq x \leq 0$

$$\frac{f(0)-f(-3)}{0-(-3)} = \frac{4-(-5)}{3} = 3.$$

$$h'(x) = 4e^{\cos x} = 3 \text{ for } -3 \leq x \leq 0. \Rightarrow x = -1.862602$$

The instantaneous rate of change of h equals the average rate of change of f over the interval $-3 \leq x \leq 0$ at $x = -1.863$ (or -1.862).

Unit 1 AP Topics 16

3. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	0	3	4
$f(x)$	5	-1	2	7
$f'(x)$	-2	4	0	1

The table above gives values of a twice-differentiable function f and its first derivative f' for selected values of x . Let g be the function defined by $g(x) = f(2x - x^2)$.

- (a) What is the value of $g'(-1)$?
- (b) It is known that $g''(0) = 0$. What is the value of $f''(0)$?
- (c) Is there a value c , for $0 < c < 3$, such that $g(c) = 2$? Justify your answer.
- (d) Let h be the function with first derivative given by $h'(x) = 4xe^x$. At what value of x in the interval $0 \leq x \leq 4$ does the instantaneous rate of change of h equal the average rate of change of f over the interval $0 \leq x \leq 4$?

Part A

Neither point may be earned for a response that does not apply the chain rule. The second point may be earned with a maximum of one error in $g'(x)$, unless the error is failing to apply the chain rule. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $g'(x)$
- ☐ $g'(-1)$

Solution:

$$g'(x) = f'(2x - x^2)(2 - 2x)$$

$$g'(-1) = f'(-3)(4) = (-2)(4) = -8$$

Part B

Points may be earned in this part for consistent and correct use of the $g'(x)$ from part (a).

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $g''(x)$ (2 points for full credit)
- ☐ answer

Solution:

$$g''(x) = f''(2x - x^2)(2 - 2x)^2 + f'(2x - x^2)(-2)$$

$$g''(0) = f''(0)(2)^2 + f'(0)(-2)$$

$$0 = 4f''(0) + 4(-2) \Rightarrow f''(0) = 2$$

Part C

Unit 1 AP Topics 16

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ g is continuous
- ☐ justification using Intermediate Value Theorem

Solution:

f is twice differentiable. $\Rightarrow f$ is continuous.

g is continuous as the composition of continuous functions.

$$g(0) = f(0) = -1 \text{ and } g(3) = f(-3) = 5.$$

$$\Rightarrow g(0) < 2 < g(3)$$

By the Intermediate Value Theorem, there is a value of c , for $0 < c < 3$, such that $g(c) = 2$.

Part D

At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.

At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

Unit 1 AP Topics 16

The average rate of change of f over the interval $0 \leq x \leq 4$ is $\frac{f(4)-f(0)}{4-0} = \frac{7-(-1)}{4} = 2$.

$$h'(x) = 4xe^x = 2 \text{ for } 0 \leq x \leq 4. \Rightarrow x = 0.351734$$

The instantaneous rate of change of h equals the average rate of change of f over the interval $0 \leq x \leq 4$ at $x = 0.352$ (or 0.351).

4.

t (sec)	0	15	25	30	35	50	60
$v(t)$ (ft/sec)	-20	-30	-20	-14	-10	0	10
$a(t)$ (ft/sec ²)	1	5	2	1	2	4	2

A car travels on a straight track. During the time interval $0 \leq t \leq 60$ seconds, the car's velocity v , measured in feet per second, and acceleration a , measured in feet per second per second, are continuous functions. The table above shows selected values of these functions.

For $0 < t < 60$, must there be a time t when $v(t) = -5$? Justify your answer.

Part C

1 point is earned $v(35) < -5 < v(50)$

1 point is earned for Yes; refers to IVT or hypotheses

Yes. Since $v(35) = -10 < -5 < 0 = v(50)$, the IVT guarantees a t in $(35, 50)$ so that $v(t) = -5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned $v(35) < -5 < v(50)$

1 point is earned for Yes; refers to IVT or hypotheses

Yes. Since $v(35) = -10 < -5 < 0 = v(50)$, the IVT guarantees a t in $(35, 50)$ so that $v(t) = -5$.

Unit 1 AP Topics 16

5. A cylindrical can of radius 10 millimeters is used to measure rainfall in Stormville. The can is initially empty, and rain enters the can during a 60-day period. The height of water in the can is modeled by the function S , where $S(t)$ is measured in millimeters and t is measured in days for $0 \leq t \leq 60$. The rate at which the height of the water is rising in the can is given by $S'(t) = 2\sin(0.03t) + 1.5$.



During the same 60-day period, rain on Monsoon Mountain accumulates in a can identical to the one in Stormville. The height of the water in the can on Monsoon Mountain is modeled by the function M , where $M(t) = \frac{1}{400}(3t^3 - 30t^2 + 330t)$. The height $M(t)$ is measured in millimeters, and t is measured in days for $0 \leq t \leq 60$. Let $D(t) = M'(t) - S'(t)$. Apply the Intermediate Value Theorem to the function D on the interval $0 \leq t \leq 60$ to justify that there exists a time t , $0 < t < 60$, at which the heights of water in the two cans are changing at the same rate.

Part D

1 point is earned if considers $D(0)$ and $D(60)$

1 point is earned for justification

$$D(0) = -0.675 < 0 \text{ and } D(60) = 69.37730 > 0$$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t , $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if considers $D(0)$ and $D(60)$

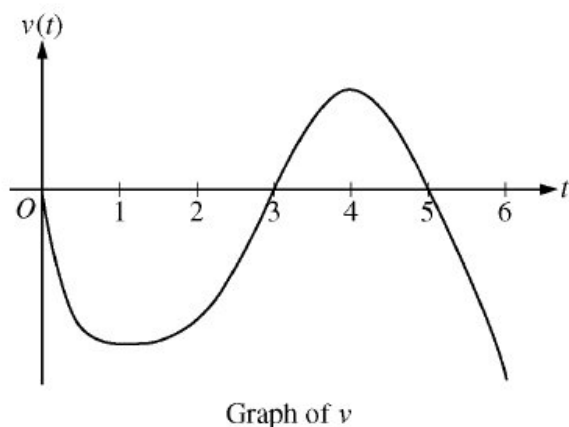
1 point is earned for justification

$$D(0) = -0.675 < 0 \text{ and } D(60) = 69.37730 > 0$$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t , $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.

Unit 1 AP Topics 16

6.



A particle moves along the x -axis so that its velocity at time t , for $0 \leq t \leq 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t -axis and the graph of v on the intervals $[0,3]$, $[3,5]$, and $[5,6]$ are 8, 3, and 2 respectively. At time $t = 0$, the particle is $x = -2$.

For how many values of t , where $0 \leq t \leq 6$, is the particle at $x = -8$? Explain your reasoning.

Part B

3 points can be earned.

1 point for positions at $t = 3$, $t = 5$ and $t = 6$.

1 point for description of motion.

1 point for conclusion.

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.



0	1	2	3
---	---	---	---

3 points can be earned.

1 point for positions at $t = 3$, $t = 5$ and $t = 6$.

1 point for description of motion.

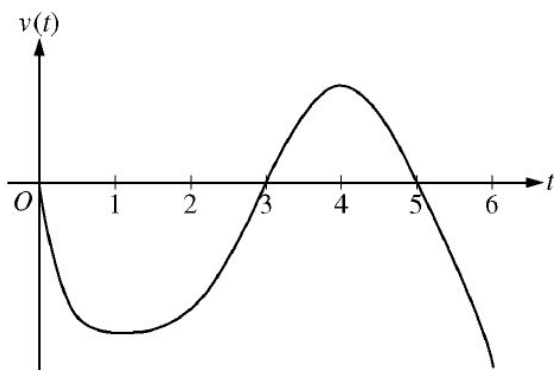
1 point for conclusion.

Unit 1 AP Topics 16

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

Unit 1 AP Topics 16

7.

Graph of v

A particle moves along the x -axis so that its velocity at time t , for $0 \leq t \leq 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t -axis and the graph of v on the intervals $[0, 3]$, $[3, 5]$, and $[5, 6]$ are 8, 3, and 2, respectively. At time $t = 0$, the particle is at $x = -2$.

For how many values of t , where $0 \leq t \leq 6$, is the particle at $x = -8$? Explain your reasoning.

Part B

The response can earn up to 3 points:

1 point: For the positions at $t = 3$, $t = 5$, and $t = 6$

1 point: For the description of motion

1 point: For the correctly conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: For the positions at $t = 3$, $t = 5$, and $t = 6$

1 point: For the description of motion

1 point: For the correctly conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

Unit 1 AP Topics 16

8.

t (hours)	0	1	3	4	7	8	9
$L(t)$ (people)	120	156	176	126	150	80	0

Concert tickets went on sale at noon ($t = 0$) and were sold out within 9 hours. The number of people waiting in line to purchase tickets at time t is modeled by a twice-differentiable function L for $0 \leq t \leq 9$. Values of $L(t)$ at various times t are shown in the table above.

For $0 \leq t \leq 9$, what is the fewest number of times at which $L'(t)$ must equal 0? Give a reason for your answer.

Part C

The response can earn up to 3 points:

1 point: For considering change in sign of L'

1 point: For the analysis

1 point: For the correct conclusion

L is differentiable $[0,9]$ so the Mean Value Theorem implies $L'(t) > 0$ for some t in $(1,3)$ and some t in $(4,7)$. Similarly, $L'(t) < 0$ for some t in $(3,4)$ and some t in $(7,8)$. Then, since L' is continuous on $[0,9]$, the Intermediate Value Theorem implies that $L'(t)=0$ for at least three values of t in $[0,9]$.

Alternate solution:

1 point: Considers relative extrema of L on $(0,9)$

1 point: For the analysis

1 point: For the correct conclusion

The continuity of L on $[1,4]$ implies that L attains a maximum value there. Since $L(3) > L(1)$ and $L(3) > L(4)$, this maximum occurs on $(1,4)$. Similarly, L attains a minimum on $(3,7)$ and a maximum on $(4,8)$. L is differentiable, so $L'(t) = 0$ at each relative extreme point on $(0,9)$. Therefore $L'(t) = 0$ for at least three values of t in $[0,9]$.

[Note: There is a function L that satisfies the given conditions with $L'(t)=0$ for exactly three values of t]



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: For considering change in sign of L'

1 point: For the analysis

Unit 1 AP Topics 16

1 point: For the correct conclusion

L is differentiable $[0,9]$ so the Mean Value Theorem implies $L'(t) > 0$ for some t in $(1,3)$ and some t in $(4,7)$. Similarly, $L'(t) < 0$ for some t in $(3,4)$ and some t in $(7,8)$. Then, since L' is continuous on $[0,9]$, the Intermediate Value Theorem implies that $L'(t)=0$ for at least three values of t in $[0,9]$.

Alternate solution:

1 point: Considers relative extrema of L on $(0,9)$

1 point: For the analysis

1 point: For the correct conclusion

The continuity of L on $[1,4]$ implies that L attains a maximum value there. Since $L(3) > L(1)$ and $L(3) > L(4)$, this maximum occurs on $(1,4)$. Similarly, L attains a minimum on $(3,7)$ and a maximum on $(4,8)$. L is differentiable, so $L'(t) = 0$ at each relative extreme point on $(0,9)$. Therefore $L'(t) = 0$ for at least three values of t in $[0,9]$.

9. Let f be a function that is continuous on the closed interval $[2, 4]$ with $f(2) = 10$ and $f(4) = 20$. Which of the following is guaranteed by the Intermediate Value Theorem?

(A) $f(x) = 13$ has at least one solution in the open interval $(2, 4)$. ✓

(B) $f(3) = 15$

(C) f attains a maximum on the open interval $(2, 4)$.

(D) $f'(x) = 5$ has at least one solution in the open interval $(2, 4)$.

(E) $f'(x) > 0$ for all x in the open interval $(2, 4)$.

10. Let f be a function that is continuous on the closed interval $[0, 2]$ with $f(0) = -2$, $f(1) = 5$, and $f(2) = 0$. Which of the following statements must be true?

(A) The maximum value of f on the closed interval $[0, 2]$ is 5.

(B) There is a number c , $0 < c < 2$, with $f(c) = -1$. ✓

(C) There is a number c , $0 < c < 2$, with $f(c) = 7$.

(D) There is a number c , $0 < c < 2$, with $f'(c) = 1$.

Answer B

Correct. Since f is continuous on the interval $[0, 2]$ and $f(0) < -1 < f(2)$, by the Intermediate Value Theorem there is a number c with $0 < c < 2$ and $f(c) = -1$.

Unit 1 AP Topics 16

11. The function f is continuous on the interval $-1 \leq x \leq 5$. If $f(-1) = 1$ and $f(5) = 13$, which of the following statements must be true?
- (A) There exists c , where $-1 < c < 5$, such that $f(c) = 0$.
 - (B) There exists c , where $-1 < c < 5$, such that $f(c) = 2$. ✓
 - (C) There exists c , where $-1 < c < 5$, such that $f'(c) = 0$.
 - (D) There exists c , where $-1 < c < 5$, such that $f'(c) = 2$.

Answer B

Correct. This is true by the Intermediate Value Theorem, since f is continuous on the interval $-1 \leq x \leq 5$ and $f(-1) < 2 < f(5)$.

12. Let f be a function that is continuous on the closed interval $[1, 3]$ with $f(1) = 10$ and $f(3) = 18$. Which of the following statements must be true?
- (A) $10 \leq f(2) \leq 18$
 - (B) f is increasing on the interval $[1, 3]$.
 - (C) $f(x) = 17$ has at least one solution in the interval $[1, 3]$. ✓
 - (D) $f'(x) = 8$ has at least one solution in the interval $(1, 3)$.
 - (E) $\int_1^3 f(x) dx > 20$

Unit 1 AP Topics 16

13.

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in giga-liters (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t=30$, the reservoir contains 125 giga-liters of water.

Explain why there must be at least one time t , other than $t = 10$, such that $W'(t) = 0.7$ GL/day.

Part C

Two points are earned for the explanation

W' is differentiable $\Rightarrow W'$ is continuous

$$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$$

By the Intermediate Value Theorem, there must be at least one time t , $22 \leq t \leq 30$, such that $W'(t) = 0.7$.



0	1	2
---	---	---

The student earns all of the following points:

Two points are earned for the explanation

W' is differentiable $\Rightarrow W'$ is continuous

$$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$$

By the Intermediate Value Theorem, there must be at least one time t , $22 \leq t \leq 30$, such that $W'(t) = 0.7$.

14. Let g be a continuous function on the closed interval $[0,1]$. Let $g(0)=1$ and $g(1)=0$. Which of the following is NOT necessarily true?

(A) There exists a number h in $[0,1]$ such that $g(h) \geq g(x)$ for all x in $[0,1]$.

(B) For all a and b in $[0,1]$, if $a=b$, then $g(a)=g(b)$

(C) There exists a number h in $[0,1]$ such that $g(h) = \frac{1}{2}$

(D) There exists a number h in $[0,1]$ such that $g(h) = \frac{3}{2}$



(E) For all h in the open interval $(0,1)$, $\lim_{x \rightarrow h} g(x) = g(h)$

Unit 1 AP Topics 16

15.

x	0	1	2
$f(x)$	1	k	2

The function f is continuous on the closed interval $[0,2]$ and has values that are given in the table above.

The equation $f(x) = \frac{1}{2}$ must have at least two solutions in the interval $[0,2]$ if $k=$

(A) 0



(B) $\frac{1}{2}$

(C) 1

(D) 2

(E) 3

Unit 1 AP Topics 16

16.

t (minutes)	0	2	5	8	12
$v_A(t)$ (meters/minute)	0	100	40	-120	-150

Train A runs back and forth on an east-west section of railroad track. Train A 's velocity, measured in meters per minute, is given by a differentiable function $v_A(t)$, where time t is measured in minutes. Selected values for $v_A(t)$ are given in the table above.

Do the data in the table support the conclusion that train A 's velocity is -100 meters per minute at some time t with $5 < t < 8$? Give a reason for your answer.

Part B

1 point is earned for $V_A(8) < -100 < V_A(5)$

1 point is earned for the conclusion, using IVT

V_A is differentiable $\Rightarrow V_A$ is continuous

$V_A(8) = -120 < -100 < 40 = V_A(5)$

Therefore, by the Intermediate Value Theorem, there is a time t , $5 < t < 8$, such that $V_A(t) = -100$.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for $V_A(8) < -100 < V_A(5)$

1 point is earned for the conclusion, using IVT

V_A is differentiable $\Rightarrow V_A$ is continuous

$V_A(8) = -120 < -100 < 40 = V_A(5)$ Therefore, by the Intermediate Value Theorem, there is a time t , $5 < t < 8$, such that $V_A(t) = -100$.

Unit 1 AP Topics 16

17.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	-4	4	2
4	-1	3	6	7

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x . The function h is given by $h(x) = f(g(x)) - 6$.

Explain why there must be a value r for $1 < r < 3$ such that $h(r) = -5$.

Part A

1 point is earned for $h(1)$ and $h(3)$

1 point is earned for the conclusion, using IVT

$$h(1) = f(g(1)) - 6 = f(2) - 6 = 9 - 6 = 3$$

$$h(3) = f(g(3)) - 6 = f(4) - 6 = -1 - 6 = -7$$

Since $h(3) < -5 < h(1)$ and h is continuous, by the Intermediate Value Theorem, there exists a value r , $1 < r < 3$, such that $h(r) = -5$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $h(1)$ and $h(3)$

1 point is earned for the conclusion, using IVT

$$h(1) = f(g(1)) - 6 = f(2) - 6 = 9 - 6 = 3$$

$$h(3) = f(g(3)) - 6 = f(4) - 6 = -1 - 6 = -7$$

Since $h(3) < -5 < h(1)$ and h is continuous, by the Intermediate Value Theorem, there exists a value r , $1 < r < 3$, such that $h(r) = -5$

Unit 1 AP Topics 16

18.

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t = 30$, the reservoir contains 125 gigaleters of water.

Explain why there must be at least one time t , other than $t = 10$, such that $W'(t) = 0.7$ GL/day.

Part C

The response can earn up to 2 points:

2 points: For the explanation

W' is differentiable $\Rightarrow W'$ is continuous.

$$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$$

By the Intermediate Value Theorem, there must be at least one time t , $22 \leq t \leq 30$, such that $W'(t) = 0.7$.



0	1	2
---	---	---

The response earns both of the following points:

2 points: For the explanation

W' is differentiable $\Rightarrow W'$ is continuous.

$$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$$

By the Intermediate Value Theorem, there must be at least one time t , $22 \leq t \leq 30$, such that $W'(t) = 0.7$.

Unit 1 AP Topics 16

19.

x	0	4	6	8	13
$f(x)$	3	4.5	3	2.5	4.4

The table above shows selected values of a continuous function f . For $0 \leq x \leq 13$, what is the fewest possible number of times $f(x) = 4$?

(A) one

(B) two

(C) three

(D) four



Answer C

This option is correct. Because f is continuous, the Intermediate Value Theorem will imply that $f(x)$ will equal 4 on an interval $[a, b]$ where 4 is between $f(a)$ and $f(b)$. According to the table, this happens for the intervals $[0, 4]$, $[4, 6]$, and $[8, 13]$. So there are at least three values of x for which the Intermediate Value Theorem applies and $f(x) = 4$.

20. Let f be a function such that $f(1) = -2$ and $f(5) = 7$. Which of the following conditions ensures that $f(c) = 0$ for some value c in the open interval $(1, 5)$?

(A) $\int_1^5 f(x) dx$ exists.(B) f is increasing on the closed interval $[1, 5]$.(C) f is continuous on the closed interval $[1, 5]$.(D) f is defined for all values of x in the closed interval $[1, 5]$.

21. Let g be a function such that $g(-1) = 0$ and $g(2) = 5$. Which of the following conditions guarantees that there is an x , $-1 < x < 2$, for which $g(x) = 3$?

(A) g is defined for all x in $(-1, 2)$.(B) g is continuous for all x in $[-1, 2]$.(C) g is increasing on $[-1, 2]$.(D) There exists an x in $(-1, 2)$ such that $g'(x) = 5$.(E) $\int_{-1}^2 g(x) dx = 3$ 

Unit 1 AP Topics 16

22. A polynomial $p(x)$ has a relative maximum at $(-2, 4)$, a relative minimum at $(1, 1)$, a relative maximum at $(5, 7)$ and no other critical points. How many zeros does $p(x)$ have?
- (A) One
- (B) Two ✓
- (C) Three
- (D) Four
- (E) Five
23. If a function f is continuous for all x and if f has a relative maximum at $(-1, 4)$ and a relative minimum at $(3, -2)$, which of the following statements must be true?
- (A) The graph of f has a point of inflection somewhere between $x = -1$ and $x = 3$.
- (B) $f'(-1) = 0$
- (C) The graph of f has a horizontal asymptote.
- (D) The graph of f has a horizontal tangent line at $x = 3$.
- (E) The graph of f intersects both axes. ✓
24. Let f be a function such that $f(1) = -2$ and $f(5) = 7$. Which of the following conditions ensures that $f(c) = 0$ for some value c in the open interval f of c , equals 0?
- (A) $\int_1^5 f(x) dx$ exists
- (B) f is increasing on the closed interval $[1, 5]$.
- (C) f is continuous on the closed interval $[1, 5]$. ✓
- (D) f is defined for all values of x in the closed interval $[1, 5]$
25. Let f be a function that is differentiable on the open interval $(1, 10)$. If $f(2) = -5$, $f(5) = 5$, and $f(9) = -5$, which of the following must be true?
- I. f has at least 2 zeros.
- II. The graph of f has at least one horizontal tangent.
- III. For some c , $2 < c < 5$, $f(c) = 3$.
- (A) None
- (B) I only
- (C) I and II only
- (D) I and III only
- (E) I, II, and III ✓

Unit 1 AP Topics 16

26. Let f be a continuous function on the closed interval $[-3, 6]$. If $f(-3) = -1$ and $f(6) = 3$, then the Intermediate Value Theorem guarantees that
- (A) $f(0) = 0$
 - (B) $f'(c) = \frac{4}{9}$ for at least one c between -3 and 6
 - (C) $-1 \leq f(x) \leq 3$ for all x between -3 and 6
 - (D) $f(c) = 1$ for at least one c between -3 and 6 ✓
 - (E) $f(c) = 0$ for at least one c between -1 and 3

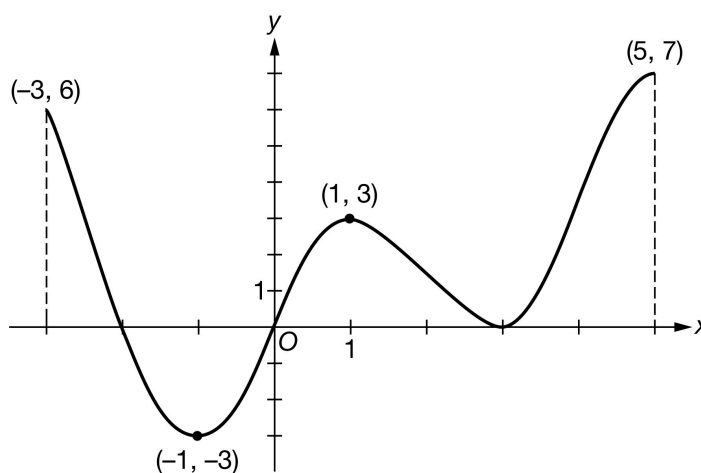
Unit 1 AP Topics 16

27. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The figure above shows the graph of g' , the derivative of a twice-differentiable function g , on the closed interval $-3 \leq x \leq 5$. The graph of g' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of g' on the intervals $[-3, -2]$, $[-2, 0]$, $[0, 3]$, and $[3, 5]$ are 3, 4, 5, and 7, respectively. The function g is defined for all real numbers and satisfies $g(1) = 3$ and $g(5) = 13$.

- (a) Find the x -coordinate of each critical point of g on the open interval $-3 < x < 5$.
- (b) Identify the critical point from part (a) which corresponds to a relative maximum of g . Give a reason for your answer.
- (c) Find the value of g at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- (d) Find the absolute maximum value of g on $-3 \leq x \leq 5$. Justify your answer.
- (e) On what open intervals is g both decreasing and concave up? Give a reason for your answer.
- (f) Explain why there must be a value of a , for $1 < a < 5$, such that $g(a) = 10$.
- (g) Find $\lim_{x \rightarrow 5^-} \frac{g(x) - 13}{3x - 15}$. Give a reason for your answer.
- (h) The function r is defined by $r(x) = (g(x))^3$. Find $r'(1)$. Show the computations that lead to your answer.

Part A

Unit 1 AP Topics 16

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -2$, $x = 0$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -2$, $x = 0$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-2, 0)$, $(0, 0)$, $(3, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = 0$ at $x = -2$, $x = 0$, and $x = 3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local maximum as -2 and give the correct reason of “ g' changes from positive to negative” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ g switches from increasing to decreasing at $x = -2$ ”
- Saying that “ $g'(-2)$ changes signs from positive to negative”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of g' , the one given in the stem of the problem.

A response indicating that “ $g'(x)$ is positive for $x < -2$ and negative for $x > -2$ ” is allowed to give this statement as a reason to earn the point.

Unit 1 AP Topics 16

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative maximum value of g occurs where $g'(x)$ changes from positive to negative values.

$x = -2$ is the only location on $-3 < x < 5$ for which this is true, so $x = -2$ is the only location of a relative maximum value of g on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = 0$, or $x = 3$ is selected as the location of the local maximum in part (b). Note that $g(-3) = 2$, $g(0) = 1$, and $g(3) = 6$. No other x -coordinates will be read for consistency.

The response $g(-2) = g(5) - (-4 + 5 + 7) = 5$ earns both points.

The response of both $13 - 8 = 5$ and $g(-2) = 5$ does not earn the first point, but earns the second point, provided no other work is presented.

The string of equalities $g(-2) = g(5) - \int_{-2}^5 g'(x) dx = 13 - 8 = 5$ earns both points.

Unit 1 AP Topics 16

Special Case: If in part (b), a response picks $x = 1$ or $x = 5$ as the location of the maximum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $g(-2) = g(5) + \int_5^{-2} g'(x) dx$

☐ Answer

Solution:

$$\begin{aligned} g(-2) &= g(5) + \int_5^{-2} g'(x) dx \\ &= g(5) - \int_{-2}^5 g'(x) dx = 13 - (7 + 5 - 4) = 5 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = -2$, the response must either state its y -value or reference it as a maximum.

For the endpoints, the y -values must be presented but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 1$), and thus the third point cannot be earned.

It is possible to not refer to $g(-3) = 2$ by indicating that g' is positive on the interval $(-3, -2)$, thus eliminating 2 as a possible absolute maximum.

A complete list of given or computable values for $g(x)$ is given in the table below.

Unit 1 AP Topics 16

x	$g(x)$
-3	2
-2	5
0	1
1	3
3	6
5	13



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = -2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$$g(-3) = g(5) + \int_5^{-3} g'(x) dx$$

$$= g(5) - \int_{-3}^5 g'(x) dx = 13 - (7 + 5 - 4 + 3) = 2$$

$$g(-2) = 5$$

$$g(5) = 13$$

So the absolute maximum value of g on $-3 \leq x \leq 5$ is $g(5) = 13$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-1, 0]$, $[-1, 0)$, $(-1, 0]$, $(-1, 0)$ all earn the first point.

The reason “ g' is increasing” can be expressed by either “ g'' is positive” or “the slope of g' is positive.”

We will overlook the false implication that $g' < 0$ on $[-1, 0]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[-1, 0]$.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where g' is negative.

g is concave up where g' is increasing.

g is both decreasing and concave up on the open interval $(-1, 0)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that g is continuous because g is differentiable or g is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[1, 5]$, or the point is not earned.

The y -values of g need to be mentioned by either referring to the numerical values of 3 and 13, or mentioning $g(1)$ and $g(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since g is differentiable it’s continuous and 10 is between 3 and 13 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

Unit 1 AP Topics 16

g is differentiable. $\Rightarrow g$ is continuous.

$$g(1) = 3 < 10 < 13 = g(5)$$

By the Intermediate Value Theorem, there must be a value of a , for $1 < a < 5$, such that $g(a) = 10$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of g' or g .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $g(5) - 13 = 0$ and $3 \cdot 5 - 15 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $g(5) = 13$ and $3 \cdot 5 = 15$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g(5)-13}{3 \cdot 5-15} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g'(x)}{3} = \frac{7}{3}$, will earn the second point, but not the third point; note that even just one of two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{0}{0}$ and $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g'(x)}{3}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{7}{3}$, or its numeric equivalent.

Unit 1 AP Topics 16



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ both g' and g are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} g(x) = g(5)$ and $\lim_{x \rightarrow 5^-} g'(x) = g'(5)$.

$$\lim_{x \rightarrow 5^-} (g(x) - 13) = g(5) - 13 = 13 - 13 = 0$$

$$\lim_{x \rightarrow 5^-} (3x - 15) = 3 \cdot 5 - 15 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{g(x) - 13}{3x - 15} &= \lim_{x \rightarrow 5^-} \frac{g'(x)}{3} \\ &= \frac{g'(5)}{3} = \frac{7}{3} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $3 \cdot (g(x))^2$ or $3 \cdot (g(1))^2$.

The chain rule point can be earned by seeing an expression involving g but not g' times $g'(1)$.

All three points will be awarded for seeing only $3 \cdot 3^2 \cdot 3$, but $3 \cdot 9 \cdot 3$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

Unit 1 AP Topics 16

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$r'(x) = 3(g(x))^2 \cdot g'(x)$$

$$r'(1) = 3(g(1))^2 \cdot g'(1) = 3 \cdot 3^2 \cdot 3 = 81$$

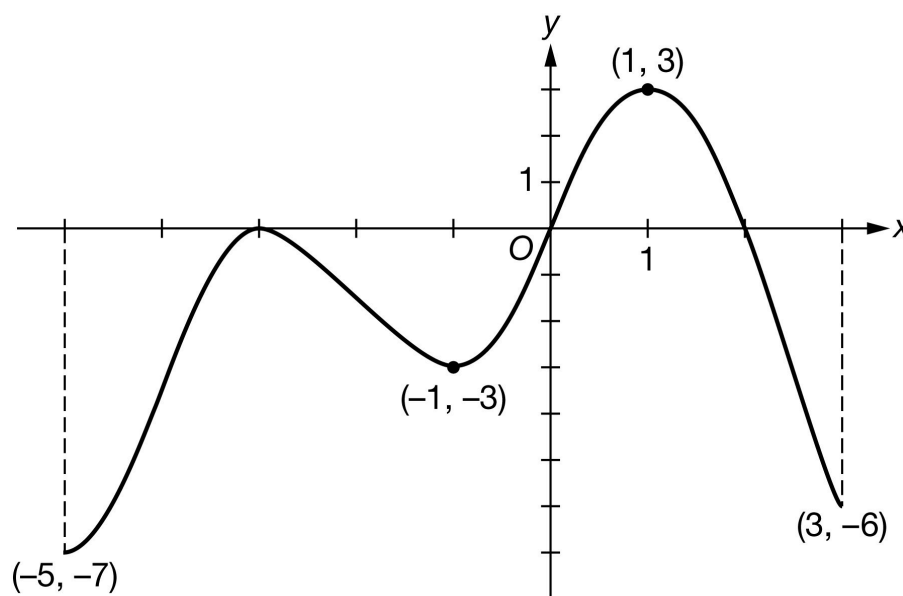
Unit 1 AP Topics 16

28. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The figure above shows the graph of h' , the derivative of a twice-differentiable function h , on the closed interval $-5 \leq x \leq 3$. The graph of h' has horizontal tangent lines at $x = -3$, $x = -1$, and $x = 1$. The areas of the regions bounded by the x -axis and the graph of h' on the intervals $[-5, -3]$, $[-3, 0]$, $[0, 2]$, and $[2, 3]$ are 7, 5, 4, and 3, respectively. The function h is defined for all real numbers and satisfies $h(1) = 12$ and $h(3) = 11$.

- Find the x -coordinate of each critical point of h on the open interval $-5 < x < 3$.
- Identify the critical point from part (a) which corresponds to a relative maximum of h . Give a reason for your answer.
- Find the value of h at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute maximum value of h on $-5 \leq x \leq 3$. Justify your answer.
- On what open intervals is h both increasing and concave down? Give a reason for your answer.
- Explain why there must be a value of b , for $1 < b < 3$, such that $h(b) = 11.5$.
- Find $\lim_{x \rightarrow 3^-} \frac{h(x) - 11}{x^2 - 9}$. Give a reason for your answer.

Unit 1 AP Topics 16

(h) The function p is defined by $p(x) = (h(x))^2$. Find $p'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = 0$, and $x = 2$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5, 3)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5, 3)$. The point is still earned provided the three critical points, $x = -3$, $x = 0$, and $x = 2$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(0, 0)$, $(2, 0)$ we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = 0$ at $x = -3$, $x = 0$, and $x = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local maximum as 2 and give the correct reason of “ h' changes from positive to negative” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ h switches from increasing to decreasing at $x = 2$ ”
- Saying that “ $h'(2)$ changes signs from positive to negative”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of h' , the one given in the stem of the problem.

Unit 1 AP Topics 16

A response indicating that “ $h'(x)$ is positive for $x < 2$ and negative for $x > 2$ ” is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative maximum value of h occurs where $h'(x)$ changes from positive to negative values.

$x = 2$ is the only location on $-3 < x < 5$ for which this is true, so $x = 2$ is the only location of a relative maximum value of h on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -5$, $x = -3$, or $x = 0$ is selected as the location of the local maximum in part (b). Note that $h(-5) = 22$, $h(-3) = 15$, and $h(0) = 10$. No other x -coordinates will be read for consistency.

The response $h(2) = h(3) - (-3) = 14$ earns both points.

The response of both $11 - (-3) = 14$ and $h(2) = 14$ does not earn the first point, but earns the second point, provided no other work is presented.

Unit 1 AP Topics 16

Special Case: If a response uses $\int_1^2 h'(x) dx = 2$ then

$\cdot h(2) = h(1) + \int_1^2 h'(x) dx = 12 + 2 = 14$ earns both points, because it's possible to find

$\int_1^2 h'(x) dx = 2$ using the areas and initial conditions given.

$\cdot$ any response using perceived symmetry about $x = 1$ can earn at most the first point, for example

$h(2) = h(1) + \int_1^2 h'(x) dx = 12 + \frac{1}{2} \cdot 4 = 14$ earns just the first point.

Special Case: If in part (b), a response picks $x = 1$ or $x = 3$ as the location of the maximum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $h(2) = h(3) + \int_3^2 h'(x) dx$

☐ Answer

Solution:

$$h(2) = h(3) + \int_3^2 h'(x) dx$$

$$= h(3) - \int_2^3 h'(x) dx = 11 - (-3) = 14$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x - value.

For $x = 2$, the response must either state its y -value or reference it as a maximum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

Unit 1 AP Topics 16

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 1$), and thus the third point cannot be earned.

It is possible to not refer to $h(3) = 11$ by indicating that h' is negative on the interval $(2, 3)$, thus eliminating 11 as a possible absolute maximum.

A complete list of given or computable values for $h(x)$ is given in the table below

x	$h(x)$
-5	22
-3	15
0	10
1	12
2	14
3	11



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$$\begin{aligned}
 h(-5) &= h(3) + \int_3^{-5} h'(x) \, dx \\
 &= h(3) - \int_{-5}^3 h'(x) \, dx = 11 - (-7 - 5 + 4 - 3) = 22
 \end{aligned}$$

$$h(2) = 14$$

$$h(3) = 11$$

So the absolute maximum value of h on $-5 \leq x \leq 3$ is $h(-5) = 22$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Unit 1 AP Topics 16

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[1, 2]$, $[1, 2)$, $(1, 2]$, and $(1, 2)$ all earn the first point.

The reason “ h' is decreasing” can be expressed by either “ h'' is negative” or “the slope of h' is negative.”

We will overlook the false implication that $h' > 0$ on $[1, 2]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[1, 2]$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where h' is positive.

h is concave down where h' is decreasing.

h is both increasing and concave down on the open interval $(1, 2)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that h is continuous because h is differentiable or h is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[1, 3]$, or the point is not earned.

The y -values of h need to be mentioned by either referring to the numerical values of 11 and 12, or mentioning $h(1)$ and $h(3)$

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since h is differentiable it’s continuous and 11.5 is between 11 and 12 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

h is differentiable. $\Rightarrow h$ is continuous.

$$h(3) = 11 < 11.5 < 12 = h(1)$$

By the Intermediate Value Theorem, there must be a value of b , for $1 < b < 3$, such that $h(b) = 11.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of h' or h .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $h(3) - 11 = 0$ and $3^2 - 9 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $h(3) = 11$ and $3^2 = 9$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h(3)-11}{3^2-9} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{0}{0}$ earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Unit 1 AP Topics 16

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h'(x)}{2x} = \frac{-6}{6}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{0}{0}$ and $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-\frac{6}{6}$ or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h is twice differentiable. $\Rightarrow$ both h' and h are continuous. $\Rightarrow \lim_{x \rightarrow 3^-} h(x) = h(3)$ and $\lim_{x \rightarrow 3^-} h'(x) = h'(3)$.

$$\lim_{x \rightarrow 3^-} (h(x) - 11) = h(3) - 11 = 11 - 11 = 0$$

$$\lim_{x \rightarrow 3^-} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} &= \lim_{x \rightarrow 3^-} \frac{h'(x)}{2x} \\ &= \frac{h'(3)}{2 \cdot 3} = \frac{-6}{6} = -1 \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $2 \cdot h(x)$ or $2 \cdot h(1)$.

Unit 1 AP Topics 16

The chain rule point can be earned by seeing an expression involving h but not h' times $h'(1)$.

All three points will be awarded for seeing only $2 \cdot 12 \cdot 3$, but $24 \cdot 3$, $2 \cdot 36$, or $6 \cdot 12$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$p'(x) = 2h(x) \cdot h'(x)$$

$$p'(1) = 2 \cdot h(1) \cdot h'(1) = 2 \cdot 12 \cdot 3 = 72$$

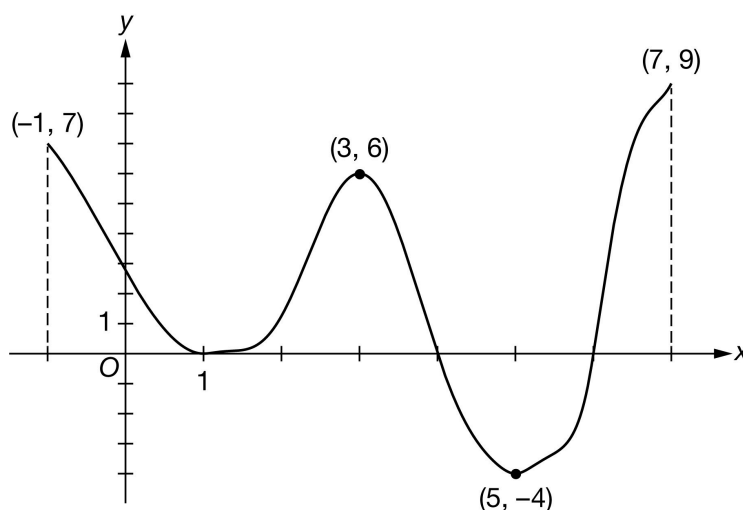
Unit 1 AP Topics 16

29. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The figure above shows the graph of h' , the derivative of a twice-differentiable function h , on the closed interval $-1 \leq x \leq 7$. The graph of h' has horizontal tangent lines at $x = 1$, $x = 3$, and $x = 5$. The areas of the regions bounded by the x -axis and the graph of h' on the intervals $[-1, 1]$, $[1, 4]$, $[4, 6]$, and $[6, 7]$ are 6, 8, 5.6, and 6, respectively. The function h is defined for all real numbers and satisfies $h(5) = 5$ and $h(7) = 8$.

- Find the x -coordinate of each critical point of h on the open interval $-1 < x < 7$.
- Identify the critical point from part (a) which corresponds to a relative minimum of h . Give a reason for your answer.
- Find the value of h at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute minimum value of h on $-1 \leq x \leq 7$. Justify your answer.
- On what open intervals is h both increasing and concave up? Give a reason for your answer.
- Explain why there must be a value of p , for $5 < p < 7$, such that $h(p) = 6.5$.
- Find $\lim_{x \rightarrow 7^-} \frac{h(x) - 8}{3x - 21}$. Give a reason for your answer.
- The function a is defined by $a(x) = (h(x))^2$. Find $a'(5)$. Show the computations that lead to your answer.

Unit 1 AP Topics 16

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = 1$, $x = 4$, and $x = 6$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1, 7)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1, 7)$. The point is still earned provided the three critical points, $x = 1$, $x = 4$, and $x = 6$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(1, 0)$, $(4, 0)$, $(6, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = 0$ at $x = 1$, $x = 4$, and $x = 6$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 6 and give the correct reason of “ h' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ h switches from decreasing to increasing at $x = 6$.”
- Saying that “ $h'(6)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of h' , the one given in the stem of the problem.

A response indicating that “ $h'(x)$ is negative for $x < 6$ and positive for $x > 6$ ” is allowed to give this statement as a reason to earn the point.

Unit 1 AP Topics 16

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative minimum value of h occurs where $h'(x)$ changes from negative to positive values.

$x = 6$ is the only location on $-1 < x < 7$ for which this is true, so $x = 6$ is the only location of a relative minimum value of h on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -1$, $x = 1$, or $x = 4$ is selected as the location of the local minimum in part (b). Note that $h(-1) = -6.4$, $h(1) = -0.4$, and $h(4) = 7.6$. No other x -coordinates will be read for consistency.

The response $h(6) = h(7) - 6 = 2$ earns both points.

The responses of both $8 - 6 = 2$ and $h(6) = 2$ do not earn the first point, but earn the second point, provided no other work is presented.

The string of equalities $h(6) = h(7) - \int_6^7 h'(x) dx = 8 - 6 = 2$ earns both points.

Unit 1 AP Topics 16

Special Case: If in part (b), a response picks $x = 5$ or $x = 7$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $h(6) = h(7) + \int_7^6 h'(x) dx$
- ☐ Answer

Solution:

$$\begin{aligned}
 h(6) &= h(7) + \int_7^6 h'(x) dx \\
 &= h(7) - \int_6^7 h'(x) dx = 8 - 6 = 2
 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 6$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 5$), and thus the third point cannot be earned.

It is possible to not refer to $h(7) = 8$ by indicating that h' is positive on the interval $(6, 7)$, thus eliminating 8 as a possible absolute minimum.

A complete list of given or computable values for $h(x)$ is given in the table below

Unit 1 AP Topics 16

x	$h(x)$
-1	-6.4
1	-0.4
4	7.6
5	5
6	2
7	8



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 6$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$h(-1) = h(7) + \int_7^{-1} h'(x) dx$$

$$= h(7) - \int_{-1}^7 h'(x) dx = 8 - (6 + 8 - 5.6 + 6) = -6.4$$

$$h(6) = 2$$

$$h(7) = 8$$

Therefore, the absolute minimum value of h on $-1 \leq x \leq 7$ is $h(-1) = -6.4$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, for example “(1, 3] and [6, 7)” earns the first point.

The reason “ h' is increasing” can be expressed by either “ h'' is positive” or “the slope of h' is positive.”

We will overlook the false implication that $h' > 0$ on $[1, 3]$ and $[6, 7]$.

Unit 1 AP Topics 16

The second point can be earned without earning the first point only if the given answer is a subset of $[1, 3] \cup [6, 7]$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where h' is positive.

h is concave up where h' is increasing.

h is both increasing and concave up on the open intervals $(1, 3)$ and $(6, 7)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that h is continuous because h is differentiable or h is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[5, 7]$, or the point is not earned.

The y -values of h need to be mentioned by either referring to the numerical values of 5 and 8, or mentioning $h(5)$ and $h(7)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since h is differentiable it’s continuous and 6.5 is between 5 and 8 such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation using Intermediate Value Theorem

Unit 1 AP Topics 16

Solution:

h is differentiable. $\Rightarrow h$ is continuous.

$$h(5) = 5 < 6.5 < 8 = h(7)$$

By the Intermediate Value Theorem, there must be a value of p , for $5 < p < 7$, such that $h(p) = 6.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of h' or h .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $h(7) - 8 = 0$ and $3 \cdot 7 - 21 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $h(7) = 8$ and $3 \cdot 7 = 21$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h(7)-8}{3 \cdot 7-21} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h'(x)}{3} = \frac{9}{3}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{0}{0}$ and

$\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h'(x)}{3}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

Unit 1 AP Topics 16

The third point is earned only for the answer of 3, or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h is twice differentiable. $\Rightarrow$ both h' and h are continuous. $\Rightarrow \lim_{x \rightarrow 7^-} h(x) = h(7)$ and $\lim_{x \rightarrow 7^-} h'(x) = h'(7)$.

$$\lim_{x \rightarrow 7^-} (h(x) - 8) = h(7) - 8 = 8 - 8 = 0$$

$$\lim_{x \rightarrow 7^-} (3x - 21) = 3 \cdot 7 - 21 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow 7^-} \frac{h(x) - 8}{3x - 21} = \lim_{x \rightarrow 7^-} \frac{h'(x)}{3}$$

$$= \frac{h'(7)}{3} = \frac{9}{3} = 3$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The Power rule point can be earned by just seeing $2 \cdot h(x)$ or $2 \cdot h(5)$.

The chain rule point can be earned by seeing an expression involving h but not h' times $h'(5)$.

All three points will be awarded for seeing only $2 \cdot 5 \cdot (-4)$, but $10 \cdot (-4)$, $2 \cdot (-20)$, or $-8 \cdot 5$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

Unit 1 AP Topics 16

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$a'(x) = 2h(x) \cdot h'(x)$$

$$a'(5) = 2 \cdot h(5) \cdot h'(5) = 2 \cdot 5 \cdot (-4) = -40$$

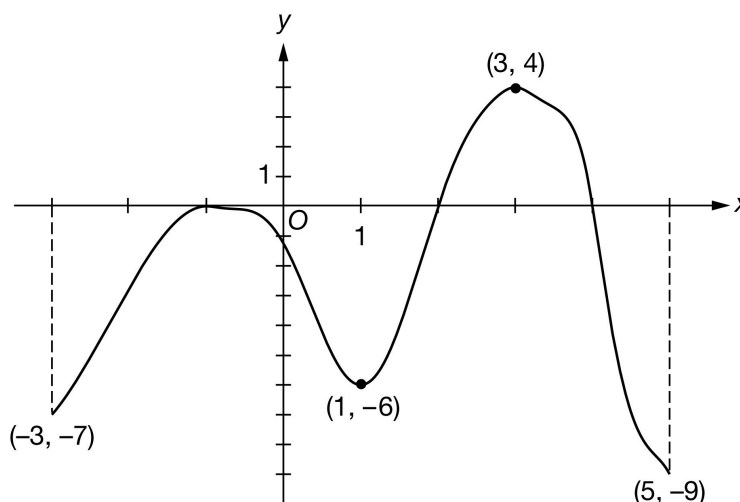
Unit 1 AP Topics 16

30. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f'

The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the closed interval $-3 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of f' on the intervals $[-3, -1]$, $[-1, 2]$, $[2, 4]$, and $[4, 5]$ are 6, 8, 5.6, and 6, respectively. The function f is defined for all real numbers and satisfies $f(3) = 10$ and $f(5) = 7$.

- Find the x -coordinate of each critical point of f on the open interval $-3 < x < 5$.
- Identify the critical point from part (a) which corresponds to a relative minimum of f . Give a reason for your answer.
- Find the value of f at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute minimum value of f on $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals is f both decreasing and concave down? Give a reason for your answer.
- Explain why there must be a value of d , for $3 < d < 5$, such that $f(d) = 8.5$.
- Find $\lim_{x \rightarrow 5^-} \frac{2f(x) - 14}{x^2 - 25}$. Give a reason for your answer.
- The function m is defined by $m(x) = (f(x))^2$. Find $m'(3)$. Show the computations that lead to your answer.

Unit 1 AP Topics 16

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -1$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(2, 0)$, $(4, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$f'(x) = 0$ at $x = -1$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 2 and give the correct reason of “ f' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ f switches from decreasing to increasing at $x = 2$ ”
- Saying that “ $f'(2)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of f' , the one given in the stem of the problem.

A response indicating that “ $f'(x)$ is negative for $x < 2$ and positive for $x > 2$ ” is allowed to give this statement as a reason to earn the point.

Unit 1 AP Topics 16

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative minimum value of f occurs where $f'(x)$ changes from negative to positive values.

$x = 2$ is the only location on $-3 < x < 5$ for which this is true, so $x = 2$ is the only location of a relative minimum value of f on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = -1$, or $x = 4$ is selected as the location of the local minimum in part (b). Note that $f(-3) = 21.4$, $f(-1) = 15.4$, and $f(4) = 13$. No other x -coordinates will be read for consistency.

The response $f(2) = f(5) - (5.6 - 6) = 7.4$ earns both points.

The responses of both $7 - (-0.4) = 7.4$ and $f(2) = 7.4$ do not earn the first point, but earn the second point, provided no other work is presented.

The string of equalities $f(2) = f(5) - \int_2^5 f'(x) dx = 7 - (-0.4) = 7.4$ earns both points.

Unit 1 AP Topics 16

Special Case: If in part (b), a response picks $x = 3$ or $x = 5$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $f(2) = f(5) + \int_5^2 f'(x) dx$
- ☐ Answer

Solution:

$$\begin{aligned} f(2) &= f(5) + \int_5^2 f'(x) dx = f(5) - \int_2^5 f'(x) dx \\ &= f(5) - \int_2^4 f'(x) dx - \int_4^5 f'(x) dx = 7 - 5.6 - (-6) = 7.4 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 2$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 3$), and thus the third point cannot be earned.

It is possible to not refer to $f(-3) = 21.4$ by indicating that f' is negative on the interval $(-3, -1)$, thus eliminating 21.4 as a possible absolute minimum.

A complete list of given or computable values for $f(x)$ is given in the table below.

Unit 1 AP Topics 16

x	$f(x)$
-3	21.4
-1	15.4
2	7.4
3	10
4	13
5	7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$f(-3) = f(5) + \int_5^{-3} f'(x) dx$$

$$= f(5) - \int_{-3}^5 f'(x) dx = 7 - (-6 + 5.6 - 8 - 6) = 21.4$$

$$f(2) = 7.4$$

$$f(5) = 7$$

So the absolute minimum value of f on $-3 \leq x \leq 5$ is

$$f(5) = 7.$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, for example “ $(-1, 1]$ and $[4, 5)$ ” earns the first point.

The reason “ f' is decreasing” can be expressed by either “ f'' is negative” or “the slope of f' is negative.”

We will overlook the false implication that $f' < 0$ on $[-1, 1]$ and $[4, 5]$.

Unit 1 AP Topics 16

The second point can be earned without earning the first point only if the given answer is a subset of $[-1, 1] \cup [4, 5]$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

f is decreasing where f' is negative.

f is concave down where f' is decreasing.

f is both decreasing and concave down on the open intervals $(-1, 1)$ and $(4, 5)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that f is continuous because f is differentiable or f is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[3, 5]$, or the point is not earned.

The y -values of f need to be mentioned by either referring to the numerical values of 7 and 10, or mentioning $f(3)$ and $f(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since f is differentiable it’s continuous and 8.5 is between 7 and 10 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation using Intermediate Value Theorem

Unit 1 AP Topics 16

Solution:

f is differentiable. $\Rightarrow f$ is continuous.

$$f(5) = 7 < 8.5 < 10 = f(3)$$

By the Intermediate Value Theorem, there must be a value of d , for $3 < d < 5$, such that $f(d) = 8.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of f' or f .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $2f(5) - 14 = 0$ and $5^2 - 25 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $2f(5) = 14$ and $5^2 = 25$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f(5)-14}{5^2-25} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f'(x)}{2x} = \frac{-9}{5}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{0}{0}$ and

$\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

Unit 1 AP Topics 16

The third point is earned only for the answer of $-\frac{9}{5}$, or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

f is twice differentiable. $\Rightarrow$ both f' and f are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} f(x) = f(5)$ and $\lim_{x \rightarrow 5^-} f'(x) = f'(5)$.

$$\lim_{x \rightarrow 5^-} (2f(x) - 14) = 2 \cdot f(5) - 14 = 14 - 14 = 0$$

$$\lim_{x \rightarrow 5^-} (x^2 - 25) = 5^2 - 25 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{2f(x) - 14}{x^2 - 25} &= \lim_{x \rightarrow 5^-} \frac{2f'(x)}{2x} \\ &= \frac{2f'(5)}{2 \cdot 5} = \frac{-9}{5} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $2 \cdot f(x)$ or $2 \cdot f(3)$.

The chain rule point can be earned by seeing an expression involving f but not f' times $f'(3)$.

All three points will be awarded for seeing only $2 \cdot 10 \cdot 4$, but $20 \cdot 4$, $2 \cdot 40$, or $8 \cdot 10$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

Unit 1 AP Topics 16

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$m'(x) = 2f(x) \cdot f'(x)$$

$$m'(3) = 2 \cdot f(3) \cdot f'(3) = 2 \cdot 10 \cdot 4 = 80$$

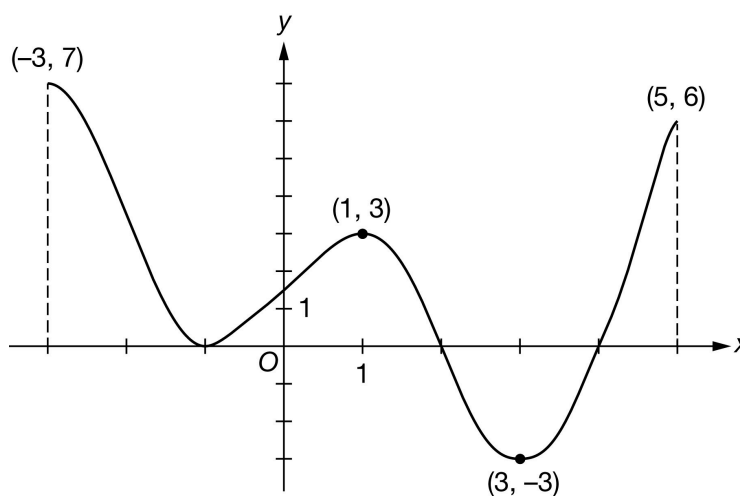
Unit 1 AP Topics 16

31. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The figure above shows the graph of g' , the derivative of a twice-differentiable function g , on the closed interval $-3 \leq x \leq 5$. The graph of g' has horizontal tangent lines at $x = -3$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of g' on the intervals $[-3, -1]$, $[-1, 2]$, $[2, 4]$, and $[4, 5]$ are 7, 5, 4, and 3, respectively. The function g is defined for all real numbers and satisfies $g(3) = 8$ and $g(5) = 9$.

- (a) Find the x -coordinate of each critical point of g on the open interval $-3 < x < 5$.
- (b) Identify the critical point from part (a) which corresponds to a relative minimum of g . Give a reason for your answer.
- (c) Find the value of g at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- (d) Find the absolute minimum value of g on $-3 \leq x \leq 5$. Justify your answer.
- (e) On what open intervals is g both decreasing and concave down? Give a reason for your answer.
- (f) Explain why there must be a value of r , for $3 < r < 5$, such that $g(r) = 8.5$.
- (g) Find $\lim_{x \rightarrow 5^-} \frac{g(x) - 9}{x^2 - 25}$. Give a reason for your answer.
- (h) The function f is defined by $f(x) = (g(x))^3$. Find $f'(3)$. Show the computations that lead to your answer.

Unit 1 AP Topics 16

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -1$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(2, 0)$, $(4, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = 0$ at $x = -1$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 4 and give the correct reason of “ g' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ g switches from decreasing to increasing at $x = 4$ ”
- Saying that “ $g'(4)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of g' , the one given in the stem of the problem.

A response indicating that “ $g'(x)$ is negative for $x < 4$ and positive for $x > 4$ ” is allowed to give this statement as a reason to earn the point.

Unit 1 AP Topics 16

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 4$ is the only location on $-3 < x < 5$ for which this is true, so $x = 4$ is the only location of a relative minimum value of g on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = -1$, or $x = 2$ is selected as the location of the local minimum in part (b). Note that $g(-3) = -2$, $g(-1) = 5$, and $g(2) = 10$. No other x -coordinates will be read for consistency.

The response $g(4) = g(5) - 3 = 6$ earns both points.

The responses of both $9 - 3 = 6$ and $g(4) = 6$ do not earn the first point, but earns the second point, provided no other work is presented.

Special Case: If a response uses $\int_4^3 g'(x) dx = 2$ then

$g(4) = g(3) - \int_4^3 g'(x) dx = 8 - 2 = 6$ earns both points, because it's possible to find $\int_4^3 g'(x) dx = 2$ using the areas and initial conditions given.

Unit 1 AP Topics 16

any response using perceived symmetry about $x = 3$ can earn at most the first point, for example

$$g(4) = g(3) - \int_4^3 g'(x) dx = 8 - \frac{1}{2} \cdot 4 = 6 \text{ earns just the first point.}$$

Special Case: If in part (b), a response picks $x = 3$ or $x = 5$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $g(4) = g(5) + \int_5^4 g'(x) dx$

☐ Answer

Solution:

$$\begin{aligned} g(4) &= g(5) + \int_5^4 g'(x) dx \\ &= g(5) - \int_4^5 g'(x) dx = 9 - 3 = 6 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 4$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 3$), and thus the third point cannot be earned.

It is possible to not refer to $g(5) = 9$ by indicating that g' is positive on the interval $(4, 5)$, thus eliminating 9 as a possible absolute minimum.

A complete list of given or computable values for $g(x)$ is given in the table below.

Unit 1 AP Topics 16

x	$g(x)$
-3	-2
1	5
2	10
3	8
4	6
5	9



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 4$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$g(-3) = g(5) + \int_5^{-3} g'(x) dx$$

$$= g(5) - \int_{-3}^5 g'(x) dx = 9 - (7 + 5 - 4 + 3) = -2$$

$$g(4) = 6$$

$$g(5) = 9$$

So the absolute minimum value of g on $-3 \leq x \leq 5$ is $g(-3) = -2$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[2, 3]$, $[2, 3)$, $(2, 3]$, and $(2, 3)$ all earn the first point.

The reason “ g' is decreasing” can be expressed by either “ g'' is negative” or “the slope of g' is negative.”

We will overlook the false implication that $g' < 0$ on $[2, 3]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[2, 3]$.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where g' is negative.

g is concave down where g' is decreasing.

g is both decreasing and concave down on the open interval $(2, 3)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that g is continuous because g is differentiable or g is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[3, 5]$, or the point is not earned.

The y -values of g need to be mentioned by either referring to the numerical values of 8 and 9, or mentioning $g(3)$ and $g(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since g is differentiable it’s continuous and 8.5 is between 8 and 9 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

Unit 1 AP Topics 16

g is differentiable. $\Rightarrow g$ is continuous.

$$g(3) = 8 < 8.5 < 9 = g(5)$$

By the Intermediate Value Theorem, there must be a value of r , for $3 < r < 5$, such that $g(r) = 8.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of g' or g .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $g(5) - 9 = 0$ and $5^2 - 25 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $g(5) = 9$ and $5^2 = 25$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g(5)-9}{5^2-25} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g'(x)}{2x} = \frac{3}{5}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{0}{0}$ and $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{3}{5}$, or its numeric equivalent.

Unit 1 AP Topics 16



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ that both g' and g are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} g(x) = g(5)$ and $\lim_{x \rightarrow 5^-} g'(x) = g'(5)$.

$$\lim_{x \rightarrow 5^-} (g(x) - 9) = g(5) - 9 = 9 - 9 = 0$$

$$\lim_{x \rightarrow 5^-} (x^2 - 25) = 5^2 - 25 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{g(x) - 9}{x^2 - 25} &= \lim_{x \rightarrow 5^-} \frac{g'(x)}{2x} \\ &= \frac{g'(5)}{2 \cdot 5} = \frac{6}{10} = \frac{3}{5} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $3 \cdot (g(x))^2$ or $3 \cdot (g(3))^2$.

The chain rule point can be earned by seeing an expression involving g but not g' times $g'(3)$.

All three points will be awarded for seeing only $3 \cdot 8^2 \cdot (-3)$, but $3 \cdot 64 \cdot (-3)$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

Unit 1 AP Topics 16

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$f'(x) = 3(g(x))^2 \cdot g'(x)$$

$$f'(3) = 3(g(3))^2 \cdot g'(3) = 3 \cdot 8^2 \cdot (-3) = -576$$

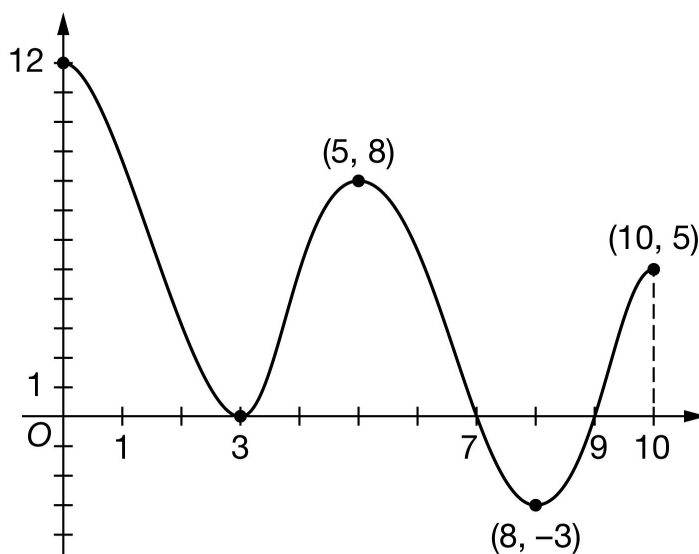
Unit 1 AP Topics 16

32. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The figure above shows a portion of the graph of the differentiable function g . The graph of g has horizontal tangents at $x = 3$, $x = 5$, and $x = 8$. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[0, 3]$, $[3, 7]$, $[7, 9]$, and $[9, 10]$ are 18, 19, 4, and 3, respectively. Let h be the function defined by $h(x) = \int_3^x g(t) \, dt$.

- For $0 < x < 10$, find all values of x at which h has a critical point.
- Classify each critical point found in part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- On what open intervals contained in $0 < x < 10$ is the graph of h both increasing and concave up? Give a reason for your answer.
- Must there exist a value of c , for $0 < c < 3$, such that $h(c) = -15$. Justify your answer.
- Find the average value of g over the interval $0 \leq x \leq 10$. Show the computations that lead to your answer.
- Evaluate $\lim_{x \rightarrow 7} \frac{h(x) - 4x + 9}{x^2 - 49}$. Give a reason for your answer.
- It is known that $\lim_{x \rightarrow \infty} g(x) = 3$. Find the value of $\int_{10}^{\infty} g'\left(\frac{x}{2}\right) \, dx$ or show that it does not exist.

Unit 1 AP Topics 16

(h) Is there a value of r such that the series $18 + 18r + 18r^2 + \cdots + 18r^n + \cdots$ equals the value of $g(5)$? Explain why or why not.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for providing all three of the values of x . No reason is required.

Points can be presented as ordered pairs with the correct h -value as the second coordinate, i.e., $(3, 0)$, $(7, 19)$, and $(9, 15)$. With any other second coordinate, the point is not earned.

Ignore any value in the response that is outside the open interval $(0, 10)$.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

h has a critical point where $h'(x) = g(x) = 0$.

$$h'(x) = g(x) = 0 \Rightarrow x = 3, x = 7, x = 9$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The three points are independent, and, for each point, a justification is required to earn the point. So, a response might classify each value of x correctly and earn no points in part (b).

Do not read any classifications for $x = 0$, $x = 10$, or any value outside the interval $[0, 10]$.

A response that uses g , rather than h' , in all three of its justifications, without having noted that $h' = g$, in either part (a) or part (b), can earn at most two of the three points.

A response that refers only to h' in all three parts of its justifications, without having noted that $h' = g$ is still eligible for all three points.

If a response presents ordered pairs, rather than values of x , to be classified, the response must either supply the correct value of $h(x)$ in the ordered pair $[(3, 0), (7, 19), \text{ and } (9, 15)]$ or refer explicitly to the values of x to earn points.

Unit 1 AP Topics 16

A sign chart, by itself, does not provide sufficient justification for the classification of any of the values of x .

The Second Derivative Test can be used to justify the classification of $x = 7$ and $x = 9$, but not for $x = 3$.

References to “it” are not equivalent to h' or g . Similarly, a statement like “the slope goes from positive to negative” is not equivalent to h' going from positive to negative.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Neither at $x = 3$ with reason
- ☐ Relative maximum at $x = 7$ with reason
- ☐ Relative minimum at $x = 9$ with reason

Solution:

$h'(x) = g(x)$ does not change sign at $x = 3$. Therefore, h has neither a relative minimum nor a relative maximum at $x = 3$.

$h'(x) = g(x)$ changes from positive to negative at $x = 7$. Therefore, h has a relative maximum at $x = 7$.

$h'(x) = g(x)$ changes from negative to positive at $x = 9$. Therefore, h has a relative minimum at $x = 9$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct answer, even if the reasoning presented in the response to support this answer is inadequate or incorrect. The intervals can be presented as either open [preferred] or closed.

A response that contains only the correct answer, but with no supporting work, earns the first point, but not the second.

A response that uses g , rather than h' , without having noted that $h' = g$, in either part (a), part (b), or part (c) cannot earn the second point.

To earn the second point, the response must only give a reason why the presented answer satisfies the required conditions. It is not necessary to list separately the intervals in which the graph of h is increasing and the intervals in which the graph is concave up.

A response can earn the second point without earning the first point if the response contains only one of the two required intervals and presents a correct reason.

Unit 1 AP Topics 16

To determine the intervals on which the graph of h is concave up, the response can contain a consideration of the sign of $h'' = g'$ rather than the increasing/decreasing behavior of g .



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where $h' = g$ is positive.

h is concave up where $h' = g$ is increasing.

$h' = g$ is positive and increasing on the open intervals $3 < x < 5$ and $9 < x < 10$.

Therefore, the graph of h is both increasing and concave up on those intervals.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

To earn the first point, the response must consider the continuity of h and give the correct values of $h(x)$ at $x = 0$ and $x = 3$.

Stating that h is continuous on the open interval $(0, 3)$, a larger interval, or simply that h is continuous with no reference as to where, is sufficient consideration of the continuity of h .

If the response presents incorrect values for $h(x)$ at $x = 0$ and $x = 3$ such that -15 does not fall in the interval with endpoints $h(0)$ and $h(3)$, the response will earn the second point by claiming that there need not exist such a value of c . This response need not consider the continuity of h .

Similar to the above, a response with incorrect values for $h(x)$ at $x = 0$ and $x = 3$ such that -15 does fall in the interval with endpoints $h(0)$ and $h(3)$ will earn the second point by claiming that there must exist such a value of c . This response must mention the continuity of h to earn the second point.

A response can earn the second point, even if the continuity of h is not considered in the response, in which case the first point would not be earned.

It is possible to provide an acceptable argument, based on “accumulating area” for the correct conclusion without invoking the Intermediate Value Theorem.

Unit 1 AP Topics 16

The response need not explicitly invoke the Intermediate Value Theorem to earn the second point. However, a response in which the conclusion is based on the Mean Value Theorem, for example, would not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable. $\Rightarrow h$ is continuous on $0 \leq x \leq 3$.

$$h(0) = \int_3^0 g(t) dt = -18$$

$$h(3) = \int_3^3 g(t) dt = 0$$

$$h(0) < -15 < h(3)$$

Therefore, by the Intermediate Value Theorem, there is a value of c , for $0 < c < 3$, such that $h(c) = -15$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response earns the first point by presenting either $\frac{1}{10} \int_0^{10} g(x) dx$ as in the model solution or by presenting the four areas [18, 19, 4, and 3] in a sum, such as “18 + 19 – 4 + 3” or “18 + 19 + 4 + 3.” Unless corrected, using the latter expression would not earn the second point.

Read to the end of the response to be sure there are no errors. For example, if the following occurs, with an incorrect value at the end of the computation, the response would not earn the second point:

$$\dots = \frac{18+19-4+3}{10} = \frac{36}{10} = \frac{19}{5}$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

Unit 1 AP Topics 16

- ☐ Average value setup – OR – Area
- ☐ Answer

Solution:

$$\frac{1}{10-0} \int_0^{10} g(x) \, dx = \frac{1}{10}(18 + 19 - 4 + 3) = \frac{1}{10} \cdot 36 = 3.6$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

To earn the first point, the response does not have to address the continuity or differentiability of g or h .

The first point is earned with the claim that the numerator and denominator separately go to 0. The first point is not earned if the response claims that the given limit equals $\frac{0}{0}$. However, for example, a response that claims that the given limit is “of the form $\frac{0}{0}$ ” or that uses arrows to show that the numerator and denominator each go to 0 will earn the first point. So, if you see $\frac{0}{0}$ in the response, read more carefully before deciding not to award the first point.

The second point is earned by applying L'Hospital's Rule, i.e., demonstrating that the limit of the quotient of the derivatives, or the quotient of the values of the derivatives, should be evaluated. A response with minor differentiation errors, e.g., having $2x - 49$ in the denominator, can still earn the second point. However, with this response the third point cannot be earned.

Only an answer of $-\frac{2}{7}$ or its equivalent earns the third point, and be sure to read to the end of the response to be sure that it is all correct. For example, a response that presented the following would not earn the third point:

$$\dots = \frac{0-4}{14} = \frac{2}{7}$$

A response can earn all three points without ever using limit notation; for example, arrows, correctly used, are acceptable. Thus, $h(x) - 4x + 9 \rightarrow 19 - 4 \cdot 7 + 9 = 0$ together with $x^2 - 49 \rightarrow 0$ would earn the first point, and $\frac{h'(x)-4}{2x} = \frac{g(x)-4}{2x} \rightarrow \frac{0-4}{14}$ would earn the second and third points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule

Unit 1 AP Topics 16

☐ Answer
Solution:
 $h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.

 $\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \rightarrow 7} h(x) = h(7)$ and $\lim_{x \rightarrow 7} h'(x) = h'(7)$.

$$\lim_{x \rightarrow 7} (h(x) - 4x + 9) = h(7) - 4 \cdot 7 + 9 = 19 - 28 + 9 = 0$$

$$\lim_{x \rightarrow 7} (x^2 - 49) = 7^2 - 49 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 7} \frac{h(x) - 4x + 9}{x^2 - 49} &= \lim_{x \rightarrow 7} \frac{h'(x) - 4}{2x} \\ &= \frac{h'(7) - 4}{2 \cdot 7} = \frac{g(7) - 4}{14} = \frac{0 - 4}{14} = -\frac{2}{7} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Only the correct antiderivative earns the first point. However, an antiderivative of the form either $K \cdot g\left(\frac{x}{2}\right)$ or $K \cdot g(x)$, where K is a constant other than 2, can earn the second point.

We will accept informal notation, like the following: $\dots = 2(g(\infty) - g(8)) = 2(3 - 8) = -10$

However, the response must, at some point, explicitly show that a limit is needed to evaluate the improper integral.

If the evaluation uses a u -substitution, be careful to check that the limits have been changed appropriately. If they have not, this point will not be earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Antiderivative
- ☐ Limit and answer

Solution:

Unit 1 AP Topics 16

$$\begin{aligned}\int_{10}^{\infty} g' \left(\frac{x}{2} \right) dx &= \lim_{b \rightarrow \infty} \int_{10}^b g' \left(\frac{x}{2} \right) dx = \lim_{b \rightarrow \infty} 2g \left(\frac{x}{2} \right) \Big|_{10}^b \\ &= \lim_{b \rightarrow \infty} 2g \left(\frac{b}{2} \right) - 2g(5) = 2 \cdot 3 - 2 \cdot 8 = -10\end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for giving the correct sum for the geometric series, but the sum must use our constant, 18, rather than a “generic” a . The response does not have to consider the radius of convergence to earn the first point. Note that a response in which a is subsequently replaced with 18 does earn the first point.

Only responses with a correct expression for the sum of the geometric series are eligible for the second point, and to earn the second point a generic a must be replaced by 18 at some point. The second point is an “algebra point,” but the response must give the value of r correctly.

The three decimal places requirement for decimal presentations does not apply to the value of r presented in the response, since this value is an “intermediate” result, and need only be presented with accuracy sufficient to determine whether its absolute value is less than 1. Thus, $r = -\frac{5}{4} = -1.3$ would earn the second point.

A response using an incorrect value for $g(5)$ cannot earn the second point, but can earn the third point.

A response with an incorrect value of r is eligible for the third point, if the conclusion for that r is correct.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Sum of geometric series
- ☐ Solves for r
- ☐ Answer with reason

Solution:

$18 + 18r + 18r^2 + \cdots + 18r^n + \cdots$ is a geometric series that converges to $\frac{18}{1-r}$ for $|r| < 1$ and diverges for $|r| \geq 1$.

$$g(5) = 8$$

$$\frac{18}{1-r} = 8 \Rightarrow 18 = 8 - 8r \Rightarrow r = -\frac{10}{8} = -\frac{5}{4}$$

Because $\left| -\frac{5}{4} \right| \geq 1$, there is no value of r such that the given series equals $g(5) = 8$.

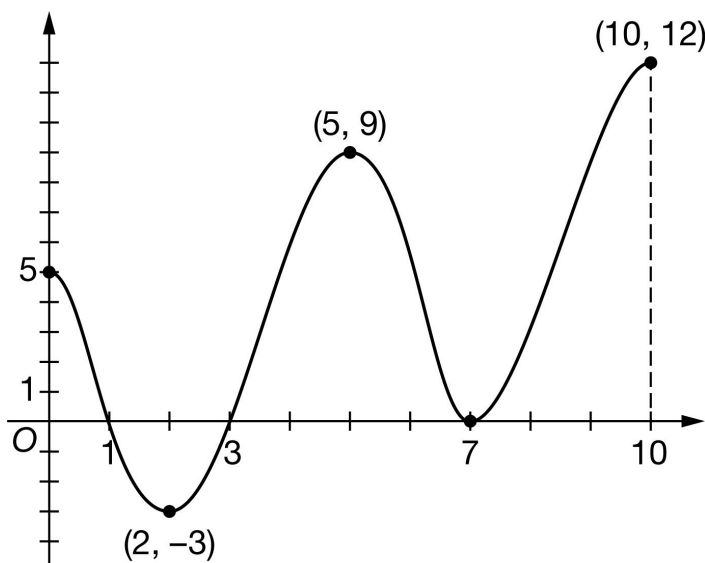
Unit 1 AP Topics 16

33. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The figure above shows a portion of the graph of the differentiable function g . The graph of g has horizontal tangents at $x = 2$, $x = 5$, and $x = 7$. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[0, 1]$, $[1, 3]$, $[3, 7]$ and $[7, 10]$ are 3, 4, 21, and 18, respectively. Let h be the function defined by $h(x) = \int_3^x g(t) \, dt$.

- For $0 < x < 10$, find all values of x at which h has a critical point.
- Classify each critical point found in part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- On what open intervals contained in $0 < x < 10$ is the graph of h both increasing and concave down? Give a reason for your answer.
- Must there exist a value of c , for $1 < c < 3$, such that $h(c) = 3.5$? Justify your answer.
- Find the average value of g over the interval $0 \leq x \leq 10$. Show the computations that lead to your answer.
- Evaluate $\lim_{x \rightarrow 7} \frac{h(x) - 3x}{x^2 - 49}$. Give a reason for your answer.
- It is known that $\lim_{x \rightarrow \infty} g(x) = 6$. Find the value of $\int_5^{\infty} g'(2x) \, dx$ or show that it does not exist.

Unit 1 AP Topics 16

(h) Is there a value of r such that the series $20 + 20r + 20r^2 + \cdots + 20r^n + \cdots$ equals the value of $g(5)$? Explain why or why not.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for providing all three of the values of x . No reason is required.

Points can be presented as ordered pairs with the correct h -value as the second coordinate, i.e., $(1, 4)$, $(3, 0)$, and $(7, 21)$. With any other second coordinate, the point is not earned.

Ignore any value in the response that is outside the open interval $(0, 10)$.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

h has a critical point where $h'(x) = g(x) = 0$.

$$h'(x) = g(x) = 0 \Rightarrow x = 1, x = 3, x = 7$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The three points are independent, and, for each point, a justification is required to earn the point. So, a response might classify each value of x correctly and earn no points in part (b).

Do not read any classifications for $x = 0$, $x = 10$, or any value outside the interval $[0, 10]$.

A response that uses g , rather than h' , in all three of its justifications, without having noted that $h' = g$, in either part (a) or part (b), can earn at most two of the three points.

A response that refers only to h' in all three parts of its justifications, without having noted that $h' = g$ is still eligible for all three points.

If a response presents ordered pairs, rather than values of x , to be classified, the response must either supply the correct value of $h(x)$ in the ordered pair $[(1, 4), (3, 0), \text{ and } (7, 21)]$ or refer explicitly to the values of x to earn points.

Unit 1 AP Topics 16

A sign chart, by itself, does not provide sufficient justification for the classification of any of the values of x .

The Second Derivative Test can be used to justify the classification of $x = 1$ and $x = 3$, but not for $x = 7$.

References to “it” are not equivalent to h' or g . Similarly, a statement like “the slope goes from positive to negative” is not equivalent to h' going from positive to negative.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum at $x = 1$ with reason
- ☐ Relative minimum at $x = 3$ with reason
- ☐ Neither at $x = 7$ with reason

Solution:

$h'(x) = g(x)$ changes from positive to negative at $x = 1$. Therefore, h has a relative maximum at $x = 1$.

$h'(x) = g(x)$ changes from negative to positive at $x = 3$. Therefore, h has a relative minimum at $x = 3$.

$h'(x) = g(x)$ does not change sign at $x = 7$. Therefore, h has neither a relative minimum nor a relative maximum at $x = 7$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct answer, even if the reasoning presented in the response to support this answer is inadequate or incorrect. The intervals can be presented as either open [preferred] or closed.

A response that contains only the correct answer, but with no supporting work, earns the first point, but not the second.

A response that uses g , rather than h' , without having noted that $h' = g$, in either part (a), part (b), or part (c) cannot earn the second point.

To earn the second point, the response must only give a reason why the presented answer satisfies the required conditions. It is not necessary to list separately the intervals in which the graph of h is decreasing and the intervals in which the graph is concave down.

A response can earn the second point without earning the first point if the response contains only one of the two required intervals and presents a correct reason.

Unit 1 AP Topics 16

To determine the intervals on which the graph of h is concave down, the response can contain a consideration of the sign of $h'' = g'$ rather than the increasing/decreasing behavior of g .



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where $h' = g$ is positive.

h is concave down where $h' = g$ is decreasing.

$h' = g$ is positive and decreasing on the open intervals $0 < x < 1$ and $5 < x < 7$.

Therefore, the graph of h is both increasing and concave down on those intervals.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

To earn the first point, the response must consider the continuity of h and give the correct values of $h(x)$ at $x = 1$ and $x = 3$.

Stating that h is continuous on the open interval $(1, 3)$, a larger interval, or simply that h is continuous with no reference as to where, is sufficient consideration of the continuity of h .

If the response presents incorrect values for $h(x)$ at $x = 1$ and $x = 3$ such that 3.5 does not fall in the interval with endpoints $h(1)$ and $h(3)$, the response will earn the second point by claiming that there need not exist such a value of c . This response need not consider the continuity of h .

Similar to the above, a response with incorrect values for $h(x)$ at $x = 1$ and $x = 3$ such that 3.5 does fall in the interval with endpoints $h(1)$ and $h(3)$ will earn the second point by claiming that there must exist such a value of c . This response must mention the continuity of h to earn the second point.

A response can earn the second point, even if the continuity of h is not considered in the response, in which case the first point would not be earned.

It is possible to provide an acceptable argument, based on “accumulating area” for the correct conclusion without invoking the Intermediate Value Theorem.

Unit 1 AP Topics 16

The response need not explicitly invoke the Intermediate Value Theorem to earn the second point. However, a response in which the conclusion is based on the Mean Value Theorem, for example, would not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable. $\Rightarrow h$ is continuous on $1 \leq x \leq 3$.

$$h(1) = \int_3^1 g(t) dt = 4$$

$$h(3) = \int_3^3 g(t) dt = 0$$

$$h(3) < 3.5 < h(1)$$

Therefore, by the Intermediate Value Theorem, there is a value of c , for $1 < c < 3$, such that $h(c) = 3.5$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response earns the first point by presenting either $\frac{1}{10} \int_0^{10} g(x) dx$ as in the model solution or by presenting the four areas [3, 4, 21, and 18] in a sum, such as “3 – 4 + 21 + 18” or “3 + 4 + 21 + 18.” Unless corrected, using the latter expression would not earn the second point.

Read to the end of the response to be sure there are no errors. For example, if the following occurs, with an incorrect value at the end of the computation, the response would not earn the second point:

$$\dots = \frac{3-4+21+18}{10} = \frac{38}{10} = \frac{17}{5}$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

Unit 1 AP Topics 16

- ☐ Average value setup – OR – Area
- ☐ Answer

Solution:

$$\frac{1}{10-0} \int_0^{10} g(x) \, dx = \frac{1}{10}(3 - 4 + 21 + 18) = \frac{1}{10} \cdot 38 = 3.8$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

To earn the first point, the response does not have to address the continuity or differentiability of g or h .

The first point is earned with the claim that the numerator and denominator separately go to 0. The first point is not earned if the response claims that the given limit equals $\frac{0}{0}$. However, for example, a response that claims that the given limit is “of the form $\frac{0}{0}$ ” or that uses arrows to show that the numerator and denominator each go to 0 will earn the first point. So, if you see $\frac{0}{0}$ in the response, read more carefully before deciding not to award the first point.

The second point is earned by applying L'Hospital's Rule, i.e., demonstrating that the limit of the quotient of the derivatives, or the quotient of the values of the derivatives, should be evaluated. A response with minor differentiation errors, e.g., having $2x - 49$ in the denominator, can still earn the second point. However, with this response the third point cannot be earned.

Only an answer of $-\frac{3}{14}$ or its equivalent earns the third point, and be sure to read to the end of the response to be sure that it is all correct. For example, a response that presented the following would not earn the third point:

$$\dots = \frac{0-3}{14} = \frac{3}{14}$$

A response can earn all three points without ever using limit notation; for example, arrows, correctly used, are acceptable. Thus, $h(x) - 3x \rightarrow 21 - 3 \cdot 7 = 0$ together with $x^2 - 49 \rightarrow 0$ would earn the first point, and

$$\frac{h'(x)-3}{2x} = \frac{g(x)-3}{2x} \rightarrow \frac{0-3}{14} \text{ would earn the second and third points.}$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule

Unit 1 AP Topics 16

☐ Answer
Solution:
 $h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.

 $\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \rightarrow 7} h(x) = h(7)$ and $\lim_{x \rightarrow 7} h'(x) = h'(7)$.

$$\lim_{x \rightarrow 7} (h(x) - 3x) = h(7) - 3 \cdot 7 = 21 - 21 = 0$$

$$\lim_{x \rightarrow 7} (x^2 - 49) = 7^2 - 49 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 7} \frac{h(x) - 3x}{x^2 - 49} &= \lim_{x \rightarrow 7} \frac{h'(x) - 3}{2x} \\ &= \frac{h'(7) - 3}{2 \cdot 7} = \frac{g(7) - 3}{14} = \frac{0 - 3}{14} = -\frac{3}{14} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Only the correct antiderivative earns the first point. However, an antiderivative of the form either $K \cdot g(2x)$ or $K \cdot g(x)$, where K is a constant other than $\frac{1}{2}$, can earn the second point.

We will accept informal notation, like the following: $\dots = \frac{g(\infty) - g(10)}{2} = \frac{6 - 12}{2} = -3$

However, the response must, at some point, explicitly show that a limit is needed to evaluate the improper integral.

If the evaluation uses a u -substitution, be careful to check that the limits have been changed appropriately. If they have not, this point will not be earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Antiderivative
- ☐ Limit and answer

Solution:

Unit 1 AP Topics 16

$$\int_5^{\infty} g'(2x) dx = \lim_{b \rightarrow \infty} \int_5^b g'(2x) dx = \lim_{b \rightarrow \infty} \left. \frac{g(2x)}{2} \right|_5^b$$

$$= \lim_{b \rightarrow \infty} \frac{g(2b)}{2} - \frac{g(10)}{2} = \frac{6}{2} - \frac{12}{2} = -3$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for giving the correct sum for the geometric series, but the sum must use our constant, 20, rather than a “generic” a . The response does not have to consider the radius of convergence to earn the first point. Note that a response in which a is subsequently replaced with 20 does earn the first point.

Only responses with a correct expression for the sum of the geometric series are eligible for the second point, and to earn the second point a generic a must be replaced by 20 at some point. The second point is an “algebra point,” but the response must give the value of r correctly.

The three decimal places requirement for decimal presentations does not apply to the value of r presented in the response, since this value is an “intermediate” result, and need only be presented with accuracy sufficient to determine whether its absolute value is less than 1. Thus, $r = -\frac{11}{9} = -1.2$ would earn the second point.

A response using an incorrect value for $g(5)$ cannot earn the second point, but can earn the third point.

A response with an incorrect value of r is eligible for the third point, if the conclusion for that r is correct.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Sum of geometric series
- ☐ Solves for r
- ☐ Answer with reason

Solution:

$20 + 20r + 20r^2 + \cdots + 20r^n + \cdots$ is a geometric series that converges to $\frac{20}{1-r}$ for $|r| < 1$ and diverges for $|r| \geq 1$.

$$g(5) = 9$$

$$\frac{20}{1-r} = 9 \Rightarrow 20 = 9 - 9r \Rightarrow r = -\frac{11}{9}$$

Unit 1 AP Topics 16

Because $\left| -\frac{11}{9} \right| \geq 1$, there is no value of r such that the given series equals $g(5) = 9$.

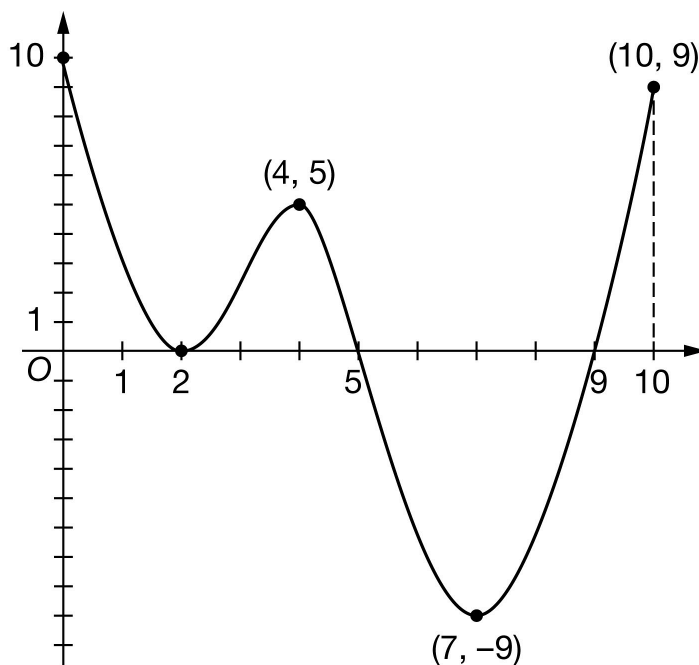
Unit 1 AP Topics 16

34. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The figure above shows a portion of the graph of the differentiable function g . The graph of g has horizontal tangents at $x = 2$, $x = 4$, and $x = 7$. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[0, 2]$, $[2, 5]$, $[5, 9]$, and $[9, 10]$ are 7, 8, 22, and 4, respectively. Let h be the function defined by $h(x) = \int_2^x g(t) \, dt$.

- For $0 < x < 10$, find all values of x at which h has a critical point.
- Classify each critical point found in part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- On what open intervals contained in $0 < x < 10$ is the graph of h both increasing and concave down? Give a reason for your answer.
- Must there exist a value of c , for $0 < c < 2$, such that $h(c) = -3.5$? Justify your answer.
- Find the average value of g over the interval $0 \leq x \leq 10$. Show the computations that lead to your answer.

Unit 1 AP Topics 16

(f) Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 25}{h(x) - 2x + 2}$. Give a reason for your answer.

(g) It is known that $\lim_{x \rightarrow \infty} g(x) = 3$. Find the value of $\int_2^{\infty} g'(5x) dx$ or show that it does not exist.

(h) Is there a value of r such that the series $12 + 12r + 12r^2 + \cdots + 12r^n + \cdots$ equals the value of $g(4)$? Explain why or why not.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for providing all three of the values of x . No reason is required.

Points can be presented as ordered pairs with the correct h -value as the second coordinate, i.e., $(2, 0)$, $(5, 8)$, and $(9, -14)$. With any other second coordinate, the point is not earned.

Ignore any value in the response that is outside the open interval $(0, 10)$.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

h has a critical point where $h'(x) = g(x) = 0$.

$$h'(x) = g(x) = 0 \Rightarrow x = 2, x = 5, x = 9$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The three points are independent, and, for each point, a justification is required to earn the point. So, a response might classify each value of x correctly and earn no points in part (b).

Do not read any classifications for $x = 0$, $x = 10$, or any value outside the interval $[0, 10]$.

A response that uses g , rather than h' , in all three of its justifications, without having noted that $h' = g$, in either part (a) or part (b), can earn at most two of the three points.

A response that refers only to h' in all three parts of its justifications, without having noted that $h' = g$ is still eligible for all three points.

Unit 1 AP Topics 16

If a response presents ordered pairs, rather than values of x , to be classified, the response must either supply the correct value of $h(x)$ in the ordered pair $[(2, 0), (5, 8), \text{ and } (9, -14)]$ or refer explicitly to the values of x to earn points.

A sign chart, by itself, does not provide sufficient justification for the classification of any of the values of x .

The Second Derivative Test can be used to justify the classification of $x = 5$ and $x = 9$, but not for $x = 2$.

References to “it” are not equivalent to h' or g . Similarly, a statement like “the slope goes from positive to negative” is not equivalent to h' going from positive to negative.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Neither at $x = 2$ with reason
- ☐ Relative maximum at $x = 5$ with reason
- ☐ Relative minimum at $x = 9$ with reason

Solution:

$h'(x) = g(x)$ does not change sign at $x = 2$. Therefore, h has neither a relative minimum nor a relative maximum at $x = 2$.

$h'(x) = g(x)$ changes from positive to negative at $x = 5$. Therefore, h has a relative maximum at $x = 5$.

$h'(x) = g(x)$ changes from negative to positive at $x = 9$. Therefore, h has a relative minimum at $x = 9$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct answer, even if the reasoning presented in the response to support this answer is inadequate or incorrect. The intervals can be presented as either open [preferred] or closed.

A response that contains only the correct answer, but with no supporting work, earns the first point, but not the second.

A response that uses g , rather than h' , without having noted that $h' = g$, in either part (a), part (b), or part (c) cannot earn the second point.

To earn the second point, the response must only give a reason why the presented answer satisfies the required conditions. It is not necessary to list separately the intervals in which the graph of h is increasing and the intervals in which the graph is concave down.

Unit 1 AP Topics 16

A response can earn the second point without earning the first point if the response contains only one of the two required intervals and presents a correct reason.

To determine the intervals on which the graph of h is concave down, the response can contain a consideration of the sign of $h'' = g'$ rather than the increasing/decreasing behavior of g .



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where $h' = g$ is positive.

h is concave down where $h' = g$ is decreasing.

$h' = g$ is positive and decreasing on the open intervals $0 < x < 2$ and $4 < x < 5$.

Therefore, the graph of h is both increasing and concave down on those intervals.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

To earn the first point, the response must consider the continuity of h and give the correct values of $h(x)$ at $x = 0$ and $x = 2$.

Stating that h is continuous on the open interval $(0, 2)$, a larger interval, or simply that h is continuous with no reference as to where, is sufficient consideration of the continuity of h .

If the response presents incorrect values for $h(x)$ at $x = 0$ and $x = 2$ such that -3.5 does not fall in the interval with endpoints $h(0)$ and $h(2)$, the response will earn the second point by claiming that there need not exist such a value of c . This response need not consider the continuity of h .

Similar to the above, a response with incorrect values for $h(x)$ at $x = 0$ and $x = 2$ such that -3.5 does fall in the interval with endpoints $h(0)$ and $h(2)$ will earn the second point by claiming that there must exist such a value of c . This response must mention the continuity of h to earn the second point.

A response can earn the second point, even if the continuity of h is not considered in the response, in which case the first point would not be earned.

Unit 1 AP Topics 16

It is possible to provide an acceptable argument, based on “accumulating area” for the correct conclusion without invoking the Intermediate Value Theorem.

The response need not explicitly invoke the Intermediate Value Theorem to earn the second point. However, a response in which the conclusion is based on the Mean Value Theorem, for example, would not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable. $\Rightarrow h$ is continuous on $0 \leq x \leq 2$.

$$h(2) = \int_2^2 g(t) dt = 0$$

$$h(0) = \int_2^0 g(t) dt = -7$$

$$h(0) < -3.5 < h(2)$$

Therefore, by the Intermediate Value Theorem, there is a value of c , for $0 < c < 2$, such that $h(c) = -3.5$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response earns the first point by presenting either $\frac{1}{10} \int_0^{10} g(x) dx$ as in the model solution or by presenting the four areas [7, 8, 22, and 4] in a sum, such as “ $7 + 8 - 22 + 4$ ” or “ $7 + 8 + 22 + 4$.” Unless corrected, using the latter expression would not earn the second point.

Read to the end of the response to be sure there are no errors. For example, if the following occurs, with an incorrect value at the end of the computation, the response would not earn the second point:

$$\dots = \frac{7+8-22+4}{10} = \frac{19-22}{10} = \frac{3}{10}$$

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Average value setup – OR – Area
- ☐ Answer

Solution:

$$\frac{1}{10-0} \int_0^{10} g(x) \, dx = \frac{1}{10}(7 + 8 - 22 + 4) = \frac{1}{10} \cdot (-3) = -0.3$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

To earn the first point, the response does not have to address the continuity or differentiability of g or h .

The first point is earned with the claim that the numerator and denominator separately go to 0. The first point is not earned if the response claims that the given limit equals $\frac{0}{0}$. However, for example, a response that claims that the given limit is “of the form $\frac{0}{0}$ ” or that uses arrows to show that the numerator and denominator each go to 0 will earn the first point. So, if you see $\frac{0}{0}$ in the response, read more carefully before deciding not to award the first point.

The second point is earned by applying L’Hospital’s Rule, i.e., demonstrating that the limit of the quotient of the derivatives, or the quotient of the values of the derivatives, should be evaluated. A response with minor differentiation errors, e.g., having $2x - 25$ in the numerator, can still earn the second point. However, with this response the third point cannot be earned.

Only an answer of -5 or its equivalent earns the third point, and be sure to read to the end of the response to be sure that it is all correct. For example, a response that presented the following would not earn the third point:

$$\dots = \frac{10}{0-2} = 5$$

A response can earn all three points without ever using limit notation; for example, arrows, correctly used, are acceptable. Thus, $h(x) - 2x + 2 \rightarrow 8 - 2 \cdot 5 + 2 = 0$ together with $x^2 - 25 \rightarrow 0$ would earn the first point,

and $\frac{2x}{h'(x)-2} = \frac{2x}{g(x)-2} \rightarrow \frac{10}{0-2}$ would earn the second and third points.

Unit 1 AP Topics 16



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.

$\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \rightarrow 5} h(x) = h(5)$ and $\lim_{x \rightarrow 5} h'(x) = h'(5)$.

$$\lim_{x \rightarrow 5} (x^2 - 25) = 5^2 - 25 = 0$$

$$\lim_{x \rightarrow 5} (h(x) - 2x + 2) = h(5) - 2 \cdot 5 + 2 = 8 - 10 + 2 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5} \frac{x^2 - 25}{h(x) - 2x + 2} &= \lim_{x \rightarrow 5} \frac{2x}{h'(x) - 2} \\ &= \frac{2 \cdot 5}{h'(5) - 2} = \frac{10}{g(5) - 2} = \frac{10}{0 - 2} = -5 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Only the correct antiderivative earns the first point. However, an antiderivative of the form either $K \cdot g(5x)$ or $K \cdot g(x)$, where K is a constant other than $\frac{1}{5}$, can earn the second point.

We will accept informal notation, like the following: Alt text: $\dots = \frac{g(\infty) - g(10)}{5} = \frac{3 - 9}{5} = \frac{-6}{5}$

However, the response must, at some point, explicitly show that a limit is needed to evaluate the improper integral.

If the evaluation uses a u-substitution, be careful to check that the limits have been changed appropriately. If they have not, this point will not be earned.



0	1	2
---	---	---

Unit 1 AP Topics 16

The student response accurately includes **both** of the criteria below.

- ☐ Antiderivative
- ☐ Limit and answer

Solution:

$$\begin{aligned}\int_2^{\infty} g'(5x) \, dx &= \lim_{b \rightarrow \infty} \int_2^b g'(5x) \, dx = \lim_{b \rightarrow \infty} \left. \frac{g(5x)}{5} \right|_2^b \\ &= \lim_{b \rightarrow \infty} \frac{g(5b)}{5} - \frac{g(10)}{5} = \frac{3}{5} - \frac{9}{5} = -\frac{6}{5}\end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for giving the correct sum for the geometric series, but the sum must use our constant, 12, rather than a “generic” a . The response does not have to consider the radius of convergence to earn the first point. Note that a response in which a is subsequently replaced with 12 does earn the first point.

Only responses with a correct expression for the sum of the geometric series are eligible for the second point, and to earn the second point a generic a must be replaced by 12 at some point. The second point is an “algebra point,” but the response must give the value of r correctly.

A response using an incorrect value for $g(4)$ cannot earn the second point, but can earn the third point.

A response with an incorrect value of r is eligible for the third point, if the conclusion for that r is correct.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Sum of geometric series
- ☐ Solves for r
- ☐ Answer with reason

Solution:

$12 + 12r + 12r^2 + \cdots + 12r^n + \cdots$ is a geometric series that converges to $\frac{12}{1-r}$ for $|r| < 1$ and diverges for $|r| \geq 1$.

$$g(4) = 5$$

Unit 1 AP Topics 16

$$\frac{12}{1-r} = 5 \Rightarrow 12 = 5 - 5r \Rightarrow r = -\frac{7}{5}$$

Because $\left| -\frac{7}{5} \right| \geq 1$, there is no value of r such that the given series equals $g(4) = 5$.

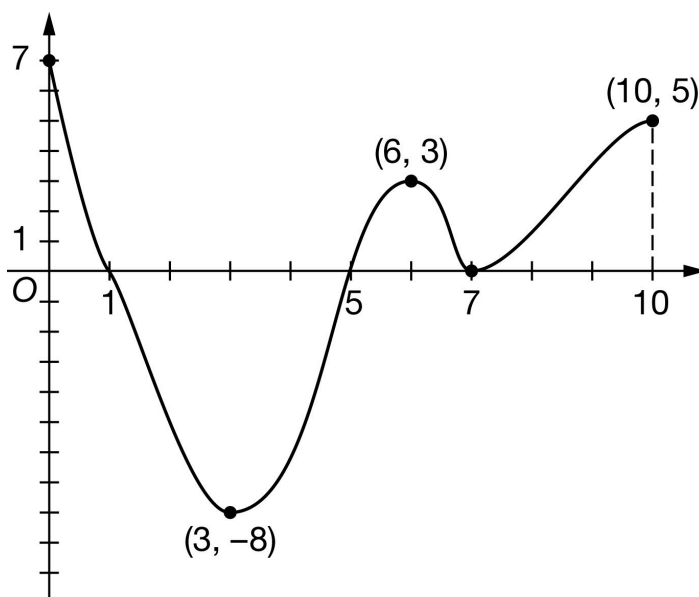
Unit 1 AP Topics 16

35. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The figure above shows a portion of the graph of the differentiable function g . The graph of g has horizontal tangents at $x = 3$, $x = 6$, and $x = 7$. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[0, 1]$, $[1, 5]$, $[5, 7]$, and $[7, 10]$ are 3, 20, 4, and 7, respectively. Let h be the

function defined by $h(x) = \int_5^x g(t) \, dt$.

- For $0 < x < 10$, find all values of x at which h has a critical point.
- Classify each critical point found in part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- On what open intervals contained in $0 < x < 10$ is the graph of h both increasing and concave up? Give a reason for your answer.
- Must there exist a value of c , for $1 < c < 5$, such that $h(c) = 12$? Justify your answer.
- Find the average value of g over the interval $0 \leq x \leq 10$. Show the computations that lead to your answer.
- Evaluate $\lim_{x \rightarrow 7} \frac{h(x) - x + 3}{x^2 - 49}$. Give a reason for your answer.

Unit 1 AP Topics 16

(g) It is known that $\lim_{x \rightarrow \infty} g(x) = 15$. Find the value of $\int_1^{\infty} g'(6x) dx$ or show that it does not exist.

(h) Is there a value of r such that the series $8 + 8r + 8r^2 + \cdots + 8r^n + \cdots$ equals the value of $g(6)$? Explain why or why not.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for providing all three of the values of x . No reason is required.

Points can be presented as ordered pairs with the correct h -value as the second coordinate, i.e., $(1, 20)$, $(5, 0)$, and $(7, 4)$. With any other second coordinate, the point is not earned.

Ignore any value in the response that is outside the open interval $(0, 10)$.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

h has a critical point where $h'(x) = g(x) = 0$.

$h'(x) = g(x) = 0 \Rightarrow x = 1, x = 5, x = 7$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The three points are independent, and, for each point, a justification is required to earn the point. So, a response might classify each value of x correctly and earn no points in part (b).

Do not read any classifications for $x = 0$, $x = 10$, or any value outside the interval $[0, 10]$.

A response that uses g , rather than h' , in all three of its justifications, without having noted that $h' = g$, in either part (a) or part (b), can earn at most two of the three points.

A response that refers only to h' in all three parts of its justifications, without having noted that $h' = g$ is still eligible for all three points.

If a response presents ordered pairs, rather than values of x , to be classified, the response must either supply the correct value of $h(x)$ in the ordered pair $[(1, 20), (5, 0), \text{ and } (7, 4)]$ or refer explicitly to the values of x to earn

Unit 1 AP Topics 16

points.

A sign chart, by itself, does not provide sufficient justification for the classification of any of the values of x .

The Second Derivative Test can be used to justify the classification of $x = 1$ and $x = 5$, but not for $x = 7$.

References to “it” are not equivalent to h' or g . Similarly, a statement like “the slope goes from positive to negative” is not equivalent to h' going from positive to negative.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum at $x = 1$ with reason
- ☐ Relative minimum at $x = 5$ with reason
- ☐ Neither at $x = 7$ with reason

Solution:

$h'(x) = g(x)$ changes from positive to negative at $x = 1$. Therefore, h has a relative maximum at $x = 1$.

$h'(x) = g(x)$ changes from negative to positive at $x = 5$. Therefore, h has a relative minimum at $x = 5$.

$h'(x) = g(x)$ does not change sign at $x = 7$. Therefore, h has neither a relative minimum nor a relative maximum at $x = 7$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct answer, even if the reasoning presented in the response to support this answer is inadequate or incorrect. The intervals can be presented as either open [preferred] or closed.

A response that contains only the correct answer, but with no supporting work, earns the first point, but not the second.

A response that uses g , rather than h' , without having noted that $h' = g$, in either part (a), part (b), or part (c) cannot earn the second point.

To earn the second point, the response must only give a reason why the presented answer satisfies the required conditions. It is not necessary to list separately the intervals in which the graph of h is increasing and the intervals in which the graph is concave up.

A response can earn the second point without earning the first point if the response contains only one of the two required intervals and presents a correct reason.

Unit 1 AP Topics 16

To determine the intervals on which the graph of h is concave up, the response can contain a consideration of the sign of $h'' = g'$ rather than the increasing/decreasing behavior of g .



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where $h' = g$ is positive.

h is concave up where $h' = g$ is increasing.

$h' = g$ is positive and increasing on the open intervals $5 < x < 6$ and $7 < x < 10$.

Therefore, the graph of h is both increasing and concave up on those intervals.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

To earn the first point, the response must consider the continuity of h and give the correct values of $h(x)$ at $x = 1$ and $x = 5$.

Stating that h is continuous on the open interval $(1, 5)$, a larger interval, or simply that h is continuous with no reference as to where, is sufficient consideration of the continuity of h .

If the response presents incorrect values for $h(x)$ at $x = 1$ and $x = 5$ such that 12 does not fall in the interval with endpoints $h(1)$ and $h(5)$, the response will earn the second point by claiming that there need not exist such a value of c . This response need not consider the continuity of h .

Similar to the above, a response with incorrect values for $h(x)$ at $x = 1$ and $x = 5$ such that 12 does fall in the interval with endpoints $h(1)$ and $h(5)$ will earn the second point by claiming that there must exist such a value of c . This response must mention the continuity of h to earn the second point.

A response can earn the second point, even if the continuity of h is not considered in the response, in which case the first point would not be earned.

It is possible to provide an acceptable argument, based on “accumulating area” for the correct conclusion without invoking the Intermediate Value Theorem.

Unit 1 AP Topics 16

The response need not explicitly invoke the Intermediate Value Theorem to earn the second point. However, a response in which the conclusion is based on the Mean Value Theorem, for example, would not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable. $\Rightarrow h$ is continuous on $1 \leq x \leq 5$.

$$h(1) = \int_5^1 g(t) dt = 20$$

$$h(5) = \int_5^5 g(t) dt = 0$$

$$h(5) < 12 < h(1)$$

Therefore, by the Intermediate Value Theorem, there is a value of c , for $1 < c < 5$, such that $h(c) = 12$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response earns the first point by presenting either $\frac{1}{10} \int_0^{10} g(x) dx$ as in the model solution or by presenting the four areas [3, 20, 4, and 7] in a sum, such as “ $3 - 20 + 4 + 7$ ” or “ $3 + 20 + 4 + 7$.” Unless corrected, using the latter expression would not earn the second point.

Read to the end of the response to be sure there are no errors. For example, if the following occurs, with an incorrect value at the end of the computation, the response would not earn the second point:

$$\dots = \frac{3-20+4+7}{10} = -\frac{6}{10} = -\frac{2}{5}$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

Unit 1 AP Topics 16

- ☐ Average value setup – OR – Area
- ☐ Answer

Solution:

$$\frac{1}{10-0} \int_0^{10} g(x) \, dx = \frac{1}{10}(3 - 20 + 4 + 7) = \frac{1}{10} \cdot (-6) = -0.6$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

To earn the first point, the response does not have to address the continuity or differentiability of g or h .

The first point is earned with the claim that the numerator and denominator separately go to 0. The first point is not earned if the response claims that the given limit equals $\frac{0}{0}$. However, for example, a response that claims that the given limit is “of the form $\frac{0}{0}$ ” or that uses arrows to show that the numerator and denominator each go to 0 will earn the first point. So, if you see $\frac{0}{0}$ in the response, read more carefully before deciding not to award the first point.

The second point is earned by applying L'Hospital's Rule, i.e., demonstrating that the limit of the quotient of the derivatives, or the quotient of the values of the derivatives, should be evaluated. A response with minor differentiation errors, e.g., having $2x - 49$ in the denominator, can still earn the second point. However, with this response the third point cannot be earned.

Only an answer of $-\frac{1}{14}$ or its equivalent earns the third point, and be sure to read to the end of the response to be sure that it is all correct. For example, a response that presented the following would not earn the third point:

$$\dots = \frac{0-1}{14} = \frac{1}{14}$$

A response can earn all three points without ever using limit notation; for example, arrows, correctly used, are acceptable. Thus, $h(x) - x + 3 \rightarrow 4 - 7 + 3 = 0$ together with $x^2 - 49 \rightarrow 0$ would earn the first point, and $\frac{h'(x)-1}{2x} = \frac{g(x)-1}{2x} \rightarrow \frac{0-1}{14}$ would earn the second and third points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule

Unit 1 AP Topics 16

☐ Answer
Solution:
 $h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.

 $\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \rightarrow 7} h(x) = h(7)$ and $\lim_{x \rightarrow 7} h'(x) = h'(7)$.

$$\lim_{x \rightarrow 7} (h(x) - x + 3) = h(7) - 7 + 3 = 4 - 7 + 3 = 0$$

$$\lim_{x \rightarrow 7} (x^2 - 49) = 7^2 - 49 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 7} \frac{h(x) - x + 3}{x^2 - 49} &= \lim_{x \rightarrow 7} \frac{h'(x) - 1}{2x} \\ &= \frac{h'(7) - 1}{2 \cdot 7} = \frac{g(7) - 1}{14} = \frac{0 - 1}{14} = -\frac{1}{14} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Only the correct antiderivative earns the first point. However, an antiderivative of the form either $K \cdot g(6x)$ or $K \cdot g(x)$, where K is a constant other than $\frac{1}{6}$, can earn the second point.

We will accept informal notation, like the following:

$$\dots = \frac{g(\infty) - g(6)}{6} = \frac{15 - 3}{6} = 2$$

However, the response must, at some point, explicitly show that a limit is needed to evaluate the improper integral.

If the evaluation uses a u -substitution, be careful to check that the limits have been changed appropriately. If they have not, this point will not be earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Antiderivative
- ☐ Limit and answer

Solution:

Unit 1 AP Topics 16

$$\begin{aligned}\int_1^{\infty} g'(6x) \, dx &= \lim_{b \rightarrow \infty} \int_1^b g'(6x) \, dx = \lim_{b \rightarrow \infty} \left. \frac{g(6x)}{6} \right|_1^b \\ &= \lim_{b \rightarrow \infty} \frac{g(6b)}{6} - \frac{g(6)}{6} = \frac{15}{6} - \frac{3}{6} = 2\end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for giving the correct sum for the geometric series, but the sum must use our constant, 8, rather than a “generic” a . The response does not have to consider the radius of convergence to earn the first point. Note that a response in which a is subsequently replaced with 8 does earn the first point.

Only responses with a correct expression for the sum of the geometric series are eligible for the second point, and to earn the second point a generic a must be replaced by 8 at some point. The second point is an “algebra point,” but the response must give the value of r correctly.

The three decimal places requirement for decimal presentations does not apply to the value of r presented in the response, since this value is an “intermediate” result, and need only be presented with accuracy sufficient to determine whether its absolute value is less than 1. Thus, $r = -\frac{5}{3} = -1.3$ would earn the second point.

A response using an incorrect value for $g(6)$ cannot earn the second point, but can earn the third point.

A response with an incorrect value of r is eligible for the third point, if the conclusion for that r is correct.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Sum of geometric series
- ☐ Solves for r
- ☐ Answer with reason

Solution:

$8 + 8r + 8r^2 + \cdots + 8r^n + \cdots$ is a geometric series that converges to $\frac{8}{1-r}$ for $|r| < 1$ and diverges for $|r| \geq 1$.

$$g(6) = 3$$

$$\frac{8}{1-r} = 3 \Rightarrow 8 = 3 - 3r \Rightarrow r = -\frac{5}{3}$$

Because $\left| -\frac{5}{3} \right| \geq 1$, there is no value of r such that the given series equals $g(6) = 3$.

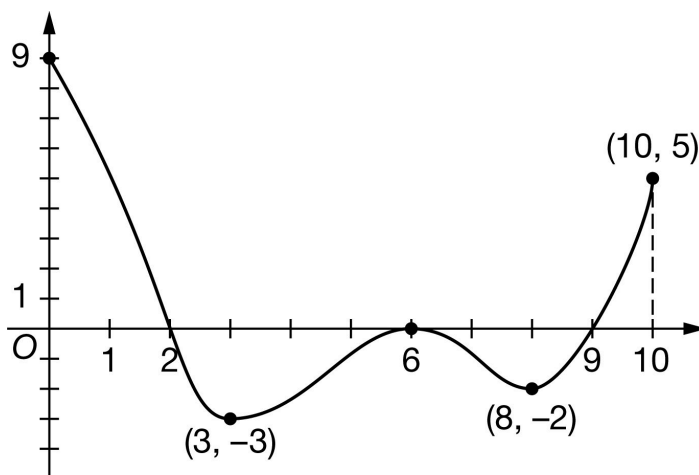
Unit 1 AP Topics 16

36. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The figure above shows a portion of the graph of the differentiable function g . The graph of g has horizontal tangents at $x = 3$, $x = 6$, and $x = 8$. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[0, 2]$, $[2, 6]$, $[6, 9]$, and $[9, 10]$ are 10, 7, 3, and 2, respectively. Let h be the function defined by $h(x) = \int_2^x g(t) \, dt$.

- For $0 < x < 10$, find all values of x at which h has a critical point.
- Classify each critical point found in part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- On what open intervals contained in $0 < x < 10$ is the graph of h both decreasing and concave up? Give a reason for your answer.
- Must there exist a value of c , for $0 < c < 2$, such that $h(c) = -4$? Justify your answer.
- Find the average value of g over the interval $0 \leq x \leq 10$. Show the computations that lead to your answer.
- Evaluate $\lim_{x \rightarrow 6} \frac{h(x) + x + 1}{x^2 - 36}$. Give a reason for your answer.
- It is known that $\lim_{x \rightarrow \infty} g(x) = 12$. Find the value of $\int_6^{\infty} g'\left(\frac{x}{2}\right) \, dx$ or show that it does not exist.
- Is there a value of r such that the series $14 + 14r + 14r^2 + \cdots + 14r^n + \cdots$ equals the value of $g(10)$? Explain why or why not.

Unit 1 AP Topics 16

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for providing all three of the values of x . No reason is required.

Points can be presented as ordered pairs with the correct h -value as the second coordinate, i.e., $(2, 0)$, $(6, -7)$, and $(9, -10)$. With any other second coordinate, the point is not earned.

Ignore any value in the response that is outside the open interval $(0, 10)$.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

h has a critical point where $h'(x) = g(x) = 0$.

$$h'(x) = g(x) = 0 \Rightarrow x = 2, x = 6, x = 9$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The three points are independent, and, for each point, a justification is required to earn the point. So, a response might classify each value of x correctly and earn no points in part (b).

Do not read any classifications for $x = 0$, $x = 10$, or any value outside the interval $[0, 10]$.

A response that uses g , rather than h' , in all three of its justifications, without having noted that $h' = g$, in either part (a) or part (b), can earn at most two of the three points.

A response that refers only to h' in all three parts of its justifications, without having noted that $h' = g$ is still eligible for all three points.

If a response presents ordered pairs, rather than values of x , to be classified, the response must either supply the correct value of $h(x)$ in the ordered pair $[(2, 0), (6, -7), \text{ and } (9, -10)]$ or refer explicitly to the values of x to earn points.

A sign chart, by itself, does not provide sufficient justification for the classification of any of the values of x .

The Second Derivative Test can be used to justify the classification of $x = 2$ and $x = 9$, but not for $x = 6$.

Unit 1 AP Topics 16

References to “it” are not equivalent to h' or g . Similarly, a statement like “the slope goes from positive to negative” is not equivalent to h' going from positive to negative.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum at $x = 2$ with reason
- ☐ Neither at $x = 6$ with reason
- ☐ Relative minimum at $x = 9$ with reason

Solution:

$h'(x) = g(x)$ changes from positive to negative at $x = 2$. Therefore, h has a relative maximum at $x = 2$.

$h'(x) = g(x)$ does not change sign at $x = 6$. Therefore, h has neither a relative minimum nor a relative maximum at $x = 6$.

$h'(x) = g(x)$ changes from negative to positive at $x = 9$. Therefore, h has a relative minimum at $x = 9$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct answer, even if the reasoning presented in the response to support this answer is inadequate or incorrect. The intervals can be presented as either open [preferred] or closed.

A response that contains only the correct answer, but with no supporting work, earns the first point, but not the second.

A response that uses g , rather than h' , without having noted that $h' = g$, in either part (a), part (b), or part (c) cannot earn the second point.

To earn the second point, the response must only give a reason why the presented answer satisfies the required conditions. It is not necessary to list separately the intervals in which the graph of h is decreasing and the intervals in which the graph is concave up.

A response can earn the second point without earning the first point if the response contains only one of the two required intervals and presents a correct reason.

To determine the intervals on which the graph of h is concave up, the response can contain a consideration of the sign of $h'' = g'$ rather than the increasing/decreasing behavior of g .

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is decreasing where $h' = g$ is negative.

h is concave up where $h' = g$ is increasing.

$h' = g$ is negative and increasing on the open intervals $3 < x < 6$ and $8 < x < 9$.

Therefore, the graph of h is both decreasing and concave up on those intervals.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

To earn the first point, the response must consider the continuity of h and give the correct values of $h(x)$ at $x = 0$ and $x = 2$.

Stating that h is continuous on the open interval $(0, 2)$, a larger interval, or simply that h is continuous with no reference as to where, is sufficient consideration of the continuity of h .

If the response presents incorrect values for $h(x)$ at $x = 0$ and $x = 2$ such that -4 does not fall in the interval with endpoints $h(0)$ and $h(2)$, the response will earn the second point by claiming that there need not exist such a value of c . This response need not consider the continuity of h .

Similar to the above, a response with incorrect values for $h(x)$ at $x = 0$ and $x = 2$ such that -4 does fall in the interval with endpoints $h(0)$ and $h(2)$ will earn the second point by claiming that there must exist such a value of c . This response must mention the continuity of h to earn the second point.

A response can earn the second point, even if the continuity of h is not considered in the response, in which case the first point would not be earned.

It is possible to provide an acceptable argument, based on “accumulating area” for the correct conclusion without invoking the Intermediate Value Theorem.

The response need not explicitly invoke the Intermediate Value Theorem to earn the second point. However, a response in which the conclusion is based on the Mean Value Theorem, for example, would not earn the second point.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable. $\Rightarrow h$ is continuous on $0 \leq x \leq 2$.

$$h(0) = \int_2^0 g(t) dt = -10$$

$$h(2) = \int_2^2 g(t) dt = 0$$

$$h(0) < -4 < h(2)$$

Therefore, by the Intermediate Value Theorem, there is a value of c , for $0 < c < 2$, such that $h(c) = -4$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response earns the first point by presenting either $\frac{1}{10} \int_0^{10} g(x) dx$ as in the model solution or by presenting the four areas [10, 7, 3, and 2] in a sum, such as “ $10 - 7 - 3 + 2$ ” or “ $10 + 7 + 3 + 2$.” Unless corrected, using the latter expression would not earn the second point.

Read to the end of the response to be sure there are no errors. For example, if the following occurs, with an incorrect value at the end of the computation, the response would not earn the second point:

$$\dots = \frac{10-7-3+2}{10} = \frac{2}{10} = 5$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Average value setup – OR – Area

Unit 1 AP Topics 16

☐ Answer
Solution:

$$\frac{1}{10-0} \int_0^{10} g(x) \, dx = \frac{1}{10}(10 - 7 - 3 + 2) = \frac{1}{10} \cdot 2 = 0.2$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

To earn the first point, the response does not have to address the continuity or differentiability of g or h .

The first point is earned with the claim that the numerator and denominator separately go to 0. The first point is not earned if the response claims that the given limit equals $\frac{0}{0}$. However, for example, a response that claims that the given limit is “of the form $\frac{0}{0}$ ” or that uses arrows to show that the numerator and denominator each go to 0 will earn the first point. So, if you see $\frac{0}{0}$ in the response, read more carefully before deciding not to award the first point.

The second point is earned by applying L'Hospital's Rule, i.e., demonstrating that the limit of the quotient of the derivatives, or the quotient of the values of the derivatives, should be evaluated. A response with minor differentiation errors, e.g., having $2x - 36$ in the denominator, can still earn the second point. However, with this response the third point cannot be earned.

Only an answer of $\frac{1}{12}$ or its equivalent earns the third point, and be sure to read to the end of the response to be sure that it is all correct. For example, a response that presented the following would not earn the third point:

$$\dots = \frac{0+1}{12} = 12$$

A response can earn all three points without ever using limit notation; for example, arrows, correctly used, are acceptable. Thus, $h(x) + x + 1 \rightarrow -7 + 6 + 1 = 0$ together with $x^2 - 36 \rightarrow 0$ would earn the first point, and $\frac{h'(x)+1}{2x} = \frac{g(x)+1}{2x} \rightarrow \frac{0+1}{12}$ would earn the second and third points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Unit 1 AP Topics 16

Solution:

$h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.

$\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \rightarrow 6} h(x) = h(6)$ and $\lim_{x \rightarrow 6} h'(x) = h'(6)$.

$$\lim_{x \rightarrow 6} (h(x) + x + 1) = h(6) + 6 + 1 = -7 + 6 + 1 = 0$$

$$\lim_{x \rightarrow 6} (x^2 - 36) = 6^2 - 36 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow 6} \frac{h(x) + x + 1}{x^2 - 36} = \lim_{x \rightarrow 6} \frac{h'(x) + 1}{2x}$$

$$= \frac{h'(6) + 1}{2 \cdot 6} = \frac{g(6) + 1}{12} = \frac{0 + 1}{12} = \frac{1}{12}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Only the correct antiderivative earns the first point. However, an antiderivative of the form either $K \cdot g\left(\frac{x}{2}\right)$ or $K \cdot g(x)$, where K is a constant other than 2, can earn the second point.

We will accept informal notation, like the following: $\dots = 2(g(\infty) - g(3)) = 2(12 + 3) = 30$

However, the response must, at some point, explicitly show that a limit is needed to evaluate the improper integral.

If the evaluation uses a u -substitution, be careful to check that the limits have been changed appropriately. If they have not, this point will not be earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Antiderivative
- ☐ Limit and answer

Solution:

$$\begin{aligned} \int_6^{\infty} g'\left(\frac{x}{2}\right) dx &= \lim_{b \rightarrow \infty} \int_6^b g'\left(\frac{x}{2}\right) dx = \lim_{b \rightarrow \infty} 2g\left(\frac{x}{2}\right) \Big|_6^b \\ &= \lim_{b \rightarrow \infty} 2g\left(\frac{b}{2}\right) - 2g(3) = 2 \cdot 12 - 2 \cdot (-3) = 30 \end{aligned}$$

Unit 1 AP Topics 16

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for giving the correct sum for the geometric series, but the sum must use our constant, 14, rather than a “generic” a . The response does not have to consider the radius of convergence to earn the first point. Note that a response in which a is subsequently replaced with 14 does earn the first point.

Only responses with a correct expression for the sum of the geometric series are eligible for the second point, and to earn the second point a generic a must be replaced by 14 at some point. The second point is an “algebra point,” but the response must give the value of r correctly.

A response using an incorrect value for $g(10)$ cannot earn the second point, but can earn the third point.

A response with an incorrect value of r is eligible for the third point, if the conclusion for that r is correct.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Sum of geometric series
- ☐ Solves for r
- ☐ Answer with reason

Solution:

$14 + 14r + 14r^2 + \cdots + 14r^n + \cdots$ is a geometric series that converges to $\frac{14}{1-r}$ for $|r| < 1$ and diverges for $|r| \geq 1$.

$$g(10) = 5$$

$$\frac{14}{1-r} = 5 \Rightarrow 14 = 5 - 5r \Rightarrow r = -\frac{9}{5}$$

Because $\left| -\frac{9}{5} \right| \geq 1$, there is no value of r such that the given series equals $g(10) = 5$.

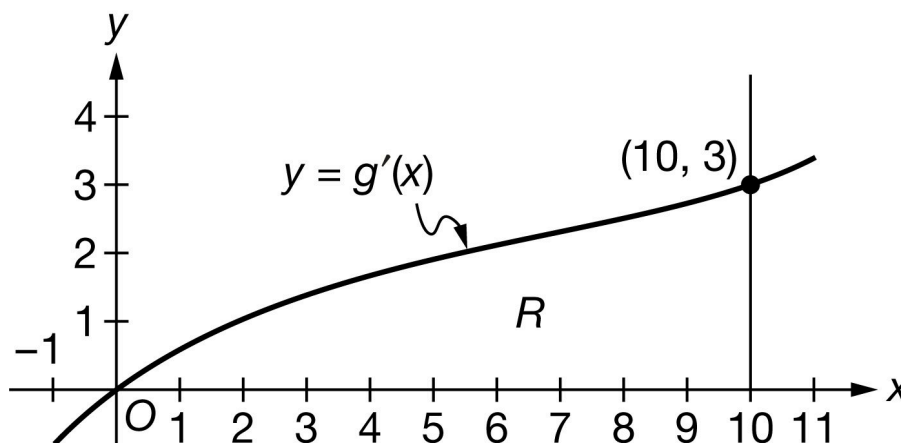
Unit 1 AP Topics 16

37. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graph of g' , the derivative of the twice-differentiable function g , is shown above for $-1 < x < 11$. The graph of g' has no horizontal tangent lines on the interval $-1 < x < 11$. Let R be the region in the first quadrant bounded by the graph of g' , the x -axis, and the vertical line $x = 10$. It is known that

$$g(0) = -3, g(10) = 15, \text{ and } \int_0^{10} g(x) \, dx = 38.$$

(a) How many points of inflection does the graph of g have on the interval $-1 < x < 11$? Give a reason for your answer.

(b) Find the average value of g' over the interval $0 \leq x \leq 10$. Show the computations that lead to your answer.

(c) Must there exist a value of c , for $0 < c < 10$, such that $g(c) = 0$? Justify your answer.

(d) Evaluate $\int_0^{10} (4g(x) + 5) \, dx$. Show the computations that lead to your answer.

(e) Evaluate $\lim_{x \rightarrow 10} \frac{g(x) - x + 2}{\sin x}$. Show the computations that lead to your answer.

(f) Let h be the function defined by $h(x) = e^x \cdot \int_{10}^x g(t) \, dt$. Find $h'(0)$. Show the computations that lead to your answer.

Unit 1 AP Topics 16

(g) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis is a rectangle with height x and base in the region R . The volume of the solid is given by

$$\int_0^{10} A(x) \, dx.$$
 Write an expression for $A(x)$.

(h) What is the volume of the solid described in part (g)? Show the computations that lead to your answer.

(i) Find the value of $\int_0^{10} \frac{g''(x)}{\sqrt{g'(x)}} \, dx$ or show that it does not exist. (Note: $\int \frac{1}{\sqrt{x}} \, dx = 2\sqrt{x} + C$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We are looking for the application of the relationship between the increasing and decreasing behavior of the first derivative and the points of inflection of the original function.

The first point is earned for “none,” with some relevant mathematical discussion.

The reasoning point is most commonly earned by correctly appealing to the increasing behavior of g' .

An appeal to g'' must address the consideration of both $g'' = 0$ and g'' undefined.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

The graph of g has a point of inflection where g' changes from increasing to decreasing or from decreasing to increasing.

g' is increasing on the interval $-1 < x < 11$.

Therefore, the graph of g has no points of inflection on the interval $-1 < x < 11$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Make sure the response addresses the average value of g' , not of g .

The Fundamental Theorem of Calculus point is earned for the expression $g(10) - g(0)$.

Unit 1 AP Topics 16

The answer point is earned only for the correct answer. The values of 15 and -3 must be shown.

$\frac{1}{10}(15 - (-3))$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\begin{aligned}\text{Average value} &= \frac{1}{10-0} \int_0^{10} g'(x) \, dx = \frac{1}{10}(g(10) - g(0)) \\ &= \frac{1}{10}(15 - (-3)) = 1.8\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The student must establish that g is a continuous function. The most common, and acceptable, response is “differentiable implies continuous.”

The student must convey that differentiability implies continuity. Stating g is differentiable and continuous, does not earn the first point.

The response that mentions continuity without establishing continuity is still eligible for the second point.

Using, or citing, the Mean Value Theorem will not earn the second point.

To earn the second point the student must explicitly convey an appropriate inequality involving 0, either mathematically or in words. For example, $-3 < 0 < 15$ or $g(0) < 0 < g(10)$.

The second point requires an answer of “yes” or an equivalent statement.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

Unit 1 AP Topics 16

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous on $0 \leq x \leq 10$.

$$g(0) = -3 < 0 < 15 = g(10)$$

By the Intermediate Value Theorem, there exists a value of c , for $0 < c < 10$, such that $g(c) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Some (incorrect) responses claim that evaluation of the integrand is equal to the value of the integral.

The first point requires demonstration of both the sum of integrals and the multiplication by a constant.

$4 \cdot 38 + 5 \cdot 10$ earns both points. This is the minimal supporting work required.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Properties of definite integrals
- ☐ Answer

Solution:

$$\int_0^{10} (4g(x) + 5) dx = 4 \int_0^{10} g(x) dx + \int_0^{10} 5 dx$$

$$= 4 \cdot 38 + 5 \cdot 10 = 202$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal supporting computations required are $\frac{15-10+2}{\sin 10}$. This alone would earn the point.

The student does not have to evaluate $\sin 10$.

Unit 1 AP Topics 16

An answer of just “ $\frac{7}{\sin 10}$ ” does not earn the point because it does not show the minimal computations.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 10} g(x) = g(10)$

$$\lim_{x \rightarrow 10} \frac{g(x) - x + 2}{\sin x} = \frac{g(10) - 10 + 2}{\sin 10} = \frac{15 - 10 + 2}{\sin 10} = \frac{7}{\sin 10}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response of only $e^x \cdot g(x)$ earns only the Fundamental Theorem of Calculus point.

Some responses may reverse the limits of integration and state this is equal to the original integral. This incorrect approach could earn only the first two points, but will not earn the answer point, unless the student resolves the limits reversal issue.

The answer point can only be earned if both of the first two points are earned, and the given answer is correct.

$1 \cdot (-38) + 1 \cdot (-3)$ earns all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Unit 1 AP Topics 16

$$h'(x) = e^x \cdot \int_{10}^x g(t) dt + e^x \cdot g(x)$$

$$h'(0) = e^0 \cdot \int_{10}^0 g(t) dt + e^0 \cdot g(0) = 1 \cdot (-38) + 1 \cdot (-3) = -41$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the student writes the integral with $x \cdot g'(x)$ as the integrand, we need to see that it is equal to the integral with $A(x)$ for the integrand.

$$\text{So, } \int_0^{10} A(x) dx = \int_0^{10} x \cdot g'(x) dx \text{ earns the point.}$$

Some responses introduce another variable, or function, for example, $R(x) = g'(x)$, before writing the final expression in terms of this other variable. This earns the point. For example, $A(x) = x \cdot R(x)$ in the presence of $R(x) = g'(x)$.

The point is not earned if the correct integrand appears only in part (h).



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$A(x) = x \cdot g'(x)$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If there is no integration by parts, no points are earned in this part.

The integration by parts point is for an “indefinite integration by parts.” The bounds are associated with the answer point.

Minimal work required to earn the integration by parts point: $10 \cdot 15 - 0 - 38$

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integration by parts
- ☐ Answer

Solution:

$$u = x \quad dv = g'(x) dx$$

$$du = dx \quad v = g(x)$$

$$\int_0^{10} x \cdot g'(x) dx = x \cdot g(x) \Big|_0^{10} - \int_0^{10} g(x) dx$$

$$= 10 \cdot g(10) - 0 \cdot g(0) - \int_0^{10} g(x) dx$$

$$= 10 \cdot 15 - 0 \cdot (-3) - 38 = 112$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Many students will write the antiderivative without any evidence of substitution. This antiderivative must be completely correct to earn the substitution point.

Limit notation must be used in order to earn the improper integral point.

The student must earn the improper integral point to be eligible for the answer point.

If the student uses an explicit substitution and makes an error in the limits, they cannot earn the answer point.

A response can earn the substitution point for working only with an indefinite integral.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Improper integral

Unit 1 AP Topics 16

☐ Answer**Solution:**

Let $u = g'(x)$. Then $du = g''(x) dx$.

When $x = 0$, $u = g'(0) = 0$.

When $x = 10$, $u = g'(10) = 3$.

$$\begin{aligned}\int_0^{10} \frac{g''(x)}{\sqrt{g'(x)}} dx &= \int_0^3 \frac{1}{\sqrt{u}} du \\&= \lim_{b \rightarrow 0^+} \int_b^3 \frac{1}{\sqrt{u}} du = \lim_{b \rightarrow 0^+} 2\sqrt{u} \Big|_b^3 = 2\sqrt{3} - \lim_{b \rightarrow 0^+} 2\sqrt{b} \\&= 2\sqrt{3} - 0 = 2\sqrt{3}\end{aligned}$$

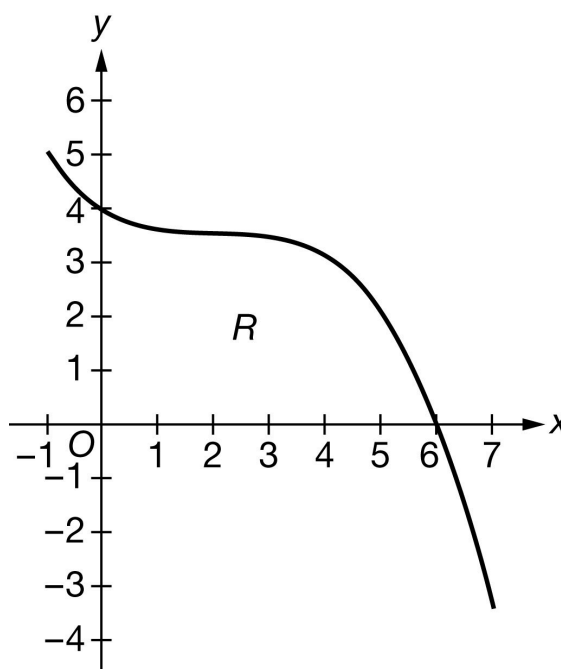
Unit 1 AP Topics 16

38. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of g' , the derivative of the twice-differentiable function g , is shown above for $-1 < x < 7$. The graph of g' has exactly one horizontal tangent line, at $x = 2$. Let R be the region in the first quadrant bounded by the graph of g' and the x -axis from $x = 0$ to $x = 6$. It is known that $g(0) = -5$, $g(6) = 13$, and $\int_0^6 g(x) \, dx = 32.4$.

(a) How many points of inflection does the graph of g have on the interval $-1 < x < 7$? Give a reason for your answer.

(b) Find the average value of g' over the interval $0 \leq x \leq 6$. Show the computations that lead to your answer.

(c) Must there exist a value of c , for $0 < c < 6$, such that $g(c) = 0$? Justify your answer.

(d) Evaluate $\int_0^6 (5g(x) + 2) \, dx$. Show the computations that lead to your answer.

Unit 1 AP Topics 16

(e) Evaluate $\lim_{x \rightarrow 0} \frac{g(x) - 3x + 1}{\cos x + 1}$. Show the computations that lead to your answer.

(f) Let h be the function defined by $h(x) = e^x \cdot \int_6^x g(t) \, dt$. Find $h'(0)$. Show the computations that lead to your answer.

(g) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis is a rectangle with height x and base in the region R . The volume of the solid is given by

$$\int_0^6 A(x) \, dx.$$

Write an expression for $A(x)$.

(h) What is the volume of the solid described in part (g)? Show the computations that lead to your answer.

(i) Find the value of $\int_0^6 \frac{g''(x)}{\sqrt{g'(x)}} \, dx$ or show that it does not exist. (Note: $\int \frac{1}{\sqrt{x}} \, dx = 2\sqrt{x} + C$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We are looking for the application of the relationship between the increasing and decreasing behavior of the first derivative and the points of inflection of the original function.

The first point is earned for “none,” with some relevant mathematical discussion.

The reasoning point is most commonly earned by correctly appealing to the decreasing behavior of g' .

An appeal to g'' prime must address the consideration of both $g'' = 0$ and g'' undefined.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

The graph of g has a point of inflection where g' changes from increasing to decreasing or from decreasing to increasing.

g' is decreasing on the interval $-1 < x < 7$.

Therefore, the graph of g has no points of inflection on the interval $-1 < x < 7$.

Part B

Unit 1 AP Topics 16

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Make sure the response addresses the average value of g' , not of g .

The Fundamental Theorem of Calculus point is earned for the expression $g(6) - g(0)$.

The answer point is earned only for the correct answer. The values of 13 and -5 must be shown.

$\frac{1}{6}(13 - (-5))$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\begin{aligned}\text{Average value} &= \frac{1}{6-0} \int_0^6 g'(x) dx = \frac{1}{6}(g(6) - g(0)) \\ &= \frac{1}{6}(13 - (-5)) = 3\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The student must establish that g is a continuous function. The most common, and acceptable, response is “differentiable implies continuous.”

The student must convey that differentiability implies continuity. Stating g is differentiable and continuous, does not earn the first point.

The response that mentions continuity without establishing continuity is still eligible for the second point.

Using, or citing, the Mean Value Theorem will not earn the second point.

To earn the second point the student must explicitly convey an appropriate inequality involving 0, either mathematically or in words. For example, $-5 < 0 < 13$ or $g(0) < 0 < g(6)$.

The second point requires an answer of “yes” or an equivalent statement.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous on $0 \leq x \leq 6$.

$$g(0) = -5 < 0 < 13 = g(6)$$

By the Intermediate Value Theorem, there exists a value of c , for $0 < c < 6$, such that $g(c) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Some (incorrect) responses claim that evaluation of the integrand is equal to the value of the integral.

The first point requires demonstration of both the sum of integrals and the multiplication by a constant.

$5 \cdot 32.4 + 2 \cdot 6$ earns both points. This is the minimal supporting work required.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Properties of definite integrals
- ☐ Answer

Solution:

$$\int_0^6 (5g(x) + 2) dx = 5 \int_0^6 g(x) dx + \int_0^6 2 dx$$

$$= 5 \cdot 32.4 + 2 \cdot 6 = 174$$

Unit 1 AP Topics 16

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal supporting computations required are $\frac{-5-0+1}{\cos 0+1}$ or $\frac{-5-0+1}{1+1}$. Either of these alone would earn the point.

The student does not have to evaluate $\cos 0$.

An answer of just “−2” does not earn the point because it does not show the minimal computations.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 0} g(x) = g(0)$

$$\lim_{x \rightarrow 0} \frac{g(x) - 3x + 1}{\cos x + 1} = \frac{g(0) - 0 + 1}{\cos 0 + 1} = \frac{-5 - 0 + 1}{1 + 1} = -2$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response of only $e^x \cdot g(x)$ earns only the Fundamental Theorem of Calculus point.

Some responses may reverse the limits of integration and state this is equal to the original integral. This incorrect approach could earn only the first two points, but will not earn the answer point, unless the student resolves the limits reversal issue.

The answer point can only be earned if both of the first two points are earned, and the given answer is correct.

$1 \cdot (-32.4) + 1 \cdot (-5)$ earns all three points.



0	1	2	3
---	---	---	---

Unit 1 AP Topics 16

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$h'(x) = e^x \cdot \int_6^x g(t) dt + e^x \cdot g(x)$$

$$h'(0) = e^0 \cdot \int_6^0 g(t) dt + e^0 \cdot g(0) = 1 \cdot (-32.4) + 1 \cdot (-5) = -37.4$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the student writes the integral with $x \cdot g'(x)$ as the integrand, we need to see that it is equal to the integral with $A(x)$ for the integrand.

So, $\int_0^6 A(x) dx = \int_0^6 x \cdot g'(x) dx$ earns the point.

Some responses introduce another variable, or function, for example, $R(x) = g'(x)$, before writing the final expression in terms of this other variable. This earns the point. For example, $A(x) = x \cdot R(x)$ in the presence of $R(x) = g'(x)$.

The point is not earned if the correct integrand appears only in part (h).



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$A(x) = x \cdot g'(x)$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 16

If there is no integration by parts, no points are earned in this part.

The integration by parts point is for an “indefinite integration by parts.” The bounds are associated with the answer point.

Minimal work required to earn the integration by parts point: $6 \cdot 13 - 0 - 32.4$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integration by parts
- ☐ Answer

Solution:

$$u = x \quad dv = g'(x) dx$$

$$du = dx \quad v = g(x)$$

$$\int_0^6 x \cdot g'(x) dx = x \cdot g(x) \Big|_0^6 - \int_0^6 g(x) dx$$

$$= 6 \cdot g(6) - 0 \cdot g(0) - \int_0^6 g(x) dx$$

$$= 6 \cdot 13 - 0 \cdot (-5) - 32.4 = 45.6$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Many students will write the antiderivative without any evidence of substitution. This antiderivative must be completely correct to earn the substitution point.

Limit notation must be used in order to earn the improper integral point.

The student must earn the improper integral point to be eligible for the answer point.

If the student uses an explicit substitution and makes an error in the limits, they cannot earn the answer point.

A response can earn the substitution point for working only with an indefinite integral.



0	1	2	3
---	---	---	---

Unit 1 AP Topics 16

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Improper integral
- ☐ Answer

Solution:

Let $u = g'(x)$. Then $du = g''(x) dx$.

When $x = 0$, $u = g'(0) = 4$.

When $x = 6$, $u = g'(6) = 0$.

$$\begin{aligned}\int_0^6 \frac{g''(x)}{\sqrt{g'(x)}} dx &= \int_4^0 \frac{1}{\sqrt{u}} du \\&= \lim_{b \rightarrow 0^+} \int_4^b \frac{1}{\sqrt{u}} du = \lim_{b \rightarrow 0^+} 2\sqrt{u} \Big|_4^b = \lim_{b \rightarrow 0^+} 2\sqrt{b} - 2\sqrt{4} \\&= 0 - 4 = -4\end{aligned}$$

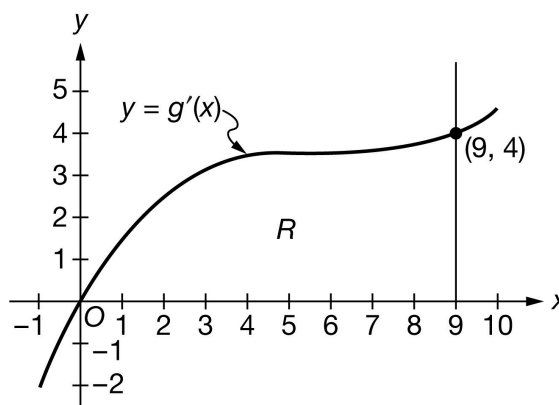
Unit 1 AP Topics 16

39. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graph of g' , the derivative of the twice-differentiable function g , is shown above for $-1 < x < 10$. The graph of g' has exactly one horizontal tangent line, at $x = 6$. Let R be the region in the first quadrant bounded by the graph of g' , the x -axis, and the vertical line $x = 9$. It is known that $g(0) = -10$,

$$g(9) = 17, \text{ and } \int_0^9 g(x) \, dx = 12.6.$$

(a) How many points of inflection does the graph of g have on the interval $-1 < x < 10$? Give a reason for your answer.

(b) Find the average value of g' over the interval $0 \leq x \leq 9$. Show the computations that lead to your answer.

(c) Must there exist a value of c , for $0 < c < 9$, such that $g(c) = 0$? Justify your answer.

(d) Evaluate $\int_0^9 (5g(x) + 6) \, dx$. Show the computations that lead to your answer.

(e) Evaluate $\lim_{x \rightarrow 9} \frac{g(x) - 2x + 2}{\cos x}$. Show the computations that lead to your answer.

(f) Let h be the function defined by $h(x) = e^x \cdot \int_9^x g(t) \, dt$. Find $h'(0)$. Show the computations that lead to your answer.

(g) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis is a rectangle with height x and base in the region R . The volume of the solid is given by

$$\int_0^9 A(x) \, dx. \text{ Write an expression for } A(x).$$

Unit 1 AP Topics 16

(h) What is the volume of the solid described in part (g)? Show the computations that lead to your answer.

(i) Find the value of $\int_0^9 \frac{g''(x)}{\sqrt{g'(x)}} dx$ or show that it does not exist. (Note: $\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We are looking for the application of the relationship between the increasing and decreasing behavior of the first derivative and the points of inflection of the original function.

The first point is earned for “none,” with some relevant mathematical discussion.

The reasoning point is most commonly earned by correctly appealing to the increasing behavior of g' .

An appeal to g'' must address the consideration of both $g'' = 0$ and g'' undefined.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

The graph of g has a point of inflection where g' changes from increasing to decreasing or from decreasing to increasing.

g' is increasing on the interval $-1 < x < 10$.

Therefore, the graph of g has no points of inflection on the interval $-1 < x < 10$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Make sure the response addresses the average value of g' , not of g .

The Fundamental Theorem of Calculus point is earned for the expression $g(9) - g(0)$.

The answer point is earned only for the correct answer. The values of 17 and -10 must be shown.

$\frac{1}{9}(17 - (-10))$ earns both points.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\begin{aligned}\text{Average value} &= \frac{1}{9-0} \int_0^9 g'(x) \, dx = \frac{1}{9}(g(9) - g(0)) \\ &= \frac{1}{9}(17 - (-10)) = 3\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The student must establish that g is a continuous function. The most common, and acceptable, response is “differentiable implies continuous.”

The student must convey that differentiability implies continuity. Stating g is differentiable and continuous, does not earn the first point.

The response that mentions continuity without establishing continuity is still eligible for the second point.

Using, or citing, the Mean Value Theorem will not earn the second point.

To earn the second point the student must explicitly convey an appropriate inequality involving 0, either mathematically or in words. For example, $-10 < 0 < 17$ or $g(0) < 0 < g(9)$.

The second point requires an answer of “yes” or an equivalent statement.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Unit 1 AP Topics 16

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous on $0 \leq x \leq 9$.

$$g(0) = -10 < 0 < 17 = g(9)$$

By the Intermediate Value Theorem, there exists a value of c , for $0 < c < 9$, such that $g(c) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Some (incorrect) responses claim that evaluation of the integrand is equal to the value of the integral.

The first point requires demonstration of both the sum of integrals and the multiplication by a constant.

$5 \cdot 12.6 + 6 \cdot 9$ earns both points. This is the minimal supporting work required.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Properties of definite integrals
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^9 (5g(x) + 6) dx &= 5 \int_0^9 g(x) dx + \int_0^9 6 dx \\ &= 5 \cdot 12.6 + 6 \cdot 9 = 117 \end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal supporting computations required are $\frac{17-18+2}{\cos 9}$. This alone would earn the point.

The student does not have to evaluate $\cos 9$.

An answer of just “sec 9” does not earn the point because it does not show the minimal computations.

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 9} g(x) = g(9)$

$$\lim_{x \rightarrow 9} \frac{g(x) - 2x + 2}{\cos x} = \frac{g(9) - 2 \cdot 9 + 2}{\cos 9} = \frac{17 - 18 + 2}{\cos 9} = \frac{1}{\cos 9}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response of only $e^x \cdot g(x)$ earns only the Fundamental Theorem of Calculus point.

Some responses may reverse the limits of integration and state this is equal to the original integral. This incorrect approach could earn only the first two points, but will not earn the answer point, unless the student resolves the limits reversal issue.

The answer point can only be earned if both of the first two points are earned, and the given answer is correct.

$1 \cdot (-12.6) + 1 \cdot (-10)$ earns all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$h'(x) = e^x \cdot \int_9^x g(t) dt + e^x \cdot g(x)$$

Unit 1 AP Topics 16

$$h'(0) = e^0 \cdot \int_9^0 g(t) dt + e^0 \cdot g(0) = 1 \cdot (-12.6) + 1 \cdot (-10) = -22.6$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the student writes the integral with $x \cdot g'(x)$ as the integrand, we need to see that it is equal to the integral with $A(x)$ for the integrand.

$$\text{So, } \int_0^9 A(x) dx = \int_0^9 x \cdot g'(x) dx \text{ earns the point.}$$

Some responses introduce another variable, or function, for example, $R(x) = g'(x)$, before writing the final expression in terms of this other variable. This earns the point. For example, $A(x) = x \cdot R(x)$ in the presence of $R(x) = g'(x)$.

The point is not earned if the correct integrand appears only in part (h).



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$A(x) = x \cdot g'(x)$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If there is no integration by parts, no points are earned in this part.

The integration by parts point is for an “indefinite integration by parts.” The bounds are associated with the answer point.

Minimal work required to earn the integration by parts point: $9 \cdot 17 - 0 - 12.6$



0	1	2
---	---	---

Unit 1 AP Topics 16

The student response accurately includes **both** of the criteria below.

- ☐ Integration by parts
- ☐ Answer

Solution:

$$u = x \quad dv = g'(x) dx$$

$$du = dx \quad v = g(x)$$

$$\int_0^9 x \cdot g'(x) dx = x \cdot g(x) \Big|_0^9 - \int_0^9 g(x) dx$$

$$= 9 \cdot g(9) - 0 \cdot g(0) - \int_0^9 g(x) dx$$

$$= 9 \cdot 17 - 0 \cdot (-10) - 12.6 = 140.4$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Many students will write the antiderivative without any evidence of substitution. This antiderivative must be completely correct to earn the substitution point.

Limit notation must be used in order to earn the improper integral point.

The student must earn the improper integral point to be eligible for the answer point.

If the student uses an explicit substitution and makes an error in the limits, they cannot earn the answer point.

A response can earn the substitution point for working only with an indefinite integral.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Improper integral
- ☐ Answer

Solution:

Let $u = g'(x)$. Then $du = g''(x) dx$.

Unit 1 AP Topics 16

When $x = 0$, $u = g'(0) = 0$.

When $x = 9$, $u = g'(9) = 4$.

$$\begin{aligned}\int_0^9 \frac{g''(x)}{\sqrt{g'(x)}} dx &= \int_0^4 \frac{1}{\sqrt{u}} du \\&= \lim_{b \rightarrow 0^+} \int_b^4 \frac{1}{\sqrt{u}} du = \lim_{b \rightarrow 0^+} 2\sqrt{u} \Big|_b^4 = 2\sqrt{4} - \lim_{b \rightarrow 0^+} 2\sqrt{b} \\&= 4 - 0 = 4\end{aligned}$$

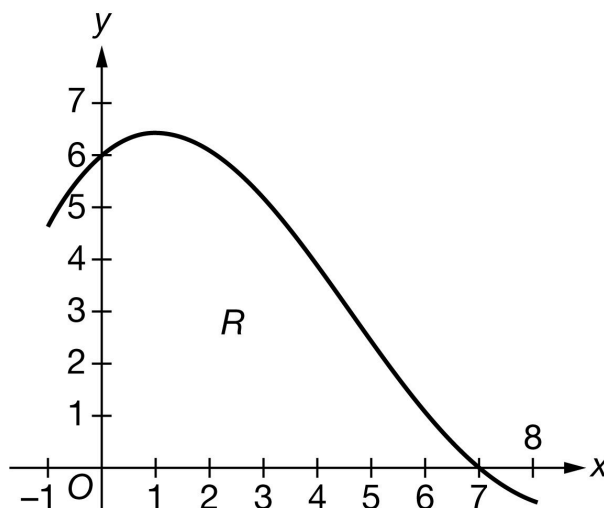
Unit 1 AP Topics 16

40. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of g' , the derivative of the twice-differentiable function g , is shown above for $-1 < x < 8$. The graph of g' has exactly one horizontal tangent line, at $x = 1$. Let R be the region in the first quadrant bounded by the graph of g' and the x -axis from $x = 0$ to $x = 7$. It is known that $g(0) = -15$,

$$g(7) = 13, \text{ and } \int_0^7 g(x) \, dx = 22.4.$$

(a) How many points of inflection does the graph of g have on the interval $-1 < x < 8$? Give a reason for your answer.

(b) Find the average value of g' over the interval $0 \leq x \leq 7$. Show the computations that lead to your answer.

(c) Must there exist a value of c , for $0 < c < 7$, such that $g(c) = 0$? Justify your answer.

(d) Evaluate $\int_0^7 (2g(x) + 3) \, dx$. Show the computations that lead to your answer.

(e) Evaluate $\lim_{x \rightarrow 0} \frac{g(x) + 4x - 4}{\sin x + 1}$. Show the computations that lead to your answer.

(f) Let h be the function defined by $h(x) = e^x \cdot \int_7^x g(t) \, dt$. Find $h'(0)$. Show the computations that lead to your answer.

Unit 1 AP Topics 16

(g) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis is a rectangle with height x and base in the region R . The volume of the solid is given by

$$\int_0^7 A(x) \, dx.$$

Write an expression for $A(x)$.

(h) What is the volume of the solid described in part (g)? Show the computations that lead to your answer.

(i) Find the value of $\int_0^7 \frac{g''(x)}{\sqrt{g'(x)}} \, dx$ or show that it does not exist. (Note: $\int \frac{1}{\sqrt{x}} \, dx = 2\sqrt{x} + C$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We are looking for the application of the relationship between the increasing and decreasing behavior of the first derivative and the points of inflection of the original function.

The first point is earned for “one,” with some relevant mathematical discussion.

The reasoning point is most commonly earned by correctly appealing to the change in behavior of g' at $x = 1$.

An appeal to g'' must address the consideration of both $g'' = 0$ and g'' undefined.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

The graph of g has a point of inflection where g' changes from increasing to decreasing or from decreasing to increasing.

g' changes from increasing to decreasing at $x = 1$.

Therefore, the graph of g has one point of inflection on the interval $-1 < x < 8$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Make sure the response addresses the average value of g' , not of g .

The Fundamental Theorem of Calculus point is earned for the expression $g(7) - g(0)$.

Unit 1 AP Topics 16

The answer point is earned only for the correct answer. The values of 13 and -15 must be shown.

$\frac{1}{7}(13 - (-15))$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\begin{aligned}\text{Average value} &= \frac{1}{7-0} \int_0^7 g'(x) \, dx = \frac{1}{7}(g(7) - g(0)) \\ &= \frac{1}{7}(13 - (-15)) = 4\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The student must establish that g is a continuous function. The most common, and acceptable, response is “differentiable implies continuous.”

The student must convey that differentiability implies continuity. Stating g is differentiable and continuous, does not earn the first point.

The response that mentions continuity without establishing continuity is still eligible for the second point.

Using, or citing, the Mean Value Theorem will not earn the second point.

To earn the second point the student must explicitly convey an appropriate inequality involving 0, either mathematically or in words. For example, $-15 < 0 < 13$ or $g(0) < 0 < g(7)$.

The second point requires an answer of “yes” or an equivalent statement.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions

Unit 1 AP Topics 16

☐ Conclusion using Intermediate Value Theorem

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous on $0 \leq x \leq 7$.

$$g(0) = -15 < 0 < 13 = g(7)$$

By the Intermediate Value Theorem, there exists a value of c , for $0 < c < 7$, such that $g(c) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Some (incorrect) responses claim that evaluation of the integrand is equal to the value of the integral.

The first point requires demonstration of both the sum of integrals and the multiplication by a constant.

$2 \cdot 22.4 + 3 \cdot 7$ earns both points. This is the minimal supporting work required.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Properties of definite integrals

☐ Answer

Solution:

$$\int_0^7 (2g(x) + 3) dx = 2 \int_0^7 g(x) dx + \int_0^7 3 dx$$

$$= 2 \cdot 22.4 + 3 \cdot 7 = 65.8$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal supporting computations required are $\frac{-15+0-4}{\sin 0+1}$ or $\frac{-15+0-4}{0+1}$. Either of these alone would earn the point.

The student does not have to evaluate 0.

Unit 1 AP Topics 16

An answer of just “−19” does not earn the point because it does not show the minimal computations.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 0} g(x) = g(0)$

$$\lim_{x \rightarrow 0} \frac{g(x) + 4x - 4}{\sin x + 1} = \frac{g(0) + 0 - 4}{\sin 0 + 1} = \frac{-15 + 0 - 4}{0 + 1} = -19$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response of only $e^x \cdot g(x)$ earns only the Fundamental Theorem of Calculus point.

Some responses may reverse the limits of integration and state this is equal to the original integral. This incorrect approach could earn only the first two points, but will not earn the answer point, unless the student resolves the limits reversal issue.

The answer point can only be earned if both of the first two points are earned, and the given answer is correct.

$1 \cdot (-22.4) + 1 \cdot (-15)$ earns all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Unit 1 AP Topics 16

$$h'(x) = e^x \cdot \int_7^x g(t) dt + e^x \cdot g(x)$$

$$h'(0) = e^0 \cdot \int_7^0 g(t) dt + e^0 \cdot g(0) = 1 \cdot (-22.4) + 1 \cdot (-15) = -37.4$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the student writes the integral with $x \cdot g'(x)$ as the integrand, we need to see that it is equal to the integral with $A(x)$ for the integrand.

$$\text{So, } \int_0^7 A(x) dx = \int_0^7 x \cdot g'(x) dx \text{ earns the point.}$$

Some responses introduce another variable, or function, for example, $R(x) = g'(x)$, before writing the final expression in terms of this other variable. This earns the point. For example, $A(x) = x \cdot R(x)$ in the presence of $R(x) = g'(x)$.

The point is not earned if the correct integrand appears only in part (h).



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$A(x) = x \cdot g'(x)$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If there is no integration by parts, no points are earned in this part.

The integration by parts point is for an “indefinite integration by parts.” The bounds are associated with the answer point.

Minimal work required to earn the integration by parts point: $7 \cdot 13 - 0 - 22.4$

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integration by parts
- ☐ Answer

Solution:

$$u = x \quad dv = g'(x) dx$$

$$du = dx \quad v = g(x)$$

$$\int_0^7 x \cdot g'(x) dx = x \cdot g(x) \Big|_0^7 - \int_0^7 g(x) dx$$

$$= 7 \cdot g(7) - 0 \cdot g(0) - \int_0^7 g(x) dx$$

$$= 7 \cdot 13 - 0 \cdot (-15) - 22.4 = 68.6$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Many students will write the antiderivative without any evidence of substitution. This antiderivative must be completely correct to earn the substitution point.

Limit notation must be used in order to earn the improper integral point.

The student must earn the improper integral point to be eligible for the answer point.

If the student uses an explicit substitution and makes an error in the limits, they cannot earn the answer point.

A response can earn the substitution point for working only with an indefinite integral.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Improper integral

Unit 1 AP Topics 16

☐ Answer**Solution:**

Let $u = g'(x)$. Then $du = g''(x) dx$.

When $x = 0$, $u = g'(0) = 6$.

When $x = 7$, $u = g'(7) = 0$.

$$\begin{aligned}\int_0^7 \frac{g''(x)}{\sqrt{g'(x)}} dx &= \int_6^0 \frac{1}{\sqrt{u}} du \\&= \lim_{b \rightarrow 0^+} \int_6^b \frac{1}{\sqrt{u}} du = \lim_{b \rightarrow 0^+} 2\sqrt{u} \Big|_6^b = \lim_{b \rightarrow 0^+} 2\sqrt{b} - 2\sqrt{6} \\&= 0 - 2\sqrt{6} = -2\sqrt{6}\end{aligned}$$

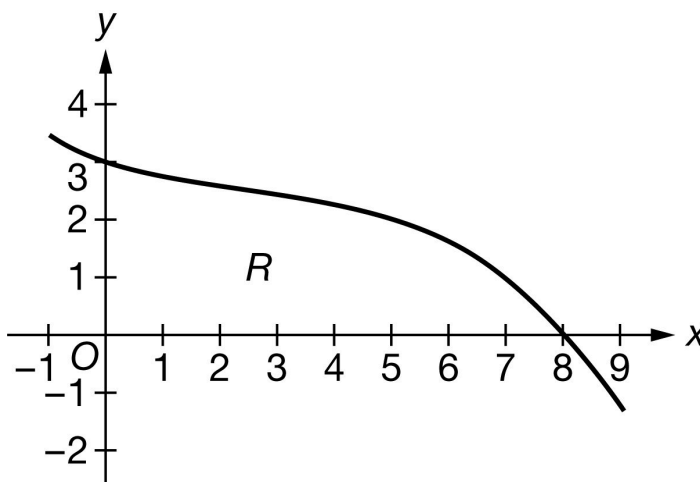
Unit 1 AP Topics 16

41. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of g' , the derivative of the twice-differentiable function g , is shown above for $-1 < x < 9$. The graph of g' has no horizontal tangent lines on the interval $-1 < x < 9$. Let R be the region in the first quadrant bounded by the graph of g' and the x -axis from $x = 0$ to $x = 8$. It is known that

$$g(0) = -6, g(8) = 10, \text{ and } \int_0^8 g(x) \, dx = 28.8.$$

(a) How many points of inflection does the graph of g have on the interval $-1 < x < 9$? Give a reason for your answer.

(b) Find the average value of g' over the interval $0 \leq x \leq 8$. Show the computations that lead to your answer.

(c) Must there exist a value of c , for $0 < c < 8$, such that $g(c) = 0$? Justify your answer.

(d) Evaluate $\int_0^8 (2g(x) + 5) \, dx$. Show the computations that lead to your answer.

(e) Evaluate $\lim_{x \rightarrow 0} \frac{g(x) + x - 1}{\sin x - 1}$. Show the computations that lead to your answer.

(f) Let h be the function defined by $h(x) = e^x \cdot \int_8^x g(t) \, dt$. Find $h'(0)$. Show the computations that lead to your answer.

Unit 1 AP Topics 16

(g) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis is a rectangle with height x and base in the region R . The volume of the solid is given by

$$\int_0^8 A(x) \, dx.$$

Write an expression for $A(x)$.

(h) What is the volume of the solid described in part (g)? Show the computations that lead to your answer.

(i) Find the value of $\int_0^8 \frac{g''(x)}{\sqrt{g'(x)}} \, dx$ or show that it does not exist. (Note: $\int \frac{1}{\sqrt{x}} \, dx = 2\sqrt{x} + C$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We are looking for the application of the relationship between the increasing and decreasing behavior of the first derivative and the points of inflection of the original function.

The first point is earned for “none,” with some relevant mathematical discussion.

The reasoning point is most commonly earned by correctly appealing to the decreasing behavior of g' .

An appeal to g'' must address the consideration of both $g'' = 0$ and g'' undefined.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

The graph of g has a point of inflection where g' changes from increasing to decreasing or from decreasing to increasing.

g' is decreasing on the interval $-1 < x < 9$.

Therefore, the graph of g has no points of inflection on the interval $-1 < x < 9$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Make sure the response addresses the average value of g' , not of g .

The Fundamental Theorem of Calculus point is earned for the expression $g(8) - g(0)$.

Unit 1 AP Topics 16

The answer point is earned only for the correct answer. The values of 10 and -6 must be shown.

$\frac{1}{8}(10 - (-6))$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\begin{aligned}\text{Average value} &= \frac{1}{8-0} \int_0^8 g'(x) \, dx = \frac{1}{8}(g(8) - g(0)) \\ &= \frac{1}{8}(10 - (-6)) = 2\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The student must establish that g is a continuous function. The most common, and acceptable, response is “differentiable implies continuous.”

The student must convey that differentiability implies continuity. Stating g is differentiable and continuous, does not earn the first point.

The response that mentions continuity without establishing continuity is still eligible for the second point.

Using, or citing, the Mean Value Theorem will not earn the second point.

To earn the second point the student must explicitly convey an appropriate inequality involving 0, either mathematically or in words. For example, $-6 < 0 < 10$ or $g(0) < 0 < g(8)$.

The second point requires an answer of “yes” or an equivalent statement.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions

Unit 1 AP Topics 16

☐ Conclusion using Intermediate Value Theorem

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous on $0 \leq x \leq 8$.

$$g(0) = -6 < 0 < 10 = g(8)$$

By the Intermediate Value Theorem, there exists a value of c , for $0 < c < 8$, such that $g(c) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Some (incorrect) responses claim that evaluation of the integrand is equal to the value of the integral.

The first point requires demonstration of both the sum of integrals and the multiplication by a constant.

$2 \cdot 28.8 + 5 \cdot 8$ earns both points. This is the minimal supporting work required.



0	1	2
---	---	---

The student response accurately includes **one** of the criteria below.

☐ Properties of definite integrals

☐ Answer

Solution:

$$\int_0^8 (2g(x) + 5) dx = 2 \int_0^8 g(x) dx + \int_0^8 5 dx$$

$$= 2 \cdot 28.8 + 5 \cdot 8 = 97.6$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal supporting computations required are $\frac{-6-1}{\sin 0-1}$ or $\frac{-6-1}{0-1}$. Either of these alone would earn the point.

The student does not have to evaluate $\sin 0$.

An answer of just “7” does not earn the point because it does not show the minimal computations.

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 0} g(x) = g(0)$

$$\lim_{x \rightarrow 0} \frac{g(x) + x - 1}{\sin x - 1} = \frac{g(0) + 0 - 1}{\sin 0 - 1} = \frac{-6 + 0 - 1}{0 - 1} = 7$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response of only $e^x \cdot g(x)$ earns only the Fundamental Theorem of Calculus point.

Some responses may reverse the limits of integration and state this is equal to the original integral. This incorrect approach could earn only the first two points, but will not earn the answer point, unless the student resolves the limits reversal issue.

The answer point can only be earned if both of the first two points are earned, and the given answer is correct.

$1 \cdot (-28.8) + 1 \cdot (-6)$ earns all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$h'(x) = e^x \cdot \int_8^x g(t) dt + e^x \cdot g(x)$$

Unit 1 AP Topics 16

$$h'(0) = e^0 \cdot \int_8^0 g(t) dt + e^0 \cdot g(0) = 1 \cdot (-28.8) + 1 \cdot (-6) = -34.8$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the student writes the integral with $x \cdot g'(x)$ as the integrand, we need to see that it is equal to the integral with $A(x)$ for the integrand.

$$\text{So, } \int_0^8 A(x) dx = \int_0^8 x \cdot g'(x) dx \text{ earns the point.}$$

Some responses introduce another variable, or function, for example, $R(x) = g'(x)$, before writing the final expression in terms of this other variable. This earns the point. For example, $A(x) = x \cdot R(x)$ in the presence of $R(x) = g'(x)$.

The point is not earned if the correct integrand appears only in part (h).



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$A(x) = x \cdot g'(x)$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If there is no integration by parts, no points are earned in this part.

The integration by parts point is for an “indefinite integration by parts.” The bounds are associated with the answer point.

Minimal work required to earn the integration by parts point: $8 \cdot 10 - 0 - 28.8$



0	1	2
---	---	---

Unit 1 AP Topics 16

The student response accurately includes **both** of the criteria below.

- ☐ Integration by parts
- ☐ Answer

Solution:

$$u = x \quad dv = g'(x) dx$$

$$du = dx \quad v = g(x)$$

$$\int_0^8 x \cdot g'(x) dx = x \cdot g(x) \Big|_0^8 - \int_0^8 g(x) dx$$

$$= 8 \cdot g(8) - 0 \cdot g(0) - \int_0^8 g(x) dx$$

$$= 8 \cdot 10 - 0 \cdot (-6) - 28.8 = 51.2$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Many students will write the antiderivative without any evidence of substitution. This antiderivative must be completely correct to earn the substitution point.

Limit notation must be used in order to earn the improper integral point.

The student must earn the improper integral point to be eligible for the answer point.

If the student uses an explicit substitution and makes an error in the limits, they cannot earn the answer point.

A response can earn the substitution point for working only with an indefinite integral.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Improper integral
- ☐ Answer

Solution:

Let $u = g'(x)$. Then $du = g''(x) dx$.

Unit 1 AP Topics 16

When $x = 0$, $u = g'(0) = 3$.

When $x = 8$, $u = g'(8) = 0$.

$$\begin{aligned}\int_0^8 \frac{g''(x)}{\sqrt{g'(x)}} dx &= \int_3^0 \frac{1}{\sqrt{u}} du \\&= \lim_{b \rightarrow 0^+} \int_3^b \frac{1}{\sqrt{u}} du = \lim_{b \rightarrow 0^+} 2\sqrt{u} \Big|_3^b = \lim_{b \rightarrow 0^+} 2\sqrt{b} - 2\sqrt{3} \\&= 0 - 2\sqrt{3} = -2\sqrt{3}\end{aligned}$$

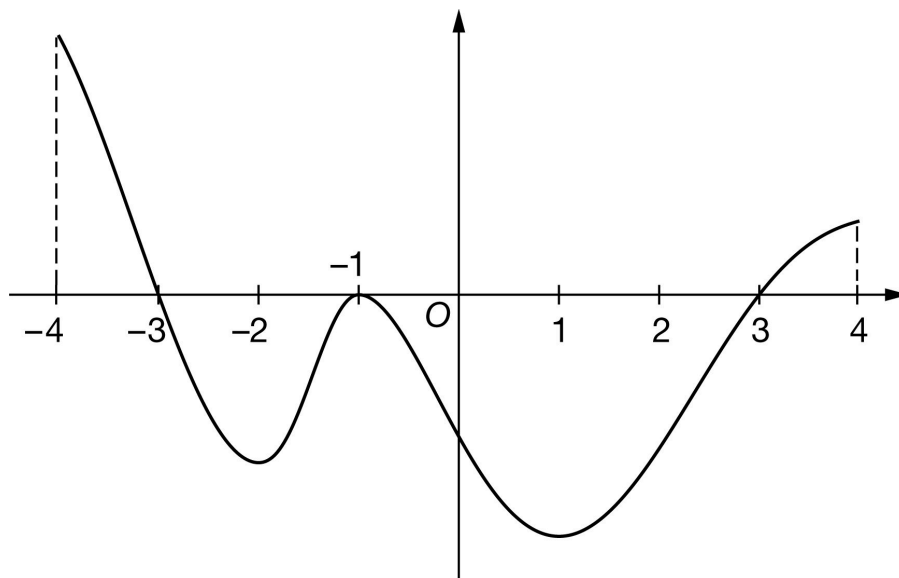
Unit 1 AP Topics 16

42. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the differentiable function f with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of f on the intervals $[-4, -3]$, $[-3, -1]$, $[-1, 3]$, and $[3, 4]$ are 12, 17, 48, and 4, respectively. The graph of f has horizontal tangents at $x = -2$, $x = -1$, and $x = 1$. Let g be the function defined by $g(x) = \int_{-1}^x f(t) dt$ for $-4 \leq x \leq 4$.

- Find the value of $g(3)$.
- Find the value of $g(-4)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of g on the interval $-4 < x < 4$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- For $-4 < x < 4$, find the x -coordinate of each point of inflection of the graph of g . Justify your answers.
- Explain why there must be a value of d , for $-1 < d < 3$, such that $g(d) = -25$.
- Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x}$. Give a reason for your answer.

Unit 1 AP Topics 16

(h) The function a is defined by $a(x) = g(x^3)$. It is known that $f(1) = -21.1$. Find $a'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -48 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(3) = \int_{-1}^3 f(t) dt = -48$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 12 and ± 17 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 12 and ± 17 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-12 - (-17)$ earns both points because numerical expressions need not be simplified.
- The expression $-12 - (-17)$ earns the first point, but not the second point.
- The expression $-12 - (-17) + 48$ earns neither point.

Note: Just the list $-12, -17$ earns neither point. However, if the values -12 and -17 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals 5 earns just the second point because not enough work has been shown.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-3}^{-4} f(t) dt \right| = 12$ and $\left| \int_{-3}^{-1} f(t) dt \right| = 17$

☐ Answer

Solution:

$$\begin{aligned} g(-4) &= \int_{-1}^{-4} f(t) dt = - \int_{-4}^{-1} f(t) dt \\ &= - \int_{-4}^{-3} f(t) dt - \int_{-3}^{-1} f(t) dt = -12 - (-17) = 5 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = -1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 4)$. The point is still earned provided the three critical points, $x = -3$, $x = -1$, and $x = 3$, are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$, and $(3, 0)$ read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = f(x) = 0 \text{ at } x = -3, x = -1, \text{ and } x = 3.$$

Unit 1 AP Topics 16

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $f(x)$ and $g'(x)$ (that is, $f(x) = g'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $f(x) = g'(x)$. Any additional error (for example, not justifying that at $x = -3$, g has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-3)$ changes from positive to negative”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = -1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

- $g(-4) = 5$

- $g(-3) = 17$

- $g(-1) = 0$

- $g(3) = -48$

- $g(4) = -44$

Unit 1 AP Topics 16

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -3$, g has a relative maximum because $g' = f$ is positive for $x < -3$ and $g' = f$ is negative for $x > -3$.
- At $x = -1$, g has neither because $g' = f$ is negative for $x < -1$ and $g' = f$ is negative for $x > -1$.
- At $x = 3$, g has a relative minimum because $g' = f$ is negative for $x < 3$ and $g' = f$ is positive for $x > 3$.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -3$, g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = -1$, g has neither a relative minimum nor a relative maximum because $g'(x) = f(x)$ does not change sign there.

At $x = 3$, g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list -2 , -1 , 1 , and only this list, earns the first point.

A response referring only to $g'(x)$ is eligible to earn the second point.

A response referring only to $f(x)$ is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $g'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- g' changes from increasing to decreasing or vice versa
- g'' changes from positive to negative or vice versa

Unit 1 AP Topics 16

· g' has a local extrema

Any incorrect statement (such as g' changes from increasing to decreasing at $x = -2$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of g occur where $g' = f$ changes from increasing to decreasing or vice versa. Therefore, the points of inflection occur at $x = -2$, $x = -1$, and $x = 1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that g is continuous in order to be eligible to earn the point. The response is not required to state that g is differentiable $\Rightarrow g$ is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $-1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since g is continuous, and $-48 < -25 < 0$, such a point exists” would earn the point.

The value of -25 must be bounded by -48 and 0 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $g(-1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $g(-1) = 0$ must be present.

A response may reference a value of $g(3)$ determined in part (a). If the value of $g(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is less than -25 .

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$g' = f$ is differentiable on $-1 \leq x \leq 3$. $\Rightarrow g$ is continuous on $-1 \leq x \leq 3$.

$$g(3) = -48 < -25 < 0 = g(-1)$$

By the Intermediate Value Theorem, there must be a value of d , for $-1 < d < 3$, such that $g(d) = -25$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that g and g' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $3^2 - 9 = 0$ and $g(3) + 48 = 0$.
 - All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $g(3) = -48$ are not enough to earn the first point.
 - The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.
 - Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} = \frac{0}{0}$ does not earn the first point. The statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} \rightarrow \frac{0}{0}$ does earn the first point.
- The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.
- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
 - Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} = \frac{2x}{g'(x) + 16}$, will earn the second point, but not the third point.

Unit 1 AP Topics 16

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 3} \frac{x^2-9}{g(x)+16x} = \frac{0}{0}$ and $\lim_{x \rightarrow 3} \frac{x^2-9}{g(x)+16x} = \frac{2x}{g'(x)+16}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{3}{8}$, or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$g' = f$ is differentiable. $\Rightarrow g' = f$ is continuous.
 $\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 3} g(x) = g(3)$ and $\lim_{x \rightarrow 3} g'(x) = g'(3)$.

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

$$\lim_{x \rightarrow 3} (g(x) + 16x) = g(3) + 16 \cdot 3 = -48 + 48 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2-9}{g(x)+16x} &= \lim_{x \rightarrow 3} \frac{2x}{g'(x)+16} \\ &= \frac{2 \cdot 3}{g'(3)+16} = \frac{6}{f(3)+16} = \frac{6}{0+16} = \frac{3}{8} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$$\cdot g'(x^3) \cdot 3x^2$$

$$\cdot f(x^3) \cdot 3x^2$$

Unit 1 AP Topics 16

$$\cdot g'(1^3) \cdot 3 \cdot 1^2$$

$$\cdot f(1^3) \cdot 3 \cdot 1^2$$

$$\cdot g'(1) \cdot 3 \cdot 1^2$$

$$\cdot f(1^3) \cdot 3 \cdot 1$$

Responses that do not earn the point include:

$$\cdot g'(x^3) \cdot 3x$$

$$\cdot f(x^3) \cdot 3x$$

$$\cdot g'(x) \cdot 3x^2$$

$$\cdot f(x) \cdot 3x^2$$

$$\cdot g'(1) \cdot 3$$

$$\cdot f(1) \cdot 3$$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking g' and f either directly or indirectly in the presented derivative of $a(x)$ or in $a'(1)$. For example, the second point is earned by $f(x^3)$ or by $f(1^3)$.

The second point may be earned without earning the first point. For example, $g'(x^3) \cdot 3x = f(x^3) \cdot 3x$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and f are linked.

The second point may be earned by going directly from $g'(1)$ to -21.1 . The function f does not have to be stated explicitly. For example, $g'(1^3) \cdot 3 \cdot 1^2 = -21.1 \cdot 3 \cdot 1$ earns the first two points. Recall, if $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f(x^3) \cdot 3x^2$ and $f(1^3) \cdot 3 \cdot 1^2$ earn the first two points.

The third point is earned for the correct answer of -63.3 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (h). Responses of the form $g'(x)^3 \cdot 3x^2$ or $(g'(x))^3 \cdot 3x^2$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
---	---	---	---

Unit 1 AP Topics 16

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$a'(x) = g'(x^3) \cdot 3x^2$$

$$= f(x^3) \cdot 3x^2$$

$$a'(1) = g'(1^3) \cdot 3 \cdot 1^2 = g'(1) \cdot 3 = f(1) \cdot 3 = (-21.1) \cdot 3 = -63.3$$

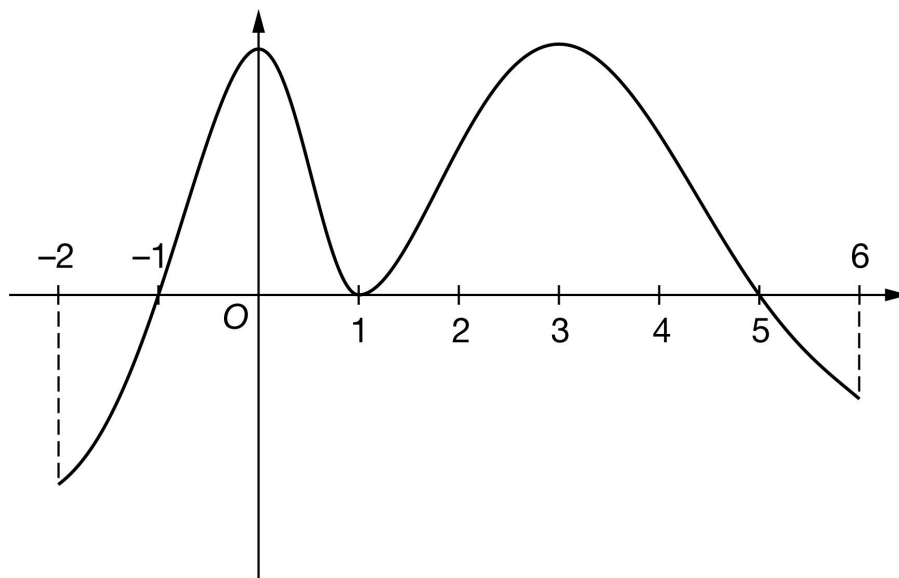
Unit 1 AP Topics 16

43. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the differentiable function h with domain $-2 \leq x \leq 6$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of h on the intervals $[-2, -1]$, $[-1, 1]$, $[1, 5]$, and $[5, 6]$ are 8, 19, 40, and 4, respectively. The graph of h has horizontal tangents at $x = 0$, $x = 1$, and $x = 3$. Let g be the function defined by $g(x) = \int_1^x h(t) \, dt$ for $-2 \leq x \leq 6$.

- Find the value of $g(5)$.
- Find the value of $g(-2)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of g on the interval $-2 < x < 6$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- For $-2 < x < 6$, find the x -coordinate of each point of inflection of the graph of g . Justify your answers.
- Explain why there must be a value of w , for $1 < w < 5$, such that $g(w) = 5$.
- Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x}$. Give a reason for your answer.
- The function z is defined by $z(x) = g(2x)$. It is known that $h(4) = 11.25$. Find $z'(2)$. Show the computations that lead to your answer.

Unit 1 AP Topics 16**Part A**

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 40 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(5) = \int_1^5 h(t) dt = 40$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 8 and ± 19 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 8 and ± 19 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-8) - 19$ earns both points because numerical expressions need not be simplified.
- The expression $(-8) - 19$ earns the first point, but not the second point.
- The expression $-(-8) - 19 - 40$ earns neither point.

Note: Just the list $-8, 19$ earns neither point. However, if the values -8 and 19 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals -11 earns just the second point because not enough work has been shown.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-2}^{-1} h(t) dt \right| = 8$ and $\left| \int_1^{-1} h(t) dt \right| = 19$

☐ Answer

Solution:

$$\begin{aligned} g(-2) &= \int_1^{-2} h(t) dt = - \int_{-2}^1 h(t) dt \\ &= - \int_{-2}^{-1} h(t) dt - \int_{-1}^1 h(t) dt = -(-8) - 19 = -11 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 1$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-2, 6)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-2, 6)$. The point is still earned provided the three critical points, $x = -1$, $x = 1$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(1, 0)$, and $(5, 0)$ read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = h(x) = 0 \text{ at } x = -1, x = 1, \text{ and } x = 5.$$

Unit 1 AP Topics 16**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justification based on g' is eligible for all three points.

If a response gives justifications based only on h , then:

- the response is eligible for all three points, if the response presents the connection between $h(x)$ and $g'(x)$ (that is, $h(x) = g'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $h(x) = g'(x)$. Any additional error (for example, not justifying that at $x = -1$, g has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $h(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-1)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -1$, $x = 1$, $x = 5$) must be included in the table or list. The correct values of relevant inputs are:

- $g(-2) = -11$
- $g(-1) = -19$
- $g(1) = 0$
- $g(5) = 40$
- $g(6) = 36$

Unit 1 AP Topics 16

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -1$, g has a relative minimum because $g' = h$ is negative for $x < -1$ and $g' = h$ is positive for $x > -1$.
- At $x = 1$, g has neither because $g' = h$ is positive for $x < 1$ and $g' = h$ is positive for $x > 1$.
- At $x = 5$, g has a relative maximum because $g' = h$ is positive for $x < 5$ and $g' = h$ is negative for $x > 5$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -1$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 1$, g has neither a relative minimum nor a relative maximum because $g'(x) = h(x)$ does not change sign there.

At $x = 5$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list 0, 1, 3, and only this list, earns the first point.

A response referring only to $g'(x)$ is eligible to earn the second point.

A response referring only to $h(x)$ is not eligible to earn the second point, unless the response has established the link between $h(x)$ and $g'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- g' changes from increasing to decreasing or vice versa
- g'' changes from positive to negative or vice versa

Unit 1 AP Topics 16

· g' has a local extrema

Any incorrect statement (such as g' changes from decreasing to increasing at $x = 0$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of g occur where $g' = h$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = 0$, $x = 1$, and $x = 3$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that g is continuous in order to be eligible to earn the point. The response is not required to state that “ g is differentiable $\Rightarrow g$ is continuous.”

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $1 < x < 5$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since g is continuous, and $0 < 5 < 40$, such a point exists” would earn the point.

The value of 5 must be bounded by 0 and 40 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $g(1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $g(1) = 0$ must be present.

A response may reference a value of $g(5)$ determined in part (a). If the value of $g(5)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 5.

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$g' = h$ is differentiable on $1 \leq x \leq 5$. $\Rightarrow g$ is continuous on $1 \leq x \leq 5$.

$$g(1) = 0 < 5 < 40 = g(5)$$

By the Intermediate Value Theorem, there must be a value of w , for $1 < w < 5$, such that $g(w) = 5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that g and g' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $5^2 - 25 = 0$ and $g(5) - 40 = 0$.
 - All arithmetic/algebraic expressions must be completely simplified to zero. For example, $5^2 = 25$ and $g(5) = 40$ are not enough to earn the first point.
 - The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.
 - Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{0}{0}$ does not earn the first point. The statement $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} \rightarrow \frac{0}{0}$ does earn the first point.
- The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.
- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
 - Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{2x}{g'(x) - 8}$, will earn the second point, but not the third point.

Unit 1 AP Topics 16

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{0}{0}$ and $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{2x}{g'(x) - 8}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-5/4$, or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$g' = h$ is differentiable. $\Rightarrow g' = h$ is continuous.
 $\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 5} g(x) = g(5)$ and $\lim_{x \rightarrow 5} g'(x) = g'(5)$.

$$\lim_{x \rightarrow 5} (x^2 - 25) = 5^2 - 25 = 0$$

$$\lim_{x \rightarrow 5} (g(x) - 8x) = g(5) - 8 \cdot 5 = 40 - 40 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} &= \lim_{x \rightarrow 5} \frac{2x}{g'(x) - 8} \\ &= \frac{2 \cdot 5}{g'(5) - 8} = \frac{10}{h(5) - 8} = \frac{10}{0 - 8} = -\frac{5}{4} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot g'(2x) \cdot 2$$

$$\cdot h(2x) \cdot 2$$

Unit 1 AP Topics 16

$$\cdot g'(2 \cdot 2) \cdot 2$$

$$\cdot h(2 \cdot 2) \cdot 2$$

Responses that do not earn the point include:

$$\cdot g'(x) \cdot 2$$

$$\cdot h(x) \cdot 2$$

$$\cdot g'(2) \cdot 2$$

$$\cdot h(2) \cdot 2$$

$$\cdot g(2x) \cdot 2$$

The second point is earned for the application of the Fundamental Theorem of Calculus as evidenced by linking g' prime and h either directly or indirectly in the presented derivative of $z(x)$ or in $z'(2)$. For example, the second point is demonstrated by $h(2x)$ or by $h(2 \cdot 2)$.

The second point may be earned without earning the first point. For example, $g'(2x) = h(2x)$ does not earn the first point because the Chain rule is applied incorrectly, but does earn the second point because g' and h are linked.

The second point may be earned by going directly from $g'(4)$ to 11.25. The function h does not have to be stated explicitly. For example, $g'(2 \cdot 2) \cdot 2 = 11.25 \cdot 2$ earns the first two points. Recall, if $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified to earn the first point.

The first two points can be earned by a single expression. For example, both $h(2x) \cdot 2$ and $h(2 \cdot 2) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of 22.5, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$z'(x) = g'(2x) \cdot 2$$

Unit 1 AP Topics 16

$$= h(2x) \cdot 2$$

$$z'(2) = h(2 \cdot 2) \cdot 2 = h(4) \cdot 2 = 11.25 \cdot 2 = 22.5$$

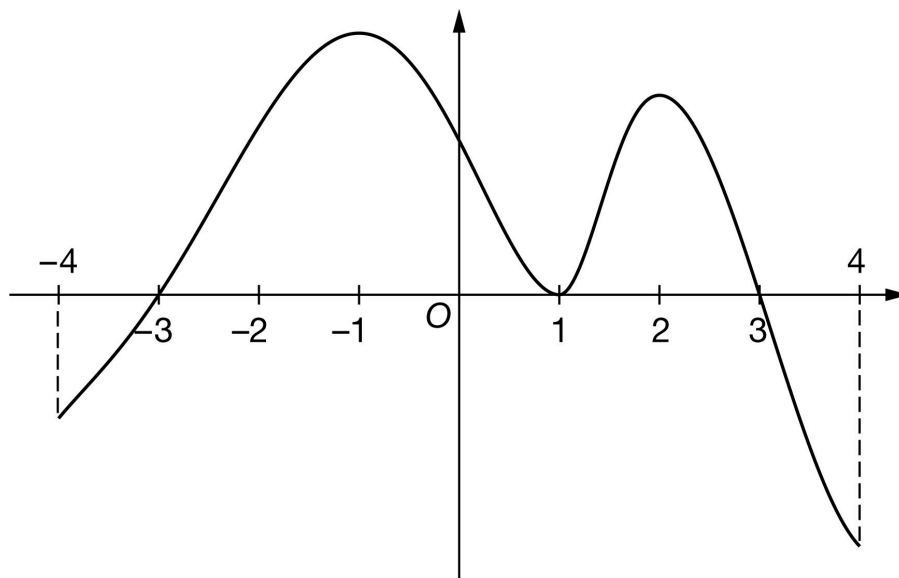
Unit 1 AP Topics 16

44. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the differentiable function f with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of f on the intervals $[-4, -3]$, $[-3, 1]$, $[1, 3]$, and $[3, 4]$ are 5, 46, 18, and 11, respectively. The graph of f has horizontal tangents at $x = -1$, $x = 1$, and $x = 2$. Let h be the function defined by $h(x) = \int_1^x f(t) dt$ for $-4 \leq x \leq 4$.

- Find the value of $h(3)$.
- Find the value of $h(-4)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of h on the interval $-4 < x < 4$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-4 < x < 4$, find the x -coordinate of each point of inflection of the graph of h . Justify your answers.
- Explain why there must be a value of c , for $1 < c < 3$, such that $h(c) = 13$.
- Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x}$. Give a reason for your answer.

Unit 1 AP Topics 16

(h) The function m is defined by $m(x) = h(x^2)$. It is known that $f(4) = -19.3$. Find $m'(2)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 18 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(3) = \int_1^3 f(t) dt = 18$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 5 and ± 46 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 5 and ± 46 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-5) - 46$ earns both points because numerical expressions need not be simplified.
- The expression $-5 - 46$ earns the first point, but not the second point.
- The expression $-(-5) - 46 + 8$ earns neither point.

Note: Just the list $-5, 46$ earns neither point. However, if the values -5 and 46 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals -41 earns just the second point because not enough work has been shown.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-4}^{-3} f(t) \, dt \right| = 5$ and $\left| \int_{-3}^1 f(t) \, dt \right| = 46$

☐ Answer

Solution:

$$\begin{aligned} h(-4) &= \int_1^{-4} f(t) \, dt = - \int_{-4}^1 f(t) \, dt \\ &= - \int_{-4}^{-3} f(t) \, dt - \int_{-3}^1 f(t) \, dt = -(-5) - 46 = -41 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 4)$. The point is still earned provided the three critical points, $x = -3$, $x = 1$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(1, 0)$, and $(3, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = f(x) = 0 \text{ at } x = -3, x = 1, \text{ and } x = 3.$$

Unit 1 AP Topics 16**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $f(x)$ and $h'(x)$ (that is, $f(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $f(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-3)$ changes from positive to negative”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = 1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

- $h(-4) = -41$

- $h(-3) = -46$

- $h(1) = 0$

- $h(3) = 18$

- $h(4) = 7$

Unit 1 AP Topics 16

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -3$, h has a relative minimum because $h' = f$ is negative for $x < -3$ and $h' = f$ is positive for $x > -3$.
- At $x = 1$, h has neither because $h' = f$ is positive for $x < 1$ and $h' = f$ is positive for $x > 1$.
- At $x = -3$, h has a relative maximum because $h' = f$ is positive for $x < 3$ and $h' = f$ is negative for $x > 3$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = 1$, h has neither a relative minimum nor a relative maximum because $h'(x) = f(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $-1, 1, 2$ and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $f(x)$ is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- h' changes from increasing to decreasing or vice versa
- h'' changes from positive to negative or vice versa

Unit 1 AP Topics 16

· h' has a local extrema

Any incorrect statement (such as h' changes from decreasing to increasing at $x = -1$) fails to earn the second point.

If the response addresses at least one correct value of x it is eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = f$ changes from increasing to decreasing or vice versa. Therefore, the points of inflection occur at $x = -1$, $x = 1$, and $x = 2$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that h is differentiable $\Rightarrow$ that h is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since h is continuous, and $0 < 13 < 18$, such a point exists” would earn the point.

The value of 13 must be bounded by 0 and 18 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $h(1) = 0$ must be present.

A response may reference a value of $h(3)$ determined in part (a). If the value of $h(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 13.

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$h' = f$ is differentiable on $1 \leq x \leq 3$. $\Rightarrow h$ is continuous on $1 \leq x \leq 3$.

$$h(1) = 0 < 13 < 18 = h(3)$$

By the Intermediate Value Theorem, there must be a value of c , for $1 < c < 3$, such that $h(c) = 13$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $3^2 - 9 = 0$ and $h(3) - 18 = 0$.
 - All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $h(3) = 18$ are not enough to earn the first point.
 - The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.
 - Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{0}{0}$ does not earn the first point. The statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x} \rightarrow \frac{0}{0}$ does earn the first point.
- The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.
- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
 - Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{2x}{h'(x) - 6}$, will earn the second point, but not the third point.

Unit 1 AP Topics 16

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-6x} = \frac{0}{0}$ and $\lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-6x} = \frac{2x}{h'(x)-6}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of -1 , or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

$h' = f$ is differentiable. $\Rightarrow h' = f$ is continuous.
 $\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \rightarrow 3} h(x) = h(3)$ and $\lim_{x \rightarrow 3} h'(x) = h'(3)$.

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

$$\lim_{x \rightarrow 3} (h(x) - 6x) = h(3) - 6 \cdot 3 = 18 - 18 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-6x} &= \lim_{x \rightarrow 3} \frac{2x}{h'(x)-6} \\ &= \frac{2 \cdot 3}{h'(3)-6} = \frac{6}{f(3)-6} = \frac{6}{0-6} = -1 \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

- $\cdot h'(x^2) \cdot 2x$
- $\cdot f(x^2) \cdot 2x$
- $\cdot h'(2^2) \cdot 2 \cdot 2$

Unit 1 AP Topics 16

$$\cdot f(2^2) \cdot 2 \cdot 2$$

$$\cdot h'(4) \cdot 2 \cdot 2$$

$$\cdot f(4) \cdot 2 \cdot 2$$

$$\cdot h'(2^2) \cdot 4$$

$$\cdot f(2^2) \cdot 4$$

Responses that do not earn the point include:

$$\cdot h'(x^2) \cdot 2$$

$$\cdot f(x^2) \cdot 2$$

$$\cdot h'(x) \cdot 2x$$

$$\cdot f(x) \cdot 2x$$

$$\cdot h'(4) \cdot 4$$

$$\cdot f(4) \cdot 4$$

$$\cdot h(x^2) \cdot 2x$$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $m(x)$ or in $m'(2)$. For example, the second point is earned by $f(x^2)$ or by $f(2^2)$.

The second point may be earned without earning the first point. For example, $h'(x^2) \cdot 2 = f(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and f are linked.

The second point may be earned by going directly from $h'(1)$ to -19.3 . The function f does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot h'(2^2) \cdot 2 \cdot 2 = -19.3 \cdot 2 \cdot 2$$

$$\cdot h'(2^2) \cdot 4 = -19.3 \cdot 4$$

Recall, if $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f(x^2) \cdot 2x$ and $f(2^2) \cdot 2 \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -77.2 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Unit 1 AP Topics 16

Be careful of parenthesis errors in part (h). Responses of the form $h'(x^2) \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$m'(x) = h'(x^2) \cdot 2x = f(x^2) \cdot 2x$$

$$m'(2) = f(2^2) \cdot 2 \cdot 2 = f(4) \cdot 4 = -19.3 \cdot 4 = -77.2$$

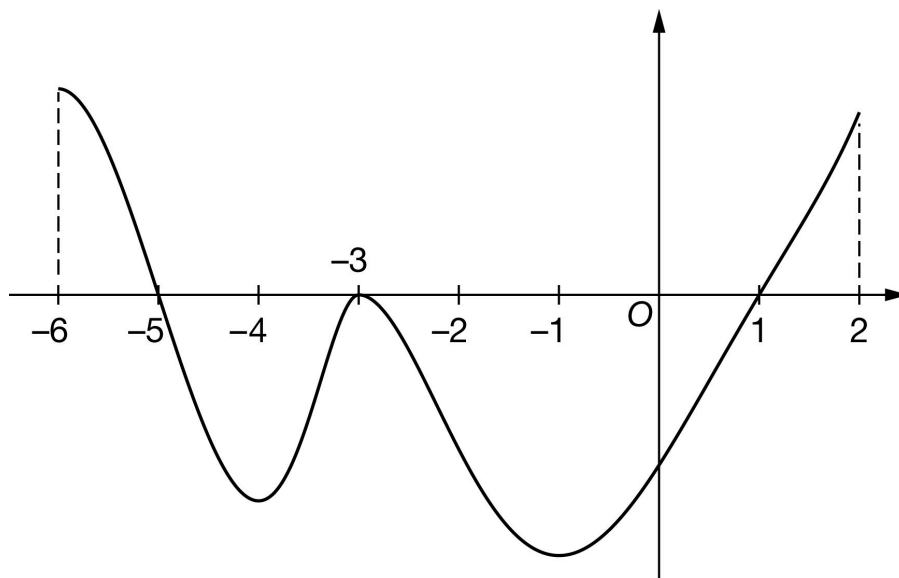
Unit 1 AP Topics 16

45. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the differentiable function g with domain $-6 \leq x \leq 2$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[-6, -5]$, $[-5, -3]$, $[-3, 1]$, and $[1, 2]$ are 9, 17, 42, and 6, respectively. The graph of g has horizontal tangents at $x = -4$, $x = -3$, and $x = -1$. Let h be the function defined by $h(x) = \int_{-3}^x g(t) dt$ for $-6 \leq x \leq 2$.

- Find the value of $h(1)$.
- Find the value of $h(-6)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of h on the interval $-6 \leq x \leq 2$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-6 \leq x \leq 2$, find the x -coordinate of each point of inflection of the graph of h . Justify your answers.
- Explain why there must be a value of k , for $-3 < k < 1$, such that $h(k) = -33$.
- Find $\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x}$. Give a reason for your answer.

Unit 1 AP Topics 16

(h) The function p is defined by $p(x) = h(2x)$. It is known that $g(-2) = -10.8$. Find $p'(-1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -42 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(1) = \int_{-3}^1 g(t) dt = -42$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 9 and ± 17 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 9 and ± 17 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-9 - (-17)$ earns both points because numerical expressions need not be simplified.
- The expression $-9 - 17$ earns the first point, but not the second point.
- The expression $-9 - (-17) + 42$ earns neither point.

Note: Just the list $-9, -17$ earns neither point. However, if the values -9 and -17 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals equal to 8 earns just the second point because not enough work has been shown.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-6}^{-5} f(t) \, dt \right| = 9$ and $\left| \int_{-5}^{-3} f(t) \, dt \right| = 17$

☐ Answer

Solution:

$$\begin{aligned} h(-6) &= \int_{-3}^{-6} g(t) \, dt = - \int_{-6}^{-3} g(t) \, dt \\ &= - \int_{-6}^{-5} g(t) \, dt - \int_{-5}^{-3} g(t) \, dt = -9 - (-17) = 8 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -5$, $x = -3$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-6, 2)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-6, 2)$. The point is still earned provided the three critical points, $x = -5$, $x = -3$, and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-5, 0)$, $(-3, 0)$, and $(1, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = g(x) = 0 \text{ at } x = -5, x = -3, \text{ and } x = 1.$$

Unit 1 AP Topics 16**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $g(x)$ and $h'(x)$ (that is, $g(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $g(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -5$, h has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-5)$ changes from positive to negative”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -5$, $x = -3$, $x = 1$) must be included in the table or list. The correct values of relevant inputs are:

- $h(-6) = 8$
- $h(-5) = 17$
- $h(-3) = 0$
- $h(1) = -42$
- $h(2) = -36$

Unit 1 AP Topics 16

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -5$, h has a relative maximum because $h' = g$ is positive for $x < -5$ and $h' = g$ is negative for $x > -5$.
- At $x = -3$, h has neither because $h' = g$ is negative for $x < -3$ and $h' = g$ is negative for $x > -3$.
- At $x = 1$, h has a relative minimum because $h' = g$ is negative for $x < 1$ and $h' = g$ is positive for $x > 1$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -5$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = -3$, h has neither a relative minimum nor a relative maximum, because $h'(x) = g(x)$ does not change sign there.

At $x = 1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list -4 , -3 , -1 , and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $g(x)$ is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- h' changes from increasing to decreasing or vice versa
- h'' changes from positive to negative or vice versa

Unit 1 AP Topics 16

· h' has a local extrema

Any incorrect statement (such as h' changes from increasing to decreasing at $x = -4$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = g$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = -4$, $x = -3$, and $x = -1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that h is differentiable $\Rightarrow h$ is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $-3 < x < 1$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since h is continuous, and $-42 < -33 < 0$, such a point exists” would earn the point.

The value of -33 must be bounded by -42 and 0 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(-3)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. We need to see $Ah(-3) = 0$.

A response may reference a value of $h(1)$ determined in part (a). If the value of $h(1)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is less than -33 .

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable on $-3 \leq x \leq 1$. $\Rightarrow h$ is continuous on $-3 \leq x \leq 1$.

$$h(1) = -42 < -33 < 0 = h(-3)$$

By the Intermediate Value Theorem, there must be a value of k , for $-3 < k < 1$, such that $h(k) = -33$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $3 \cdot 1^2 - 3 = 0$ and $h(1) + 42 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3 \cdot 1^2 = 3$ and $h(1) = -42$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

· Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x} = \frac{0}{0}$ does not earn the first point. The statement

$\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x} \rightarrow \frac{0}{0}$ does earn the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x} = \frac{6x}{h'(x) + 42}$, will earn the second point, but not the third point.

Unit 1 AP Topics 16

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 1} \frac{3x^2-3}{h(x)+42x} = \frac{0}{0}$ and $\lim_{x \rightarrow 1} \frac{3x^2-3}{h(x)+42x} = \frac{6x}{h'(x)+42}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{1}{7}$, or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.
 $\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \rightarrow 1} h(x) = h(1)$ and $\lim_{x \rightarrow 1} h'(x) = h'(1)$.

$$\lim_{x \rightarrow 1} (3x^2 - 3) = 3 \cdot 1^2 - 3 = 0$$

$$\lim_{x \rightarrow 1} (h(x) + 42x) = h(1) + 42 \cdot 1 = -42 + 42 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{3x^2-3}{h(x)+42x} &= \lim_{x \rightarrow 1} \frac{6x}{h'(x)+42} \\ &= \frac{6 \cdot 1}{h'(1)+42} = \frac{6}{g(1)+42} = \frac{6}{0+42} = \frac{1}{7} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot h'(2x) \cdot 2$$

$$\cdot g(2x) \cdot 2$$

Unit 1 AP Topics 16

$$\cdot h'(2 \cdot (-1)) \cdot 2$$

$$\cdot g(2 \cdot (-1)) \cdot 2$$

Responses that do not earn the point include:

$$\cdot h'(x) \cdot 2$$

$$\cdot g(x) \cdot 2$$

$$\cdot h'(-2) \cdot 2$$

$$\cdot g(-2) \cdot 2$$

$$\cdot h(2x) \cdot 2$$

The second point is earned for the application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $p(x)$ or in $p'(-1)$. For example, the second point is demonstrated by $g(2x)$ or by $g(2 \cdot 2)$.

The second point may be earned without earning the first point. For example, $h'(2x) = g(2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(-2)$ to -10.8 . The function g does not have to be stated explicitly. For example, $h'(2 \cdot (-1)) \cdot 2 = -10.8 \cdot 2$ earns the first two points. Recall, if $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point.

The first two points can be earned by a single expression. For example, both $g(2x) \cdot 2$ and $g(2 \cdot (-1)) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -21.6 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$p'(x) = h'(2x) \cdot 2$$

Unit 1 AP Topics 16

$$= g(2x) \cdot 2$$

$$p'(-1) = g(2 \cdot (-1)) \cdot 2 = g(-2) \cdot 2 = (-10.8) \cdot 2 = -21.6$$

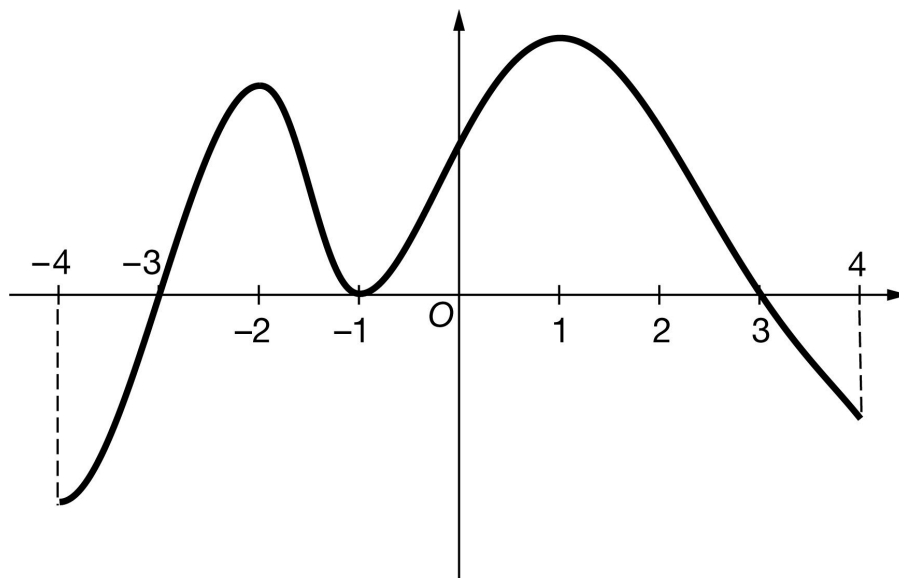
Unit 1 AP Topics 16

46. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the differentiable function g with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[-4, -3]$, $[-3, -1]$, $[-1, 3]$, and $[3, 4]$ are 10, 18, 45, and 5, respectively. The graph of g has horizontal tangents at $x = -2$, $x = -1$, and $x = 1$. Let h be the function defined by $h(x) = \int_{-1}^x g(t) dt$ for $-4 \leq x \leq 4$.

- (a) Find the value of $h(3)$.
- (b) Find the value of $h(-4)$. Show the computations that lead to your answer.
- (c) Find the x -coordinate of each critical point of h on the interval $-4 < x < 4$.
- (d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- (e) For $-4 < x < 4$, find the x -coordinate of each point of inflection of the graph of h . Justify your answers.
- (f) Explain why there must be a value of r , for $-1 < r < 3$, such that $h(r) = 20$.
- (g) Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x}$. Give a reason for your answer.

Unit 1 AP Topics 16

(h) The function k is defined by $k(x) = h(x^2)$. It is known that $g(1) = 19.7$. Find $k'(-1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 45 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(3) = \int_{-1}^3 g(t) dt = 45$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 10 and ± 18 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 10 and ± 18 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-10) - 18$ earns both points because numerical expressions need not be simplified.
- The $(-10) - 18$ expression earns the first point, but not the second point.
- The expression $-(-10) - 18 - 45$ earns neither point.

Note: Just the list $-10, 18$ earns neither point. However, if the values -10 and 18 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals $= -8$ earns just the second point because not enough work has been shown.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-4}^{-3} g(t) dt \right| = 10$ and $\left| \int_{-3}^{-1} g(t) dt \right| = 18$

☐ Answer

Solution:

$$\begin{aligned} h(-4) &= \int_{-1}^{-4} g(t) dt = - \int_{-4}^{-1} g(t) dt \\ &= - \int_{-4}^{-3} g(t) dt - \int_{-3}^{-1} g(t) dt = -(-10) - 18 = -8 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = -1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 4)$. The point is still earned provided the three critical points, $x = -3$, $x = -1$, and $x = 3$, are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$, and $(3, 0)$, read the x -coordinates and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = g(x) = 0 \text{ at } x = -3, x = -1, \text{ and } x = 3.$$

Unit 1 AP Topics 16**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $g(x)$ and $h'(x)$ (that is, $g(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $g(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = -1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

- $h(-4) = -8$

- $h(-3) = -18$

- $h(-1) = 0$

- $h(3) = 45$

- $h(4) = 40$

Unit 1 AP Topics 16

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -3$, h has a relative minimum because $h' = g$ is negative for $x < -3$ and $h' = g$ is positive for $x > -3$.
- At $x = -1$, h has neither because $h' = g$ is positive for $x < -1$ and $h' = g$ is positive for $x > -1$.
- At $x = 3$, h has a relative maximum because $h' = g$ is positive for $x < 3$ and $h' = g$ is negative for $x > 3$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = -1$, h has neither a relative minimum nor a relative maximum, because $h'(x) = g(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list -2 , -1 , 1 , and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $g(x)$ is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- “ h' changes from increasing to decreasing or vice versa”
- “ h'' changes from positive to negative or vice versa”

Unit 1 AP Topics 16

· “ h' has a local extrema”

Any incorrect statement (such as h' changes from decreasing to increasing at $x = -2$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = g$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = -2$, $x = -1$, and $x = 1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that “ h is differentiable $\Rightarrow h$ is continuous.”

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval stated must contain the interval $-1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since h is continuous, and $0 < 20 < 45$, such a point exists” would earn the point.

The value of 20 must be bounded by 0 and 45 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(-1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. We need to see $h(-1) = 0$.

A response may reference a value of $h(3)$ determined in part (a). If the value of $h(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 20.

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable on $-1 \leq x \leq 3$. $\Rightarrow h$ is continuous on $-1 \leq x \leq 3$.

$$h(-1) = 0 < 20 < 45 = h(3)$$

By the Intermediate Value Theorem, there must be a value of r , for $-1 < r < 3$, such that $h(r) = 20$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $3^2 - 9 = 0$ and $h(3) - 45 = 0$.
- All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $h(3) = 45$ are not enough to earn the first point.

- The statement that the limit gives the indeterminate form $0/0$ earns the first point.

- Indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $0/0$. For example, the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{0}{0}$ does not earn the first point, but the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} \rightarrow \frac{0}{0}$ earns the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
- Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{2x}{h'(x) - 15}$, will earn the second point, but not the third point.

Unit 1 AP Topics 16

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{0}{0}$ and $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{2x}{h'(x) - 15}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-\frac{2}{5}$, or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.
 $\Rightarrow h$ is continuous $\Rightarrow \lim_{x \rightarrow 3} h(x) = h(3)$ and $\lim_{x \rightarrow 3} h'(x) = h'(3)$.

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

$$\lim_{x \rightarrow 3} (h(x) - 15x) = h(3) - 15 \cdot 3 = 45 - 45 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} &= \lim_{x \rightarrow 3} \frac{2x}{h'(x) - 15} \\ &= \frac{2 \cdot 3}{h'(3) - 15} = \frac{6}{g(3) - 15} = \frac{6}{0 - 15} = -\frac{2}{5} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$$\cdot h'(x^2) \cdot 2x$$

Unit 1 AP Topics 16

$$\cdot g(x^2) \cdot 2x$$

$$\cdot h'((-1)^2) \cdot 2 \cdot (-1)$$

$$\cdot g((-1)^2) \cdot 2 \cdot (-1)$$

$$\cdot h'(1) \cdot 2 \cdot (-1)$$

$$\cdot g(1) \cdot 2 \cdot (-1)$$

$$\cdot h'((-1)^2) \cdot (-2)$$

$$\cdot g((-1)^2) \cdot (-2)$$

Responses that do not earn the point include:

$$\cdot h'(x^2) \cdot 2$$

$$\cdot g(x^2) \cdot 2$$

$$\cdot h'(x) \cdot 2x$$

$$\cdot g(x) \cdot 2x$$

$$\cdot h'(1) \cdot (-2)$$

$$\cdot g(1) \cdot (-2)$$

$$\cdot h(x^2) \cdot 2x$$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(-1)$. For example, the second point is earned by $g(x^2)$ or by $g((-1)^2)$.

The second point may be earned without earning the first point. For example, $h'(x^2) \cdot 2 = g(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(1)$ to 19.7. The function g does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot h'((-1)^2) \cdot 2 \cdot (-1) = 19.7 \cdot 2 \cdot (-1)$$

$$\cdot h'((-1)^2) \cdot (-2) = 19.7 \cdot (-2)$$

Unit 1 AP Topics 16

Recall, if $x = -1$ is substituted for x , $x = -1$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $g(x^2) \cdot 2x$ and $g((-1)^2) \cdot 2 \cdot (-1)$ earn the first two points.

The third point is earned for the correct answer of -39.4 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (h). Responses of the form $h'(x)^2 \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = h'(x^2) \cdot 2x$$

$$= g(x^2) \cdot 2x$$

$$k'(-1) = g((-1)^2) \cdot 2 \cdot (-1) = g(1) \cdot (-2) = 19.7 \cdot (-2) = -39.4$$

Unit 1 AP Topics 16

47.  A cylindrical can of radius 10 millimeters is used to measure rainfall in Stormville. The can is initially empty, and rain enters the can during a 60-day period. The height of water in the can is modeled by the function S , where $S(t)$ is measured in millimeters and t is measured in days for $0 \leq t \leq 60$. The rate at which the height of the water is rising in the can is given by $S'(t) = 2\sin(0.03t) + 1.5$.
- (a) According to the model, what is the height of the water in the can at the end of the 60-day period?
- (b) According to the model, what is the average rate of change in the height of water in the can over the 60-day period? Show the computations that lead to your answer. Indicate units of measure.
- (c) Assuming no evaporation occurs, at what rate is the volume of water in the can changing at time $t = 7$? Indicate units of measure.
- (d) During the same 60-day period, rain on Monsoon Mountain accumulates in a can identical to the one in Stormville. The height of the water in the can on Monsoon Mountain is modeled by the function M , where $M(t) = 1/400(3t^3 - 30t^2 + 330t)$. The height $M(t)$ is measured in millimeters, and t is measured in days for $0 \leq t \leq 60$. Let $D(t) = M'(t) - S'(t)$. Apply the Intermediate Value Theorem to the function D on the interval $0 \leq t \leq 60$ to justify that there exists a time t , $0 < t < 60$, at which the heights of water in the two cans are changing at the same rate.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$S(60) = \int_0^{60} S'(t) dt = 171.813 \text{ mm}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$S(60) = \int_0^{60} S'(t) dt = 171.813 \text{ mm}$$

Part B

Unit 1 AP Topics 16

1 point is earned for the answer

1 point earned for units in (b) or (c)

$$\frac{S(60) - S(0)}{60} = 2.863 \text{ or } 2.864 \text{ mm/day}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point earned for units in (b) or (c)

$$\frac{S(60) - S(0)}{60} = 2.863 \text{ or } 2.864 \text{ mm/day}$$

Part C

1 point is earned for relationship between V and S

1 point is earned for the answer

1 point earned for units in (b) or (c)

$$V(t) = 100\pi S(t)$$

$$V'(7) = 100\pi S'(7) = 602.218$$

The volume of water in the can is increasing at a rate of $602.218 \text{ mm}^3/\text{day}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for relationship between V and S

1 point is earned for the answer

1 point earned for units in (b) or (c)

Unit 1 AP Topics 16

$$V(t) = 100\pi S(t)$$

$$V'(7) = 100\pi S'(7) = 602.218$$

The volume of water in the can is increasing at a rate of $602.218 \text{ mm}^3/\text{day}$.

Part D

1 point is earned if considers $D(0)$ and $D(60)$

1 point is earned for justification

$$D(0) = -0.675 < 0 \text{ and } D(60) = 69.37730 > 0$$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t , $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if considers $D(0)$ and $D(60)$

1 point is earned for justification

$$D(0) = -0.675 < 0 \text{ and } D(60) = 69.37730 > 0$$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t , $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.

Unit 1 AP Topics 16

48.

x	-2	-1	0	1	2	3
$f(x)$	-2	5	2	-4	-1	3

Selected values of a continuous function f are given in the table above. What is the fewest possible number of zeros of f in the interval $[-2, 3]$?

- (A) Zero
(B) One
(C) Two
(D) Three

**Answer D**

Correct. The Intermediate Value Theorem guarantees that if a function is continuous on a closed interval $[a, b]$ and c is a value between $f(a)$ and $f(b)$, then there is a value of x in the interval $[a, b]$ where $f(x) = c$. Since f is continuous and $f(-2) < 0 < f(-1)$, $f(1) < 0 < f(0)$, and $f(2) < 0 < f(3)$, there are at least three zeros for f in the interval $[-2, 3]$.

49.

x	2	3	4
$g(x)$	-2	1	-4

Selected values of the continuous function g are given in the table above. Which of the following statements must be true?

- (A) The minimum value of g on the interval $[2, 4]$ is -4 .
(B) $g'(x) = -1$ has at least one solution in the interval $[2, 4]$.
(C) $g'(x) = 0$ has at least two solutions in the interval $[2, 4]$.
(D) $g(x) = 0$ has at least two solutions in the interval $[2, 4]$.

**Answer D**

Correct. Because g is continuous and $g(2) < 0 < g(3)$, the Intermediate Value Theorem guarantees a c_1 in the open interval $(2, 3)$ where $g(c_1) = 0$. Because $g(4) < 0 < g(3)$, the Intermediate Value Theorem also guarantees a c_2 in the open interval $(3, 4)$ where $g(c_2) = 0$. Therefore it is true that $g(x) = 0$ has at least two solutions in the interval $[2, 4]$.

Unit 1 AP Topics 16

50. The function f is continuous on the closed interval $[a, b]$, where a and b are constants and $a < b$. If $f(a) < 0 < f(b)$, which of the following statements must be true?

- (A) There is at least one value of x in the interval (a, b) such that $f(x) = 0$. ✓
- (B) There is at least one value of x in the interval (a, b) such that $f'(x) = \frac{f(b)-f(a)}{b-a}$.
- (C) $f'(x) > 0$ for all x in the interval (a, b) .
- (D) $f(a) < f(x) < f(b)$ for all x in the interval (a, b) .

Answer A

Correct. This statement is true by the Intermediate Value Theorem, since f is continuous on the closed interval $[a, b]$ and $f(a) < 0 < f(b)$.

51. Let f be a function of x . Which of the following statements, if true, would guarantee that there is a number c in the interval $[-5, 4]$ such that $f(c) = 12$?

- (A) f is increasing on the interval $[-5, 4]$, where $f(-5) = 0$ and $f(4) = 20$.
- (B) f is increasing on the interval $[-5, 4]$, where $f(-5) = 15$ and $f(4) = 30$.
- (C) f is continuous on the interval $[-5, 4]$, where $f(-5) = 0$ and $f(4) = 20$. ✓
- (D) f is continuous on the interval $[-5, 4]$, where $f(-5) = 15$ and $f(4) = 30$.

Answer C

Correct. If f is continuous on the interval $[-5, 4]$ and $f(-5) = 0 < 12 < 20 = f(4)$, then by the Intermediate Value Theorem there exists a number c in the interval $[-5, 4]$ such that $f(c) = 12$.

Unit 1 AP Topics 16

52. Let f be a function such that $f(3) < 4 < f(5)$. Which of the following statements provides sufficient additional information to conclude that there is a value $x = c$ in the interval $[3, 5]$ such that $f(c) = 4$?
- (A) f is defined for all x .
- (B) f is increasing for all x .
- (C) f is continuous for all x . ✓
- (D) There is a value $x = c$ in the interval $[3, 5]$ such that $\lim_{x \rightarrow c} f(x) = 4$.

Answer C

Correct. If f is continuous on the closed interval $[3, 5]$ and $f(3) < 4 < f(5)$, then by the Intermediate Value Theorem $f(x)$ is guaranteed to equal 4 at some point c in the interval $[3, 5]$.

53.  Let f be the function given by $f(x) = \frac{9+2xe^{-\frac{x}{4}}}{\cos(\frac{x}{2})}$. The Intermediate Value Theorem applied to f on the closed interval $[24, 28]$ guarantees a solution in $[24, 28]$ to which of the following equations?
- (A) $f(x) = 0$
- (B) $f(x) = 9.090$
- (C) $f(x) = 12.235$ ✓
- (D) $f(x) = 76.999$

Answer C

Correct. The function f is continuous on the closed interval $[24, 28]$ because the numerator and denominator are continuous everywhere and the denominator does not equal zero on the interval $[24, 28]$. Since $f(24) = 10.806$ and $f(28) = 66.193$, the Intermediate Value Theorem guarantees that $f(c) = 12.235$ for some c in the interval $[24, 28]$.

Unit 1 AP Topics 16

54.  Let f be the function given by $f(x) = 2x + \tan\left(\frac{x}{5}\right) - 15$. The Intermediate Value Theorem applied to f on the closed interval $[10, 15]$ guarantees a solution in $[10, 15]$ to which of the following equations?
- (A) $f(x) = -15$
- (B) $f(x) = 0$
- (C) $f(x) = 5$ ✓
- (D) $f(x) = 15$

Answer C

Correct. Note that $10 \leq x \leq 15$ implies $2 \leq \frac{x}{5} \leq 3$, which is an interval where the graph of the tangent function has no vertical asymptote. Each term of f is continuous on the closed interval $[10, 15]$; therefore, the function f is continuous on the closed interval $[10, 15]$. Since $f(10) \approx 2.815 \leq 5 \leq 14.857 \approx f(15)$, the Intermediate Value Theorem guarantees that $f(c) = 5$ for some c in the interval $[10, 15]$.

55.  $f(x) = \frac{x^4 + x^3 + x^2 + x + 1}{380 - \ln\left(\frac{x^2 + 1}{2}\right)}$

Let f be the function defined above. The Intermediate Value Theorem applied to f on the closed interval $[10, 12]$ guarantees a solution in $[10, 12]$ to which of the following equations?

- (A) $f(x) = 0$
- (B) $f(x) = 27.372$
- (C) $f(x) = 42.421$ ✓
- (D) $f(x) = 67.205$

Answer C

Correct. The function f is continuous on the closed interval $[10, 12]$ because the numerator and denominator are continuous everywhere and the denominator does not equal zero on the interval $[10, 12]$. Since $f(10) = 29.544$ and $f(12) = 66.208$, the Intermediate Value Theorem guarantees that $f(c) = 42.421$ for some c in the interval $[10, 12]$.

Unit 1 AP Topics 16

56.

x	0	1	2	3	4	5
$f(x)$	1	-5	-4	2	-10	-15

Selected values of a continuous function f are given in the table above. What is the fewest possible number of zeros of f in the interval $[0, 5]$?

- (A) Zero, because $f(x)$ is not equal to 0 for any of the values in the table.
- (B) One, because f is continuous on the interval $[0, 5]$ and $f(0) > 0 > f(5)$.
- (C) Two, because the values for $f(x)$ in the table change from positive to negative twice.
- (D) Three, because f is continuous on the interval $[0, 5]$ and $f(0) > 0 > f(1)$, $f(1) < 0 < f(3)$, and $f(3) > 0 > f(5)$. ✓

Answer D

Correct. The Intermediate Value Theorem guarantees that if a function is continuous on a closed interval $[a, b]$ and c is a value between $f(a)$ and $f(b)$, then there is a value of x in the interval $[a, b]$ where $f(x) = c$. Because f is continuous on the interval $[0, 5]$, the Intermediate Value Theorem can be applied on the subintervals $[0, 1]$, $[1, 3]$, and $[3, 5]$ to guarantee at least three zeros of f .

Unit 1 AP Topics 16

57.

x	-2	-1	0	1	2
$g(x)$	-4	-1	4	11	20

The table above gives values of a function g at selected values of x . Which of the following statements, if true, would be individually sufficient to conclude that there exists a number c in the interval $[-2, 2]$ such that $g(c) = 6$?

- I. g is defined for all x in the interval $[-2, 2]$.
- II. g is increasing on the interval $[-2, 2]$.
- III. g is continuous on the interval $[-2, 2]$.

(A) II only

(B) III only

(C) I and III only

(D) I, II, and III

**Answer B**

Correct. Since $g(0) = 4 < 6 < g(1) = 11$, if g is continuous on the interval $[-2, 2]$, then by the Intermediate Value Theorem there is at least one number c in the interval $[-2, 2]$ such that $g(c) = 6$.

Unit 1 AP Topics 16

58.

x	0	1	3	7
$g(x)$	24	35	42	68

The table above gives values of a function g at selected values of x . Which of the following statements, if true, would be sufficient to conclude that there exists a number c in the interval $[0, 7]$ such that $g(c) = 50$?

- I. g is defined for all x in the interval $[0, 7]$.
- II. g is increasing for all x in the interval $[0, 7]$.
- III. g is continuous for all x in the interval $[0, 7]$.

(A) II only

(B) III only

(C) I and III only

(D) I, II, and III

**Answer B**

Correct. Since $g(0) = 24 < 50 < 68 = g(7)$, if g is continuous on the interval $[0, 7]$, then by the Intermediate Value Theorem there is at least one number c in the interval $[0, 7]$ such that $g(c) = 50$.

59.  Let f be the function given by $f(x) = x + \tan\left(\frac{x}{5}\right) - 10$. The Intermediate Value Theorem applied to f on the closed interval $[12, 15]$ guarantees a solution in $[12, 15]$ to which of the following equations?

(A) $f(x) = -10$ (B) $f(x) = 0$ (C) $f(x) = 4$ (D) $f(x) = 14$ **Answer C**

Correct. The function f is continuous on the closed interval $[12, 15]$. Note that $12 \leq x \leq 15$ implies $2.4 \leq \frac{x}{5} \leq 3$, an interval on which the graph of the tangent function has no vertical asymptote. Since $f(12) = 1.084 < 4 < 4.857 = f(15)$, the Intermediate Value Theorem guarantees that $f(c) = 4$ for some c in the interval $[12, 15]$.

Unit 1 AP Topics 16

60. Let f be a function such that $f(5) < 6 < f(7)$. Which of the following statements provides sufficient additional information to conclude that there is a value $x = c$ in the interval $[5, 7]$ such that $f(c) = 6$?
- (A) f is defined for all x .
 - (B) f is increasing for all x .
 - (C) f is continuous for all x . ✓
 - (D) There is a value $x = c$ in the interval $[5, 7]$ such that $\lim_{x \rightarrow c} f(x) = 6$.

Answer C

Correct. If f is continuous on the closed interval $[5, 7]$ and $f(5) < 6 < f(7)$, then by the Intermediate Value Theorem $f(x)$ is guaranteed to equal 6 at some point c in the interval $[5, 7]$.

61. Let f be a function of x . Which of the following statements, if true, would guarantee that there is a number c in the interval $[-2, 3]$ such that $f(c) = 10$?
- (A) f is increasing on the interval $[-2, 3]$, where $f(-2) = 0$ and $f(3) = 20$.
 - (B) f is increasing on the interval $[-2, 3]$, where $f(-2) = 15$ and $f(3) = 30$.
 - (C) f is continuous on the interval $[-2, 3]$, where $f(-2) = 0$ and $f(3) = 20$. ✓
 - (D) f is continuous on the interval $[-2, 3]$, where $f(-2) = 15$ and $f(3) = 30$.

Answer C

Correct. If f is continuous on the interval $[-2, 3]$ and $f(-2) = 0 < 10 < 20 = f(3)$, then by the Intermediate Value Theorem there exists a number c in the interval $[-2, 3]$ such that $f(c) = 10$.

Unit 1 AP Topics 16

62.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	3	8	12
$H(t)$ (degrees Celsius)	90	80	67	58

As a bowl of soup cools, the temperature of the soup is given by the twice-differentiable function H for $0 \leq t \leq 12$, where $H(t)$ is measured in degrees Celsius ($^{\circ}\text{C}$) and time t is measured in minutes. Values of $H(t)$ at selected values of time t are shown in the table above.

- Use the data in the table to approximate $H'(5)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $H'(5)$ in the context of the problem.
- Is there a time t , for $0 \leq t \leq 12$, at which the temperature of the soup is 60°C ? Justify your answer.
- The temperature of the soup is also modeled by the twice-differentiable function C for $0 \leq t \leq 12$, where $C(t)$ is measured in degrees Celsius and time t is measured in minutes. It is known that $C(3) = 80$ and the derivative of C is given by $C'(t) = -3.6e^{-0.05t}$. Write an equation for the line tangent to the graph of C at $t = 3$ and use it to approximate $C(5)$.
- Based on the model given in part (c), at what rate, in degrees Celsius per minute per minute, is the rate of change of the temperature of the soup changing at time $t = 3$ minutes?

Part A

The first point requires a correct substitution of values. A simplified answer is not required.

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Unit 1 AP Topics 16

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ interpretation with units

Solution:

$$H'(5) \approx \frac{H(8) - H(3)}{8 - 3} = \frac{67 - 80}{5} = -2.6$$

At time $t = 5$ minutes, the temperature of the soup is changing at a rate of -2.6 degrees Celsius per minute.

Part B

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ H is continuous
- ☐ justification using Intermediate Value Theorem

Solution:

H is twice differentiable. $\Rightarrow H$ is continuous.

$$H(12) = 58 < 60 < 90 = H(0)$$

By the Intermediate Value Theorem, there is a value of t , for $0 \leq t \leq 12$, such that $H(t) = 60$.

Part C

The second point may be earned if using an incorrect value for slope, provided the value is labeled as $C'(3)$.

The third point does not require a simplified answer. Substitution of function values is required.

Unit 1 AP Topics 16

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ slope
- ☐ equation for tangent line
- ☐ approximation

Solution:

$$C'(3) = -3.098549$$

An equation for the tangent line is $y = 80 - 3.098549(t - 3)$.

$$C(5) \approx 80 - C'(3)(5 - 3) = 73.802903$$

$$C(5) = 73.803 \text{ (or } 73.802)$$

Part D

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $C''(3)$
- ☐ interpretation with units

Solution:

$$C''(3) = 0.154927$$

At time $t = 3$ minutes, the rate of change of the temperature is changing by 0.155 (or 0.154) degree Celsius per minute per minute.

Unit 1 AP Topics 16

63. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	3	8	12	14
$S'(t)$ (centimeters per hour)	2.3	2.1	1.9	1.7	1.6

The depth of snow in a field is given by the twice-differentiable function S for $0 \leq t \leq 14$, where $S(t)$ is measured in centimeters and time t is measured in hours. Values of $S'(t)$, the derivative of S , at selected values of time t are shown in the table above. It is known that the graph of S is concave down for $0 \leq t \leq 14$.

- Use the data in the table to approximate $S''(10)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $S''(10)$ in the context of the problem.
- Is there a time t , for $0 \leq t \leq 14$, at which the depth of snow is changing at a rate of 2 centimeters per hour? Justify your answer.
- At time $t = 8$, the depth of snow is 45 centimeters. Use the line tangent to the graph of S at $t = 8$ to approximate the depth of snow at time $t = 10$. Is the approximation an underestimate or an overestimate of the actual depth of snow at time $t = 10$? Justify your answer.
- The depth of snow, in centimeters, is also modeled by the twice-differentiable function D for $0 \leq t \leq 14$, where $D(t) = 120 - 92e^{-\frac{t}{40}}$ and time t is measured in hours. Use the model to find $D'(10)$. Using correct units, explain the meaning of $D'(10)$ in the context of the problem.

Part A

The first point requires a correct substitution of values. A simplified answer is not required.

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Unit 1 AP Topics 16

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ interpretation with units

Solution:

$$S''(10) \approx \frac{S'(12) - S'(8)}{12 - 8} = \frac{1.7 - 1.9}{4} = -0.05$$

At time $t = 10$ hours, the rate of change of the depth of snow is changing at a rate of -0.05 centimeter per hour per hour.

Part B

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ S' is continuous
- ☐ justification using Intermediate Value Theorem

Solution:

S is twice differentiable. $\Rightarrow S'$ is differentiable and thus continuous.

$$S'(8) = 1.9 < 2 < 2.1 = S'(3)$$

By the Intermediate Value Theorem, there is a value of t , for $0 \leq t \leq 14$, such that $S'(t) = 2$.

Part C

The second point may be earned if using an incorrect value for slope, provided the value is labeled as $S'(8)$.

Unit 1 AP Topics 16

The second point does not require a simplified answer. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ equation for tangent line
- ☐ approximation
- ☐ overestimate with justification using concavity

Solution:

$$S'(8) = 1.9 \text{ from table}$$

An equation for the tangent line is $y = 45 + 1.9(t - 8)$.

$$S(10) \approx 45 + S'(8)(10 - 8) = 48.8 \text{ centimeters}$$

This approximation is an overestimate because the graph of S is concave down for $8 \leq t \leq 10$.

Part D

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $D'(10)$
- ☐ interpretation with units

Solution:

$$D'(10) = 1.791$$

Unit 1 AP Topics 16

At time $t = 10$ hours, the depth of snow is changing by 1.791 centimeters per hour.

Unit 1 AP Topics 16

64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$N(t) = \begin{cases} f(t) & \text{for } 0 \leq t < 6 \\ 25t + 150 & \text{for } 6 \leq t < 8 \\ \frac{200+80t}{2+0.05t} & \text{for } t \geq 8 \end{cases}$$

The number of fish in a pond at time t years is modeled by the function N defined above, where f is a continuous function such that $f(0) = 80$.

- (a) Find $\lim_{t \rightarrow \infty} N(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} N(t)$ in the context of the problem.
- (b) Is the function N continuous at $t = 8$? Justify your answer.
- (c) The function N is continuous at $t = 6$. Is there a time t , for $0 \leq t \leq 6$, at which $N(t) = 250$? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the number of fish.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} N(t)$
- ☐ interpretation

Solution:

Unit 1 AP Topics 16

$$\lim_{t \rightarrow \infty} N(t) = \lim_{t \rightarrow \infty} \frac{200+80t}{2+0.05t} = \frac{80}{0.05} = 1600$$

As time increases, the number of fish in the pond approaches 1600.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 8^-} N(t)$
- ☐ $\lim_{t \rightarrow 8^+} N(t)$
- ☐ answer with justification

Solution:

$$\lim_{t \rightarrow 8^-} N(t) = \lim_{t \rightarrow 8^-} (25t + 150) = 350$$

$$\lim_{t \rightarrow 8^+} N(t) = \lim_{t \rightarrow 8^+} \frac{200+80t}{2+0.05t} = \frac{840}{2.4} = 350$$

$$N(8) = 350$$

Yes, N is continuous at $t = 8$ because

$$\lim_{t \rightarrow 8} N(t) = N(8).$$

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

Unit 1 AP Topics 16

- ☐ answer with justification
- ☐ specific reference to IVT

Solution:

N is continuous for $0 \leq t \leq 6$.

$$N(0) = 80 < 250 < 300 = N(6)$$

By the Intermediate Value Theorem, there must be at least one time t , for $0 \leq t \leq 6$, such that

$$N(t) = 250.$$

Unit 1 AP Topics 16

65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$F(t) = \begin{cases} 2^{t+1} - 2 & \text{for } 0 \leq t < 7 \\ \frac{800t - 4800}{t - 4} & \text{for } t \geq 7 \end{cases}$$

The amount of money raised during a fund-raising campaign is modeled by the function F defined above, where $F(t)$ is measured in United States dollars and t is the time in days since the campaign began.

- (a) Find $\lim_{t \rightarrow \infty} F(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} F(t)$ in the context of the problem.
- (b) Is the function F continuous at $t = 7$? Justify your answer.
- (c) The amount of money raised during a competing fund-raising campaign is modeled by the function M defined by $M(t) = \frac{240(2^t - 1)}{(2^t + 36)}$, where $M(t)$ is measured in United States dollars and t is the time in days since that campaign began. According to this model, is there a time t , for $0 \leq t \leq 2$, at which the amount of money raised is 10 dollars? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the amount raised.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} F(t)$
- ☐ interpretation

Solution:

Unit 1 AP Topics 16

$$\lim_{t \rightarrow \infty} F(t) = \lim_{t \rightarrow \infty} \frac{800t - 4800}{t - 4} = \frac{800}{1} = 800$$

As time increases, the amount raised by the fund-raising campaign approaches \$800.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 7^-} F(t)$
- ☐ $\lim_{t \rightarrow 8^+} F(t)$
- ☐ answer with justification

Solution:

$$\lim_{t \rightarrow 7^-} F(t) = \lim_{t \rightarrow 7^-} (2^{t+1} - 2) = 2^8 - 2 = 254$$

$$\lim_{t \rightarrow 7^+} F(t) = \lim_{t \rightarrow 7^+} \frac{800t - 4800}{t - 4} = \frac{800 \cdot 7 - 4800}{7 - 4} = \frac{800}{3}$$

No, F is not continuous at $t = 7$ because $\lim_{t \rightarrow 7^-} F(t) \neq \lim_{t \rightarrow 7^+} F(t)$.

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

- ☐ answer with justification
- ☐ specific reference to IVT

Unit 1 AP Topics 16**Solution:**

M is continuous for $0 \leq t \leq 2$.

$$M(0) = \frac{240(2^0 - 1)}{2^0 + 36} = \frac{240(1 - 1)}{1 + 36} = 0$$

$$M(2) = \frac{240(2^2 - 1)}{2^2 + 36} = \frac{240 \cdot 3}{40} = 18$$

$$M(0) = 0 < 10 < 18 = M(2)$$

By the Intermediate Value Theorem, there must be at least one time t , for $0 \leq t \leq 2$, such that $M(t) = 10$.

Unit 1 AP Topics 16

66. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$Y(t) = \begin{cases} 10e^{0.05t} & \text{for } 0 \leq t \leq 10 \\ f(t) & \text{for } 10 < t \leq 12 \\ \frac{600}{20+10e^{-0.05(t-12)}} & \text{for } t > 12 \end{cases}$$

The mass of yeast in a culture, measured in milligrams, at time t hours is modeled by the function Y defined above, where f is a continuous function such that $f(12) = 20$.

Find $\lim_{t \rightarrow \infty} Y(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} Y(t)$ in the context of the problem.

Is the function Y continuous at $t = 12$? Justify your answer.

The function Y is continuous at $t = 10$. Is there a time t , for $0 < t < 12$, at which $Y(t) = 18$? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the mass of the yeast in the culture.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} Y(t)$
- ☐ interpretation

Solution:

$$\lim_{t \rightarrow \infty} Y(t) = \lim_{t \rightarrow \infty} \frac{600}{20+10e^{-0.05(t-12)}} = \frac{600}{20+0} = 30$$

Unit 1 AP Topics 16

As time increases, the mass of the yeast in the culture approaches 30 milligrams.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 12^-} Y(t)$
- ☐ $\lim_{t \rightarrow 12^+} Y(t)$
- ☐ answer with justification

Solution

Because f is a continuous function and $f(12) = 20$, $\lim_{t \rightarrow 12^-} Y(t) = \lim_{t \rightarrow 12^-} f(t) = f(12) = 20$.

$$\lim_{t \rightarrow 12^+} Y(t) = \lim_{t \rightarrow 12^+} \frac{600}{20 + 10e^{-0.05(t-12)}} = \frac{600}{20 + 10} = 20$$

$$Y(12) = f(12) = 20$$

Yes, Y is continuous at $t = 12$ because $\lim_{t \rightarrow 12} Y(t) = Y(12)$.

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

- ☐ answer with justification
- ☐ specific reference to *IVT*

Unit 1 AP Topics 16**Solution:**

Y is continuous for $0 \leq t \leq 12$.

$$Y(0) = 10 < 18 < 20 = Y(12)$$

By the Intermediate Value Theorem, there must be at least one time t , $0 \leq t \leq 12$, such that $Y(t) = 18$.

Unit 1 AP Topics 16

67. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$R(t) = \begin{cases} 41\sqrt{\frac{t}{20}} & \text{for } 0 \leq t < 20 \\ \frac{100t^2 + 1000}{t^2 + 600} & \text{for } t \geq 20 \end{cases}$$

The total accumulated revenue a company has received up to time t is modeled by the function R defined above, where $R(t)$ is measured in millions of United States dollars and t is the time in years since 1980.

- Find $\lim_{t \rightarrow \infty} R(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} R(t)$ in the context of the problem.
- Is the function R continuous at $t = 20$? Justify your answer.
- The company's total accumulated expenses up to time t is modeled by the function E defined by $E(t) = 3\log_2(12t + 4)$, where $E(t)$ is measured in millions of United States dollars and t is the time in years since 1980. According to these models, is there a time t , for $0 \leq t \leq 5$, at which the total accumulated profit, $R(t) - E(t)$, is equal to 0? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the revenue of the company.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} R(t)$
- ☐ interpretation

Solution:

Unit 1 AP Topics 16

$$\lim_{t \rightarrow \infty} R(t) = \lim_{t \rightarrow \infty} \frac{100t^2 + 1000}{t^2 + 600} = \frac{100}{1} = 100$$

As time increases, the total accumulated revenue of the company approaches \$100 million.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 20^-} R(t)$
- ☐ $\lim_{t \rightarrow 20^+} R(t)$
- ☐ answer with justification

Solution:

$$\lim_{t \rightarrow 20^-} R(t) = \lim_{t \rightarrow 20^-} 41\sqrt{\frac{t}{20}} = 41\sqrt{\frac{20}{20}} = 41$$

$$\lim_{t \rightarrow 20^+} R(t) = \lim_{t \rightarrow 20^+} \frac{100t^2 + 1000}{t^2 + 600} = \lim_{t \rightarrow 20^+} \frac{100(t^2 + 10)}{t^2 + 600} = \frac{100(20^2 + 10)}{20^2 + 600} = \frac{100(410)}{1000} = 41$$

$$R(20) = 41$$

Yes, R is continuous at $t = 20$ because $\lim_{t \rightarrow 20} R(t) = R(20)$.

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

Unit 1 AP Topics 16

- ☐ answer with justification
- ☐ specific reference to IVT.

Solution:

R and E are continuous for $0 \leq t \leq 5$.

Therefore, the function defined by $R(t) - E(t)$ is continuous for $0 \leq t \leq 5$.

$$R(0) - E(0) = 41\sqrt{\frac{0}{20}} - 3\log_2(12 \cdot 0 + 4) = 0 - 6 = -6 < 0$$

$$\begin{aligned} R(5) - E(5) &= 41\sqrt{\frac{5}{20}} - 3\log_2(12 \cdot 5 + 4) \\ &= \frac{41}{2} - 3\log_2 64 = 20.5 - 18 > 0 \end{aligned}$$

$$R(0) - E(0) < 0 < R(5) - E(5)$$

By the Intermediate Value Theorem, there must be at least one time t , for $0 \leq t \leq 5$, such that $R(t) - E(t) = 0$.

Unit 1 AP Topics 16

68. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	3	7	10	12
$r'(t)$ (centimeters per day)	-6.1	-5.0	-4.4	-3.8	-3.5

An ice sculpture melts in such a way that it can be modeled as a cone that maintains a conical shape as it decreases in size. The radius of the base of the cone is given by a twice-differentiable function r , where $r(t)$ is measured in centimeters and t is measured in days. The table shown gives selected values of $r'(t)$, the rate of change of the radius, over the time interval $0 \leq t \leq 12$.

- Approximate $r''(8.5)$ using the average rate of change of r' over the interval $7 \leq t \leq 10$. Show the computations that lead to your answer, and indicate units of measure.
- Is there a time t , $0 \leq t \leq 3$, for which $r'(t) = -6$? Justify your answer.
- Use a right Riemann sum with the four subintervals indicated in the table to approximate the value of $\int_0^{12} r'(t) \, dt$.
- The height of the cone decreases at a rate of 2 centimeters per day. At time $t = 3$ days, the radius is 100 centimeters and the height is 50 centimeters. Find the rate of change of the volume of the cone with respect to time, in cubic centimeters per day, at time $t = 3$ days. (The volume V of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.)

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the supporting work must include at least a difference and a quotient.

Simplification is not required to earn the first point. If the numerical value is simplified, it must be correct.

The second point can be earned with an incorrect approximation for $r''(8.5)$ but cannot be earned without some value for $r''(8.5)$ presented.

Unit 1 AP Topics 16

Units may be written in any equivalent form (such as cm/day^2).



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $r''(8.5)$ with supporting work
- ☐ Units

Solution:

$$r''(8.5) \approx \frac{r'(10) - r'(7)}{10 - 7} = \frac{-3.8 - (-4.4)}{10 - 7} \text{ centimeter per day per day}$$

$$= \frac{0.6}{3} = 0.2$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, the response must establish that -6 is between $r'(0)$ and $r'(3)$ (or -6.1 and -5). This statement may be represented symbolically (with or without including one or both endpoints in an inequality) or verbally. A response of “ $r'(t) = -6$ because $r'(0) = -6.1$ and $r'(3) = -5$ ” does not state that -6 is between -6.1 and -5 . Thus this response does not earn the first point.

To earn the second point:

- The response must state that $r'(t)$ is continuous because $r'(t)$ is differentiable (or because $r(t)$ is twice differentiable).
- The response must have earned the first point. Exception: A response of “ $r'(t) = -6$ because $r'(0) = -6.1$ and $r'(3) = -5$ ” does not earn the first point because of imprecise communication but may nonetheless earn the second point if all other criteria for the second point are met.
- The response must conclude that there is a time t such that $r'(t) = -6$ (A statement of “yes” would be sufficient.)

To earn the second point, the response need not explicitly name the Intermediate Value Theorem, but if a theorem is named, it must be correct.



0	1	2
---	---	---

Unit 1 AP Topics 16

The student response accurately includes **both** of the criteria below.

- ☐ $r'(0) < -6 < r'(3)$
- ☐ Conclusion using Intermediate Value Theorem

Solution:

$r(t)$ is twice-differentiable. $\Rightarrow r'(t)$ is differentiable.
 $\Rightarrow r'(t)$ is continuous.

$$r'(0) = -6.1 < -6 < -5.0 = r'(3)$$

Therefore, by the Intermediate Value Theorem, there is a time t , $0 < t < 3$, such that $r'(t) = -6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, at least seven of the eight factors in the Riemann sum must be correct. If there is any error in the Riemann sum, the response does not earn the second point.

A response of $3(-5.0) + 4(-4.4) + 3(-3.8) + 2(-3.5)$ earns both the first and second points, unless there is a subsequent error in simplification, in which case the response would earn only the first point.

A response that presents the correct answer, with accompanying work that shows the four products in the Riemann sum (without explicitly showing all of the factors and/or the sum process) does not earn the first point but earns the second point. For example, $-15 + 4(-4.4) + 3(-3.8) + -7$ does not earn the first point but earns the second point. Similarly, $-15, -17.6, -11.4, -7 \rightarrow -51$ does not earn the first point but earns the second point.

A response that presents the correct answer (-51) with no supporting work earns no points.

A response that provides a completely correct left Riemann sum and approximation $\int_0^{12} r'(t) dt$ (i.e., $3r'(0) + 4r'(3) + 3r'(7) + 2r'(10) = 3(-6.1) + 4(-5.0) + 3(-4.4) + 2(-3.8) = -59.1$) earns 1 of the 2 points.

A response that has any error in a left Riemann sum or evaluation for $\int_0^{12} r'(t) dt$ earns no points.

Units are not required or read in this part.



0	1	2
---	---	---

Unit 1 AP Topics 16

The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\begin{aligned}\int_0^{12} r'(t) dt &\approx 3r'(3) + 4r'(7) + 3r'(10) + 2r'(12) \\ &= 3(-5.0) + 4(-4.4) + 3(-3.8) + 2(-3.5) \\ &= -51\end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first 2 points could be earned in either order.

A response with a completely correct product rule, missing one or both of the correct differentials, earns the product rule point, but not the chain rule point.

For example, $\frac{dV}{dt} = \frac{2}{3}\pi r h + \frac{1}{3}\pi r^2$ earns the first point, but not the second.

A response that treats r or h (but not both) as a constant is eligible for the chain rule point but not the product rule point. For example, $\frac{dV}{dt} = \frac{2}{3}\pi r h \frac{dr}{dt}$ is correct if h is constant, and thus earns the chain rule point.

Note: Neither $\frac{dV}{dt} = \frac{2}{3}\pi r \frac{dh}{dt}$ nor $\frac{dV}{dt} = \frac{2}{3}\pi r h \frac{dr}{dt} \frac{dh}{dt}$ earns any points.

A response that assumes a functional relationship between r and h (such as $r = 2h$), and uses this relationship to create a function for volume in terms of one variable, is eligible for at most the chain rule point. For example,

$$r = 2h \rightarrow V = \frac{1}{3}\pi(2h)^2 h = \frac{4}{3}\pi h^3 \rightarrow \frac{dV}{dt} = 4\pi h^2 \frac{dh}{dt} \text{ earns only the chain rule point.}$$

A response that mishandles the constant $\frac{1}{3}\pi$ cannot earn the third point but is eligible for the first 2 points.

The third point cannot be earned without both of the first 2 points.

$$\frac{dV}{dt} = \frac{2}{3}\pi(100)(50)(-5) + \frac{1}{3}\pi(100)^2(-2) \text{ earns all 3 points.}$$

Units are not required or read in this part.



0	1	2	3
---	---	---	---

Unit 1 AP Topics 16

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$\frac{dV}{dt} = \frac{2}{3}\pi r h \frac{dr}{dt} + \frac{1}{3}\pi r^2 \frac{dh}{dt}$$

$$\frac{dV}{dt} \Big|_{t=3} = \frac{2}{3}\pi (100)(50)(-5) + \frac{1}{3}\pi (100)^2(-2) = -\frac{70,000\pi}{3}$$

The rate of change of the volume of the sculpture at $t = 3$ is approximately $-\frac{70,000\pi}{3}$ cubic centimeters per day.

Unit 1 AP Topics 16

69. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	15	45	70	90	100
$A(t)$ (automobiles)	0	30	190	250	405	600

Prior to a sporting event, the number of automobiles that have entered a stadium parking lot is modeled by the differentiable function A , where t is the number of minutes since the parking lot opened. Values of $A(t)$ for selected values of t are given in the table above.

According to the model, what is the average rate at which automobiles enter the parking lot, in automobiles per minute, over the time interval $45 \leq t \leq 90$ minutes?

Write $A'(95)$ as the limit of a difference quotient. Use the data in the table to approximate $A'(95)$. Show the computations that lead to your answer.

What is the shortest time interval during which it is guaranteed that $A(t) = 400$ for some time t in the interval? Justify your answer.

For $0 \leq t \leq 45$, the function f defined by $f(t) = \frac{2}{27}t^2 + \frac{8}{9}t$ models the number of automobiles that have entered the parking lot, where t is the number of minutes since the parking lot opened. Find $f'(10)$, the rate at which automobiles enter the parking lot, in automobiles per minute, at time $t = 10$ minutes.

Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct average rate of change.

Unit 1 AP Topics 16

Solution:

$$\frac{A(90) - A(45)}{90 - 45} = \frac{405 - 190}{90 - 45} = \frac{215}{45} = \frac{43}{9}$$

The average rate at which automobiles enter the parking the time interval $45 \leq t \leq 90$ minutes is $\frac{43}{9}$ automobiles per minute.

Part B

The second point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ limit expression
- ☐ approximation

Solution:

$$A'(95) = \lim_{t \rightarrow 95} \frac{A(t) - A(95)}{t - 95}$$

$$\text{OR } A'(95) = \lim_{h \rightarrow 0} \frac{A(95+h) - A(95)}{h}$$

$$A'(95) \approx \frac{A(100) - A(90)}{100 - 90} = \frac{600 - 405}{100 - 90} = \frac{195}{10} = \frac{39}{2}$$

Part C

1 out of 2 points if correct justification does not specifically reference IVT.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

Unit 1 AP Topics 16

- ☐ time interval
- ☐ justification, using Intermediate Value Theorem

Solution:

The shortest time interval for which it is guaranteed that $A(t) = 400$ is $70 \leq t \leq 90$.

Because A is differentiable, A is continuous.

$$A(70) = 250 < 400 < 405 = A(90)$$

By the Intermediate Value Theorem, there must be at least one time t , for $70 \leq t \leq 90$, such that $A(t) = 400$.

Part D

The first point is earned for demonstration of a correct derivative for f and does not require a separate statement of $f'(t)$. The derivative and evaluation of the derivative can be expressed in one line to earn both points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $f'(t)$
- ☐ $f'(10)$

Solution:

$$f'(t) = \frac{4}{27}t + \frac{8}{9}$$

$$f'(10) = \frac{4}{27} \cdot 10 + \frac{8}{9} = \frac{40}{27} + \frac{24}{27} = \frac{64}{27}$$

At time $t = 10$ minutes, automobiles enter the parking lot at a rate of $\frac{64}{27}$ automobiles per minute.

Unit 1 AP Topics 16

70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	3	6	8
$P(t)$ (passengers)	0	35	204	600	728

The number of passengers who have boarded a ship is modeled by the differentiable function P , where t is the number of hours since boarding began. Values of $P(t)$ for selected values of t are given in the table above.

- According to the model, what is the average rate at which passengers board the ship, in passengers per hour, over the time interval $1 \leq t \leq 8$ hours?
- Write $P'(4.5)$ as the limit of a difference quotient. Use the data in the table to approximate $P'(4.5)$. Show the computations that lead to your answer.
- Must there be a time t , for $3 \leq t \leq 6$, at which $P(t) = 500$? Justify your answer.
- The total number of gallons of water used by the passengers on the ship is modeled by the function G defined by $G(t) = 120t\sqrt{t}$ for $0 \leq t \leq 8$, where t is the number of hours since boarding began. Find $G'(4)$, the rate at which passengers use water, in gallons per hour, at time $t = 4$ hours.

Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct average rate of change.

Solution:

Unit 1 AP Topics 16

$$\frac{P(8)-P(1)}{8-1} = \frac{728-35}{7} = 99$$

The average rate at which passengers board the ship over the time interval $1 \leq t \leq 8$ hours is 99 passengers per hour.

Part B

The second point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ limit expression
- ☐ approximation

Solution:

$$P'(4.5) = \lim_{t \rightarrow 4.5} \frac{P(t) - P(4.5)}{t - 4.5}$$

OR

$$P'(4.5) = \lim_{h \rightarrow 0} \frac{P(4.5+h) - P(4.5)}{h}$$

$$P'(4.5) \approx \frac{P(6) - P(3)}{6 - 3} = \frac{600 - 204}{3} = 132$$

Part C

1 out of 2 points if correct justification does not specifically reference IVT .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

Unit 1 AP Topics 16

- ☐ P is continuous with reason
- ☐ justification, using Intermediate Value Theorem

Solution:

Because P is differentiable, P is continuous.

$$P(3) = 204 < 500 < 600 = P(6)$$

By the Intermediate Value Theorem, there must be at least one time t , for $3 \leq t \leq 6$, such that

$$P(t) = 500.$$

Part D

The first point is earned for demonstration of a correct derivative for G and does not require a separate statement of $G'(t)$.

The derivative and evaluation of the derivative can be expressed in one line to earn both points.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $G'(t)$
- ☐ $G'(4)$

Solution:

$$G(t) = 120t\sqrt{t} = 120t^{\frac{3}{2}}$$

$$G'(t) = 120 \cdot \frac{3}{2}t^{\frac{1}{2}} = 180\sqrt{t}$$

$$G'(4) = 180\sqrt{4} = 180 \cdot 2 = 360$$

At time $t = 4$ hours, passengers use water at a rate of 360 gallons per hour.

Unit 1 AP Topics 16

71. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	2	3	4
$B(t)$ (miles per hour)	1	8	1.5	-5	11

Brandon and Chloe ride their bikes for 4 hours along a flat, straight road. Brandon's velocity, in miles per hour, at time t hours is given by a differentiable function B for $0 \leq t \leq 4$. Values of $B(t)$ for selected times t are given in the table above. Chloe's velocity, in miles per hour, at time t hours is given by the piecewise function C defined by

$$C(t) = \begin{cases} te^{4-t^2} & \text{for } 0 \leq t \leq 2 \\ 12 - 3t - t^2 & \text{for } 2 < t \leq 4. \end{cases}$$

- How many miles did Chloe travel from time $t = 0$ to time $t = 2$?
- At time $t = 3$, is Chloe's speed increasing or decreasing? Give a reason for your answer.
- Is there a time t , for $0 \leq t \leq 4$, at which Brandon's acceleration is equal to 2.5 miles per hour per hour? Justify your answer.
- Is there a time t , for $0 \leq t \leq 2$, at which Brandon's velocity is equal to Chloe's velocity? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned with a connection to the correct part of the piecewise defined function. $\int_0^2 C(t) dt$ by itself is not sufficient to earn the point.

The second point is earned with $-\frac{1}{2}e^{4-t^2}$ or with $-\frac{1}{2}e^u$ for $u = 4 - t^2$. Limits of integration are part of the third point.

The third point does not require units.

Unit 1 AP Topics 16



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ integral
- ☐ antiderivative
- ☐ answer

Solution:

$$\int_0^2 te^{4-t^2} dt = -\frac{1}{2}e^{4-t^2} \Big|_{t=0}^{t=2} = -\frac{1}{2} + \frac{1}{2}e^4$$

Chloe traveled $-\frac{1}{2} + \frac{1}{2}e^4$ miles from time $t = 0$ to time $t = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by showing both $C(3)$ and $C'(3)$ are less than zero or that both have the same sign.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $C(3) < 0$ and $C'(3) < 0$
- ☐ answer with reason

Solution:

$$C(3) = -6 < 0$$

$$C'(3) = -9 < 0$$

Chloe's speed is increasing at time $t = 3$ because her velocity and acceleration have the same sign.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 16

The second point is earned with the justification that B is continuous and a numerical difference quotient equal to 2.5. There is no requirement for the response to indicate that B is differentiable, nor to refer to the Mean Value Theorem by name.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $\frac{B(4)-B(0)}{4-0}$
- ☐ answer with justification

Solution:

B is differentiable $\Rightarrow B$ is continuous on $[0, 4]$.

$$\frac{B(4)-B(0)}{4-0} = \frac{11-1}{4-0} = 2.5$$

By the Mean Value Theorem, there is a time t , for $0 < t < 4$, such that $B'(t) = 2.5$ miles per hour per hour.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by comparing B and C , either by a subtraction statement or by comparing $B(t)$ and $C(t)$ for some t in $\{0, 1, 2\}$.

The second point is earned with a justification that includes asserting the continuity of B and C or of $B - C$, and showing that $B(0) > C(0)$ and $B(2) < C(2)$. It is not required for the response to specifically refer to the Intermediate Value Theorem.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ considers $B(t) - C(t)$
- ☐ answer with justification

Solution:

B and C are continuous on $[0, 2]$, therefore $B - C$ is continuous on $[0, 2]$.

Unit 1 AP Topics 16

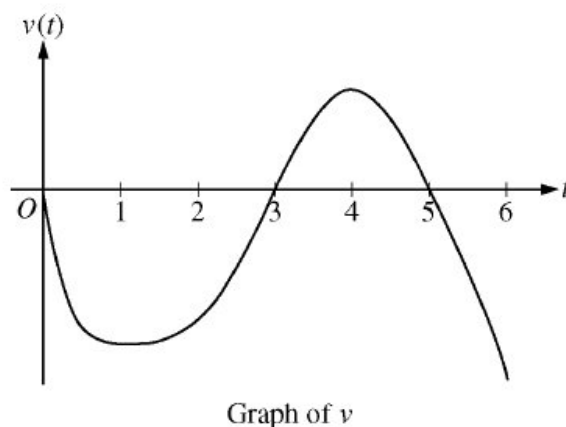
$$B(0) - C(0) = 1 - 0 > 0$$

$$B(2) - C(2) = 1.5 - 2 < 0$$

By the Intermediate Value Theorem, there is a time t , for $0 < t < 2$, such that $B(t) - C(t) = 0$, or $B(t) = C(t)$.

Unit 1 AP Topics 16

72.



A particle moves along the x -axis so that its velocity at time t , for $0 \leq t \leq 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t -axis and the graph of v on the intervals $[0,3]$, $[3,5]$, and $[5,6]$ are 8, 3, and 2 respectively. At time $t = 0$, the particle is $x = -2$.

- (a) For $0 \leq t \leq 6$, find *both the time and the position of the particle when the particle is farthest to the left. Justify your answer.*
- (b) For how many values of t , where $0 \leq t \leq 6$, is the particle at $x = -8$? Explain your reasoning.
- (c) On the interval $2 < t < 3$, is the speed of the particle increasing or decreasing? Give a reason for your answer.
- (d) During what time intervals, if any, is the acceleration of the particle negative? Justify your answer.

Part A

1 point is earned for identifying $t = 3$ as a candidate

1 point is earned for considering $\int_0^6 v(t) dt$

1 point is earned for conclusion

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

$$x(3) = -2 + \int_0^3 v(t) dt = -2 - 8 = -10$$

$$x(6) = -2 + \int_0^6 v(t) dt = -2 - 8 + 3 - 2 = -9$$

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

Unit 1 AP Topics 16



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for identifying $t = 3$ as a candidate

1 point is earned for considering $\int_0^6 v(t) dt$

1 point is earned for conclusion

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

$$x(3) = -2 + \int_0^3 v(t) dt = -2 - 8 = -10$$

$$x(6) = -2 + \int_0^6 v(t) dt = -2 - 8 + 3 - 2 = -9$$

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

Part B

1 point for positioning at $t = 3$, $t = 5$, and $t = 6$

1 point for description of motion

1 point for conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point for positioning at $t = 3$, $t = 5$, and $t = 6$

1 point for description of motion

Unit 1 AP Topics 16

1 point for conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.

Part C

1 point is earned for answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.

Part D

1 point for answer

1 point for justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point for answer

1 point for justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.

Unit 1 AP Topics 16

73. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	2	8	14
$v(t)$ (meters per second)	15	12	6	0

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

- Use the data in the table to approximate $v'(6)$ using the average rate of change of $v(t)$ over the interval $2 \leq t \leq 8$. Show the computations that lead to your answer. Indicate units of measure.
- Interpret the meaning of $v'(6)$ in the context of the problem.
- Justify why there must be a time $t = k$, for $8 \leq k \leq 14$, when the velocity of the particle is 3 meters per second.
- Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{14} v(t) \, dt$. Show the computations that lead to your answer.
- Find $\int_3^{21} v' \left(\frac{t}{3} + 7 \right) \, dt$. Show the computations that lead to your answer.
- Let $p(x) = \int_0^{4x} v(2t) \, dt$. Find $p'(1)$. Show the computations that lead to your answer.
- The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(4) = 5$, $g'(4) = 2$, and $g''(4) = -6$. Is the speed of particle Q increasing or decreasing at time $t = 4$? Give a reason for your answer.
- Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x + 3y^2$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

Unit 1 AP Topics 16

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include

$$\frac{6-12}{8-2}, \frac{6-12}{6}, \text{ and } -\frac{6}{8-2}.$$

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(8)-v(2)}{8-2}$ alone will not earn the point as it does not pull values from the table (yet).

Note: the reciprocal of the quotient gives the same numerical answer. This does not earn the point.

Also: $\frac{v(14)-v(8)}{14-8}$ happens to equal -1 as well. A response using an incorrect pair of points does not earn the point.

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$v'(6) \approx \frac{6-12}{8-2} = -1 \text{ meter per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Unit 1 AP Topics 16

First point: Interpretation

There are two components to look for:

- “rate of change of velocity” or “acceleration” of the particle
- at $t = 6$

“The rate at which the velocity is decreasing at $t = 6$ ” is acceptable. If the answer to part (a) was reported to be positive, “the rate at which the velocity is increasing at $t = 6$ is acceptable. We will not penalize for the same error twice.

Responses that say “velocity is decreasing at a rate of $-1 \frac{m}{s^2}$ ” or similar do not earn the point. This is interpreted to mean that the velocity is increasing.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(6)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 6$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values 0 and 6 pulled from the table and correctly compared with 3, in words or using inequality symbols.

Unit 1 AP Topics 16

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always decreasing on the interval $[8, 14]$. This point will not be earned if the response contains such errors in reasoning.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding 3 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(14) = 0 < 3 < 6 = v(8)$$

By the Intermediate Value Theorem, there must be a time $t = k$, for $8 < k < 14$, such that the velocity of the particle is 3 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

Must see a sum of three terms, each of which is a product of two factors:

$$(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$$

If exactly 5 out of 6 of these numbers ($b_1, b_2, b_3, h_1, h_2, h_3$) are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(0) \cdot (2 - 0) + v(2) \cdot (8 - 2) + v(8) \cdot (14 - 8)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

138 or equivalent.

Unit 1 AP Topics 16

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(30 + 72 + 36)$, or a graph with areas labeled and summed to 138 or equivalent, can earn this point.

The response of only $15 \cdot 2 + 12 \cdot 6 + 6 \cdot 6$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$$\begin{aligned}\int_0^{14} v(t) dt &\approx v(0)(2-0) + v(2)(8-2) + v(8)(14-8) \\ &= 15 \cdot 2 + 12 \cdot 6 + 6 \cdot 6 = 138\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

1st point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

· $u = \frac{t}{3} + 7$

· $du = \frac{dt}{3}$ or equivalent

· Appropriate limits of integration throughout (8 and 14 for u , 3 and 21 for t).

Substitution can be implicit. A coefficient of 3 accompanying a correct antiderivative convinces us of this:

$$\int_3^{21} v' \left(\frac{t}{3} + 7 \right) dt = 3v \left(\frac{t}{3} + 7 \right) \Big|_3^{21}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (3 and 21), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{3} + 7$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled

Unit 1 AP Topics 16

correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Expressions/equations that earn this point are:

· $k(v(14) - v(8))$ or $k(v(\frac{21}{3} + 7) - v(\frac{3}{3} + 7))$, where k is a nonzero constant

· $kv(\frac{t}{3} + 7) \Big|_3^{21}$ or $k v(u) \Big|_8^{14}$, where k is a nonzero constant

· $\int v'(\frac{t}{3} + 7) dt = kv(\frac{t}{3} + 7)$ or $k \int v'(u) du = kv(u)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral

· $\int_3^{21} v'(\frac{t}{3} + 7) dt = k v(\frac{t^2}{6} + 7t)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

–18 or equivalent, with credible supporting work, earns this point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{3} + 7$.

Then $du = \frac{dt}{3}$.

When $t = 3$, $u = 8$.

When $t = 21$, $u = 14$.

$$\int_3^{21} v' \left(\frac{t}{3} + 7 \right) dt = 3 \int_8^{14} v'(u) du$$

Unit 1 AP Topics 16

$$= 3(v(14) - v(8))$$

$$= 3(0 - 6) = -18$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus (FTC)

Substitutes $4x$ into $v(2t)$. We must see either $v(2 \cdot 4x)$ or $v(8x)$.

The expression $v(2 \cdot 4x) - v(2(0))$ or $v(2 \cdot 4x) \cdot 4 - v(2(0))$ will earn this point. The error of subtracting $v(2(0))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2}du = dt$ and rewrites: $\int_0^{4x} v(2t) dt = \frac{1}{2} \int_0^{8x} v(u) du$.

· Or Alt text: $u = 4x$, $\frac{du}{dx} = 4$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by 4

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 8 multiplied separately.

Third point: Answer

24 or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Unit 1 AP Topics 16

Solution:

$$p'(x) = \frac{d}{dx} \int_0^{4x} v(2t) dt = v(2 \cdot 4x) \cdot 4$$

$$p'(1) = v(8) \cdot 4 = 6 \cdot 4 = 24$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “decreasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “decreasing because velocity and acceleration have different signs” earns neither of the two points. It is not clear that the response is considering g' and g'' .

A response that calls out all three given values ($g(4)$, $g'(4)$, and $g''(4)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “slowing down” will be interpreted as “decreasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(4) = 2 > 0$, the velocity of the particle is positive. Because $g''(4) = -6 < 0$, the acceleration of the particle is negative. Therefore, because the signs of the velocity and acceleration are different, the speed of the

Unit 1 AP Topics 16

particle is decreasing at time $t = 4$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

$$y - 2 = (-2(3) + 3(2)^2)(x - 3)$$

$$y - 2 = 6(x - 3)$$

$$y = 6x - 16$$

$y - 2 = (-2x + 3y^2)(x - 3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = 6(x - 3) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(3,2)} = -2 \cdot 3 + 3 \cdot 2^2 = 6$$

$$y = 6(x - 3) + 2$$

Unit 1 AP Topics 16

74. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	2	5	11
$v(t)$ (meters per second)	3	5.5	7	13

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

- Use the data in the table to approximate $v'(8)$ using the average rate of change of $v(t)$ over the interval $5 \leq t \leq 11$. Show the computations that lead to your answer. Indicate units of measure.
- Interpret the meaning of $v'(8)$ in the context of the problem.
- Justify why there must be a time $t = b$, for $0 \leq b \leq 2$, when the velocity of the particle is 5 meters per second.
- Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t) \, dt$. Show the computations that lead to your answer.
- Find $\int_0^9 v' \left(\frac{t}{3} + 2 \right) \, dt$. Show the computations that lead to your answer.
- Let $h(x) = \int_1^{\frac{5}{2}x} v(2t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.
- The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(2) = 5.5$, $g'(2) = -4$, and $g''(2) = -3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.
- Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -3x^2y^3$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

Unit 1 AP Topics 16

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include

$$\frac{13-7}{11-5}, \frac{13-7}{6}, \text{ and } \frac{6}{11-5}.$$

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(11)-v(5)}{11-5}$ alone will not earn the point as it does not pull values from the table (yet).

Note: the reciprocal of this quotient gives the same numerical value. This does not earn the point.

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$v'(8) \approx \frac{13-7}{11-5} = 1 \text{ meter per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

Unit 1 AP Topics 16

· “rate of change of velocity” or “acceleration” of the particle

· at $t = 8$

“The rate at which the velocity is increasing at $t = 8$ ” is acceptable. If the answer to part (a) was reported to be negative, “the rate at which the velocity is decreasing at $t = 8$ ” is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(8)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 8$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values 3 and 5.5 pulled from the table and correctly compared to 5, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always increasing on the interval $[0, 2]$. This point will not be earned if the response contains such errors in reasoning.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding 5 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(0) = 3 < 5 < 5.5 = v(2)$$

By the Intermediate Value Theorem, there must be a time $t = b$, for $0 \leq b \leq 2$, such that the velocity of the particle is 5 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

If exactly 5 out of 6 of these numbers (b_1 , b_2 , b_3 , h_1 , h_2 , h_3) are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(0) \cdot (2 - 0) + v(2) \cdot (5 - 2) + v(5) \cdot (11 - 5)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

64.5 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(6 + 16.5 + 42)$, or a graph with areas labeled and summed to 64.5 or equivalent, can earn this point.

The response of only $3 \cdot 2 + 5.5 \cdot 3 + 7 \cdot 6$ earns both points.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{11} v(t) dt \approx v(0)(2-0) + v(2)(5-2) + v(5)(11-5)$$

$$= 3 \cdot 2 + 5.5 \cdot 3 + 7 \cdot 6 = 64.5$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

· $u = \frac{t}{3} + 2$

· $du = \frac{dt}{3}$ or equivalent

· Appropriate limits of integration throughout (2 and 5 for u , 0 and 9 for t).

Substitution can be implicit. A coefficient of 3 accompanying a correct antiderivative convinces us of this:

$$\int_0^9 v' \left(\frac{t}{3} + 2 \right) dt = 3v \left(\frac{t}{3} + 2 \right) \Big|_0^9$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (0 and 9), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{3} + 2$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Unit 1 AP Topics 16

Expressions/equations that earn this point are:

· $k(v(5) - v(2))$ or $k\left(v\left(\frac{9}{3} + 2\right) - v\left(\frac{0}{3} + 2\right)\right)$, where k is a nonzero constant.

· $kv\left(\frac{t}{3} + 2\right)\bigg|_0^9$ or $k v(u)\bigg|_2^5$, where k is a nonzero constant.

· $\int v'\left(\frac{t}{3} + 2\right) dt = kv\left(\frac{t}{3} + 2\right)$ or $k \int v'(u) du = kv(u)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

· $\int_0^9 v'\left(\frac{t}{3} + 2\right) dt = k v\left(\frac{t^2}{6} + 2t\right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

4.5 or equivalent, with credible supporting work, earns this point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{3} + 2$. Then $du = \frac{dt}{3}$.

When $t = 0$, $u = 2$.

When $t = 9$, $u = 5$.

$$\int_0^9 v'\left(\frac{t}{3} + 2\right) dt = 3 \int_2^5 v'(u) du$$

$$= 3(v(5) - v(2))$$

$$= 3(7 - 5.5) = 4.5$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 16

First point: Fundamental Theorem of Calculus (FTC)

Substitutes $\frac{5}{2}x$ into $v(2t)$. We must see either $v\left(2 \cdot \frac{5}{2}x\right)$ or $v(5x)$.

The expression $v\left(2 \cdot \frac{5}{2}x\right) - v(2(1))$ or $v\left(2 \cdot \frac{5}{2}x\right) \cdot \frac{5}{2} - v(2(1))$ will earn this point. The error of subtracting $v(2(1))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2} du = dt$ and rewrite $\int_1^{\frac{5}{2}x} v(2t) dt = \frac{1}{2} \int_2^{5x} v(u) du$.

· Or $u = \frac{5}{2}x$, $\frac{du}{dx} = \frac{5}{2}$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{5}{2}$

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 5 multiplied separately.

Third point: Answer

$\frac{35}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{\frac{5}{2}x} v(2t) dt = v\left(2 \cdot \frac{5}{2}x\right) \cdot \frac{5}{2}$$

Unit 1 AP Topics 16

$$h'(1) = v(5) \cdot \frac{5}{2} = 7 \cdot \frac{5}{2} = \frac{35}{2}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “increasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “increasing because velocity and acceleration are both negative” earns neither of the two points. It is not clear that the response is considering g' and g'' . A response that calls out all three given values ($g(2)$, $g'(2)$, and $g''(2)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “speeding up” will be interpreted as “increasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(2) = -4 < 0$, the velocity of the particle is negative. Because $g''(2) = -3 < 0$, the acceleration of the particle is negative. Therefore, because both the velocity and acceleration are negative, the speed of the particle is increasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Unit 1 AP Topics 16

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

$$y - 2 = -3(3)^2(2)^3(x - 3)$$

$$y - 2 = -216(x - 3)$$

$$y = -216x + 650$$

$y - 2 = -3x^2y^3(x - 3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = -216(x - 3) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(3,2)} = -3 \cdot 3^2 \cdot 2^3 = -216$$

$$y = -216(x - 3) + 2$$

Unit 1 AP Topics 16

75. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	8	11
$v(t)$ (meters per second)	-2	-5	-11	-12

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

- Use the data in the table to approximate $v'(5)$ using the average rate of change of $v(t)$ over the interval $3 \leq t \leq 8$. Show the computations that lead to your answer. Indicate units of measure.
- Interpret the meaning of $v'(5)$ in the context of the problem.
- Justify why there must be a time $t = d$, for $0 \leq d \leq 3$, when the velocity of the particle is -4 meters per second.
- Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t) \, dt$. Show the computations that lead to your answer.
- Find $\int_0^{10} v' \left(\frac{t}{2} + 3 \right) \, dt$. Show the computations that lead to your answer.
- Let $r(x) = \int_0^{\frac{3}{2}x} v(2t) \, dt$. Find $r'(1)$. Show the computations that lead to your answer.
- The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = -4$, $g'(3) = -2$, and $g''(3) = 6$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.
- Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 3x^2 - y^3$ with initial condition $f(-1) = 2$. Write an equation for the line tangent to the graph of f at the point $(-1, 2)$.

Part A

Unit 1 AP Topics 16

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include

$$\frac{-11 - (-5)}{8 - 3}, \frac{-11 - (-5)}{5}, \text{ and } \frac{-6}{8 - 3}.$$

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(8) - v(3)}{8 - 3}$ alone will not earn the point as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$v'(5) \approx \frac{-11 - (-5)}{8 - 3} = -\frac{6}{5} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

Unit 1 AP Topics 16

· “rate of change of velocity” or “acceleration” of the particle

· at $t = 5$

“The rate at which the velocity is decreasing at $t = 5$ ” is acceptable. If the answer to part (a) was reported to be positive, “the rate at which the velocity is increasing at $t = 5$ ” is acceptable. We will not penalize for the same error twice.

Responses that say “velocity is decreasing at a rate of $-\frac{6}{5} \frac{m}{s^2}$ ” or similar do not earn the point. This is interpreted to mean that the velocity is increasing.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(5)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 5$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values -5 and -2 pulled from the table and correctly compared with -4 , in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Unit 1 AP Topics 16

Responses often “say too much”; especially common are responses that indicate that v is always decreasing on the interval $[0, 3]$. This point will not be earned if the response contains such errors in reasoning.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding -4 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(3) = -5 < -4 < -2 = v(0)$$

By the Intermediate Value Theorem, there must be a time $t = d$, $0 < d < 3$, such that the velocity of the particle is -4 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$

If exactly 5 out of 6 of these numbers $(b_1, b_2, b_3, h_1, h_2, h_3)$ are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(0) \cdot (3 - 0) + v(3) \cdot (8 - 3) + v(8) \cdot (11 - 8)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

-64 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(-6 + -25 + -33)$ or a graph with areas labeled and summed to -64 or equivalent, can earn this point.

Unit 1 AP Topics 16

The response of only $(-2 \cdot 3) + (-5 \cdot 5) + (-11 \cdot 3)$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{11} v(t) dt \approx v(0)(3-0) + v(3)(8-3) + v(8)(11-8)$$

$$= -2 \cdot 3 + (-5) \cdot 5 + (-11) \cdot 3 = -64$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

· $u = \frac{t}{2} + 3$

· $du = \frac{dt}{2}$ or equivalent

· Appropriate limits of integration throughout (3 and 8 for u , 0 and 10 for t).

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_0^{10} v' \left(\frac{t}{2} + 3 \right) dt = 2v \left(\frac{t}{2} + 3 \right) \Big|_0^{10}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (0 and 10), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2} + 3$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Unit 1 AP Topics 16

Expressions/equations that earn this point are:

$$k(v(8) - v(3)) \text{ or } k\left(v\left(\frac{10}{2} + 3\right) - v\left(\frac{0}{2} + 3\right)\right),$$

· where k is a nonzero constant.

$$\cdot kv\left(\frac{t}{2} + 3\right)\Big|_0^{10} \text{ or } kv(u)\Big|_3^8, \text{ where } k \text{ is a nonzero constant.}$$

$$\cdot \int v'\left(\frac{t}{2} + 3\right) dt = kv\left(\frac{t}{2} + 3\right) \text{ or } k \int v'(u) du = kv(u), \text{ where } k \text{ is a nonzero constant, with or without appropriate limits of integration displayed on either integral.}$$

$$\cdot \int_0^{10} v'\left(\frac{t}{2} + 3\right) dt = kv\left(\frac{t^2}{4} + 3t\right) \text{ where } k \text{ is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.}$$

Third point: Answer

−12 or equivalent, with credible supporting work, earns this point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\text{Let } u = \frac{t}{2} + 3. \text{ Then } du = \frac{dt}{2}.$$

$$\text{When } t = 0, u = 3.$$

$$\text{When } t = 10, u = 8.$$

$$\int_0^{10} v'\left(\frac{t}{2} + 3\right) dt = 2 \int_3^8 v'(u) du$$

$$= 2(v(8) - v(3))$$

$$= 2(-11 - (-5)) = -12$$

Part F

Unit 1 AP Topics 16

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus

Substitutes $\frac{3}{2}x$ into $v(2t)$. We must see either $v\left(2 \cdot \frac{3}{2}x\right)$ or $v(3x)$.

The expression $v\left(2 \cdot \frac{3}{2}x\right) - v(2(0))$ or $v\left(2 \cdot \frac{3}{2}x\right) \cdot \frac{3}{2} - v(2(0))$ will earn this point. The error of subtracting $v(2(0))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2}du = dt$ and rewrites: $\int_0^{\frac{3}{2}x} v(2t) dt = \frac{1}{2} \int_0^{3x} v(u) du$.

· Or $u = \frac{3}{2}x$, $\frac{du}{dx} = \frac{3}{2}$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{3}{2}$.

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 3 multiplied separately.

Third point: Answer

$-\frac{15}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

Unit 1 AP Topics 16

$$r'(x) = \frac{d}{dx} \int_0^{\frac{3}{2}x} v(2t) dt = v\left(2 \cdot \frac{3}{2}x\right) \cdot \frac{3}{2}$$

$$r'(1) = v(3) \cdot \frac{3}{2} = -5 \cdot \frac{3}{2} = -\frac{15}{2}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “decreasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “decreasing because velocity and acceleration have different signs” earns neither of the two points. It is not clear that the response is considering g' and g''

A response that calls out all three given values ($g(3)$, $g'(3)$, and $g''(3)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “slowing down” will be interpreted as “decreasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(3) = -2 < 0$, the velocity of the particle is negative. Because $g''(3) = 6 > 0$, the acceleration of the particle is positive. Because the signs of the velocity and acceleration are different, the speed of the particle is decreasing at time $t = 3$.

Unit 1 AP Topics 16

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

$$y - 2 = (3(-1)^2 - 2^3)(x + 1)$$

$$y - 2 = -5(x + 1)$$

$$y = -5x - 3$$

$y - 2 = (3x^2 - y^3)(x + 1)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = -5(x + 1) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(-1, 2)} = 3 \cdot (-1)^2 - 2^3 = -5$$

$$y = 2 - 5(x + 1)$$

Unit 1 AP Topics 16

76. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	8	11
$v(t)$ (meters per second)	-15	-12	-8	-3

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

- Use the data in the table to approximate $v'(10)$ using the average rate of change of $v(t)$ over the interval $8 \leq t \leq 11$. Show the computations that lead to your answer. Indicate units of measure.
- Interpret the meaning of $v'(10)$ in the context of the problem.
- Justify why there must be a time $t = k$, for $0 \leq k \leq 3$, when the velocity of the particle is -13 meters per second.
- Use a right Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t) \, dt$. Show the computations that lead to your answer.
- Find $\int_4^{14} v'\left(\frac{t}{2} + 1\right) \, dt$. Show the computations that lead to your answer.
- Let $h(x) = \int_3^{\frac{11}{2}x} v(2t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.
- The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(2) = -5.5$, $g'(2) = 4$, and $g''(2) = 3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.
- Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = y^2 - xy$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

Unit 1 AP Topics 16

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include

$$\frac{-3+8}{11-8}, \frac{-3-(-8)}{3}, \text{ and } \frac{5}{11-8}.$$

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(11)-v(8)}{11-8}$ alone will not earn the point as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ and fraction or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$v'(10) \approx \frac{-3-(-8)}{11-8} = \frac{5}{3} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

Unit 1 AP Topics 16

- “rate of change of velocity” or “acceleration” of the particle

- at $t = 10$

“The rate at which the velocity is increasing at $t = 10$ ” is acceptable. If the answer to part (a) was reported to be negative, “the rate at which the velocity is decreasing at $t = 10$ ” is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(10)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 10$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values -15 , and -12 pulled from the table and correctly compared to -13 , in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always increasing on the interval $[0, 3]$. This point will not be earned if the response contains such errors in reasoning.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding -13 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(0) = -15 < -13 < -12 = v(3)$$

By the Intermediate Value Theorem, there must be a time $t = k$, for $0 \leq k \leq 3$, such that the velocity of the particle is -13 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of right Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

If exactly 5 out of 6 of these numbers $(b_1, b_2, b_3, h_1, h_2, h_3)$ are correctly placed in the above expression, we are convinced that the response carries the form of a right Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression

$$v(3) \cdot (3 - 0) + v(8) \cdot (8 - 3) + v(11) \cdot (11 - 8) \text{ is sufficient.}$$

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

second point: Answer

-85 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a right Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(-36 + (-40) + (-9))$, or a graph with areas labeled and summed to -85 or equivalent, can earn this point.

The response of only $(-12) \cdot 3 + (-8) \cdot 5 + (-3) \cdot 3$ earns both points.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{11} v(t) dt \approx v(3)(3-0) + v(8)(8-3) + v(11)(11-8)$$

$$= -12 \cdot 3 + (-8) \cdot 5 + (-3) \cdot 3 = -85$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

· $u = \frac{t}{2} + 1$

· $du = \frac{dt}{2}$ or equivalent

· Appropriate limits of integration throughout (3 and 8 for u , 4 and 14 for t).

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_4^{14} v' \left(\frac{t}{2} + 1 \right) dt = 2v \left(\frac{t}{2} + 1 \right) \Big|_4^{14}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (4 and 14), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2} + 1$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Unit 1 AP Topics 16

Expressions/equations that earn this point are:

· $k(v(8) - v(3))$ or $k\left(v\left(\frac{14}{2} + 1\right) - v\left(\frac{4}{2} + 1\right)\right)$, where k is a nonzero constant.

· $kv\left(\frac{t}{2} + 1\right)\Big|_4^{14}$ or $k v(u)\Big|_3^8$, where k is a nonzero constant.

· $\int v'\left(\frac{t}{2} + 1\right) dt = kv\left(\frac{t}{2} + 1\right)$ or $k \int v'(u) du = kv(u)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

· $\int_4^{14} v'\left(\frac{t}{2} + 1\right) dt = k v\left(\frac{t^2}{4} + t\right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

8 or equivalent, with credible supporting work, earns this point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{2} + 1$. Then $du = \frac{dt}{2}$.

When $t = 4$, $u = 3$.

When $t = 14$, $u = 8$.

$$\int_4^{14} v'\left(\frac{t}{2} + 1\right) dt = 2 \int_3^8 v'(u) du$$

$$= 2(v(8) - v(3))$$

$$= 2(-8 - (-12)) = 8$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 16

First point: Fundamental Theorem of Calculus (FTC)

Substitutes $\frac{11}{2}x$ into $v(2t)$. We must see either $v\left(2 \cdot \frac{11}{2}x\right)$ or $v(11x)$.

The expression $v\left(2 \cdot \frac{11}{2}x\right) - v(2(3))$ or $v\left(2 \cdot \frac{11}{2}x\right) \cdot \frac{11}{2} - v(2(3))$ will earn this point. The error of subtracting $v(2(3))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2}du = dt$ and rewrites: $\int_3^{\frac{11}{2}x} v(2t)dt = \frac{1}{2} \int_6^{11x} v(u)du$.

· Or $u = \frac{11}{2}x$, $\frac{du}{dx} = \frac{11}{2}$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{11}{2}$

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 11 multiplied separately.

Third point: Answer

$-\frac{33}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_3^{\frac{11}{2}x} v(2t) dt = v\left(2 \cdot \frac{11}{2}x\right) \cdot \frac{11}{2}$$

Unit 1 AP Topics 16

$$h'(1) = v(11) \cdot \frac{11}{2} = -3 \cdot \frac{11}{2} = -\frac{33}{2}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “increasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “increasing because velocity and acceleration are both negative” earns neither of the two points. It is not clear that the response is considering g' and g'' . A response that calls out all three given values ($g(2)$, $g'(2)$, and $g''(2)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “speeding up” will be interpreted as “increasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(2) = 4 > 0$, the velocity of the particle is positive. Because $g''(2) = 3 > 0$, the acceleration of the particle is positive. Therefore, because both the velocity and acceleration are positive, the speed of the particle is increasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Unit 1 AP Topics 16

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

$$y - 2 = ((2)^2 - 3 \cdot 2)(x - 3)$$

$$y - 2 = -2(x - 3)$$

$$y = -2x + 8$$

$y - 2 = (y^2 - xy)(x - 3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = -2(x - 3) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(3,2)} = 2^2 - 3 \cdot 2 = -2$$

$$y = -2(x - 3) + 2$$

Unit 1 AP Topics 16

77. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	10	12
$v(t)$ (meters per second)	2	6	15	20

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

- Use the data in the table to approximate $v'(11)$ using the average rate of change of $v(t)$ over the interval $10 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.
- Interpret the meaning of $v'(11)$ in the context of the problem.
- Justify why there must be a time $t = c$, for $4 \leq c \leq 10$, when the velocity of the particle is 8 meters per second.
- Use a right Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{12} v(t) \, dt$. Show the computations that lead to your answer.
- Find $\int_4^{12} v' \left(\frac{t}{2} - 2 \right) \, dt$. Show the computations that lead to your answer.
- Let $h(x) = \int_2^{4x} v(3t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.
- The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(4) = 7$, $g'(4) = -3$, and $g''(4) = -2$. Is the speed of particle Q increasing or decreasing at time $t = 4$? Give a reason for your answer.
- Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 3x^2 - y^3$ with initial condition $f(2) = 1$. Write an equation for the line tangent to the graph of f at the point $(2, 1)$.

Part A

Unit 1 AP Topics 16

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{20-15}{12-10}$, $\frac{20-15}{2}$, and $\frac{5}{12-10}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(12)-v(10)}{12-10}$ alone is not sufficient as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$v'(11) \approx \frac{20-15}{12-10} = \frac{5}{2} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

· “rate of change of velocity” or “acceleration” of the particle

Unit 1 AP Topics 16

· at $t = 11$

“The rate at which the velocity is increasing at $t = 11$ ” is acceptable. If the answer to part (a) was reported to be negative, “the rate at which the velocity is decreasing at $t = 11$ ” is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

v' (11) is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 11$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values 6 and 15 pulled from the table and correctly compared with 8, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always increasing on the interval $[6, 15]$. This point will not be earned if the response contains such errors in reasoning.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding 8 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(4) = 6 < 8 < 15 = v(10)$$

By the Intermediate Value Theorem, there must be a time $t = c$, for $4 < c < 10$, such that the velocity of the particle is 8 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of right Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

If exactly 5 out of 6 of these numbers (b_1 , b_2 , b_3 , h_1 , h_2 , h_3) are correctly placed in the above expression, we are convinced that the response carries the form of a right Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(4) \cdot (4 - 0) + v(10) \cdot (10 - 4) + v(12) \cdot (12 - 10)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

154 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a right Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(24 + 90 + 40)$, or a graph with areas labeled and summed to 154 or equivalent, can earn this point.

The response of only $6 \cdot 4 + 15 \cdot 6 + 20 \cdot 2$ earns both points.

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{12} v(t) dt \approx v(4)(4-0) + v(10)(10-4) + v(12)(12-10)$$

$$= 6 \cdot 4 + 15 \cdot 6 + 20 \cdot 2 = 154$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

· $u = \frac{t}{2} - 2$

· $du = \frac{dt}{2}$ or equivalent

· Appropriate limits of integration throughout (0 and 4 for u , 4 and 12 for t).

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_4^{12} v' \left(\frac{t}{2} - 2 \right) dt = 2v \left(\frac{t}{2} - 2 \right) \Big|_4^{12}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (4 and 12), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2} - 2$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Unit 1 AP Topics 16

Expressions/equations that earn this point are:

· $k(v(4) - v(0))$ or $k\left(v\left(\frac{12}{2} - 2\right) - v\left(\frac{4}{2} - 2\right)\right)$, where k is a nonzero constant.

· $kv\left(\frac{t}{2} - 2\right)\Big|_4^{12}$ or $k v(u)\Big|_0^4$, where k is a nonzero constant.

· $\int v'\left(\frac{t}{2} - 2\right) dt = kv\left(\frac{t}{2} - 2\right)$ or $k \int v'(u) du = kv(u)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

· $\int_4^{12} v'\left(\frac{t}{2} - 2\right) dt = k v\left(\frac{t^2}{4} - 2t\right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

8 or equivalent, with credible supporting work, earns this point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{2} - 2$. Then $du = \frac{dt}{2}$.

When $t = 4$, $u = 0$.

When $t = 12$, $u = 4$.

$$\int_4^{12} v'\left(\frac{t}{2} - 2\right) dt = 2 \int_0^4 v'(u) du$$

$$= 2(v(4) - v(0))$$

$$= 2(6 - 2) = 8$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 16

First point: Fundamental Theorem of Calculus

Substitutes $4x$ into $v(3t)$. We must see either $v(3 \cdot 4x)$ or $v(12x)$.

The expression $v(3 \cdot 4x) - v(3(2))$ or $v(3 \cdot 4x) \cdot 4 - v(3(2))$ will earn this point. The error of subtracting $v(3(2))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 3t$, $\frac{1}{3}du = dt$ and rewrites: $\int_2^{4x} v(3t) dt = \frac{1}{3} \int_6^{12x} v(u) du$.

· Or $u = 4x$, $\frac{du}{dx} = 4$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by 4

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{3}$ and 12 multiplied separately.

Third point: Answer

80 or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_2^{4x} v(3t) dt = v(3 \cdot 4x) \cdot 4$$

$$h'(1) = v(12) \cdot 4 = 20 \cdot 4 = 80$$

Unit 1 AP Topics 16

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “increasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “increasing because velocity and acceleration are both negative” earns neither of the two points. It is not clear that the response is considering g' and g'' . A response that calls out all three given values ($g(4)$, $g'(4)$, and $g''(4)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “speeding up” will be interpreted as “increasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(4) = -3 < 0$, the velocity of the particle is negative. Because $g''(4) = -2 < 0$, the acceleration of the particle is negative. Therefore, because both the velocity and acceleration are negative, the speed of the particle is increasing at time $t = 4$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

Unit 1 AP Topics 16

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

$$y - 1 = (3(2)^2 - (1)^3)(x - 2)$$

$$y - 1 = 11(x - 2)$$

$$y = 11x - 21$$

f does not earn the point. This is not the equation of a line. The response can go on present one of the correct linear forms, and then would earn the point.

$f(x) = 11(x - 2) + 1$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(2,1)} = 3 \cdot 2^2 - 1^3 = 11$$

$$y = 11(x - 2) + 1$$

Unit 1 AP Topics 16

78. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	6	12	18	24
$x'(t)$ (feet per second)	2.4	-1.4	4.0	5.4	6.2

The position of a particle moving along a straight line is modeled by a twice-differentiable function $x(t)$, where time t is measured in seconds and $x(t)$ is measured in feet. Selected values of $x'(t)$, the derivative of $x(t)$, are shown in the table above. The position of the particle at time $t = 0$ is $x(0) = 1$.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to

approximate $\int_0^{24} x'(t) \, dt$. Show the computations that lead to your answer.

(b) Using correct units, explain the meaning of $\int_0^{24} x'(t) \, dt$ in the context of the problem.

(c) Must there be a time t , for $0 < t < 24$, when the velocity of the particle is zero? Justify your answer.

(d) Use Euler's method, starting at $t = 0$ with two steps of equal size, to approximate the position of the particle at time $t = 12$. Show the computations that lead to your answer.

(e) It is known that $x''(0) = -6.8$. Write the second-degree Taylor polynomial for x about $t = 0$.

(f) Let s be a function such that the sixth derivative of s with respect to t is $(t + 5)^7$. Write the sixth-degree term of the sixth-degree Taylor polynomial for s about $t = -2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The answer need not be simplified but must be equivalent to 48.

If errors are made in simplification, the point is not earned.

The answer must have accompanying work, specifically the sum of two products.

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\int_0^{24} x'(t) dt \approx x'(6)(12 - 0) + x'(18)(24 - 12)$$

$$= (-1.4) \cdot 12 + 5.4 \cdot 12 = 48$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

“Displacement” can be written as “change in position.”

“Distance traveled” or “total feet” is not acceptable.

The time interval must be specific, i.e., $t = 0$ to $t = 24$ seconds; at $t = 24$ seconds is not acceptable.

“Over 24 seconds” is not acceptable.

Second point:

“Feet” may be stated as the units in part (a) or in part (b). The point is earned in either case, unless some other unit is re(stated) in part (b).

“Seconds” need not be stated as the variable t was defined in the problem stem.

Units point may be earned without the interpretation point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Interpretation

☐ Units

Unit 1 AP Topics 16

Solution:

$\int_0^{24} x'(t) dt$ is the displacement of the particle, in feet, from $t = 0$ to $t = 24$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The connection of velocity to $x'(t)$ need not be explicit; any specific reference to $x'(t)$ or specific values in the table will indicate the connection and earn the point.

Second point:

Both conditions must be stated, i.e., the continuity of $x'(t)$ and the bounding of 0 by two values of $x'(t)$ of opposite signs.

The conditions point requires $x'(t)$ is differentiable and therefore continuous. The point is not earned for just $x'(t)$ is differentiable and continuous.

The interval upon which $x'(t)$ is continuous and/or differentiable does not need to be stated. If stated, it must be a correct interval with the possible exception of inclusion/exclusion of the endpoints.

The values of $x'(t)$ must explicitly bound 0, either verbally or symbolically.

Intervals need not be presented using mathematical notation; for example, “ $x'(0) > 0$ and $x'(6) < 0$ ” or “ $x'(t)$ changes signs between $t = 0$ and $t = 6$ ” are both acceptable.

A general statement such as “ $x'(t)$ changes signs” is not acceptable.

Third point:

Intermediate Value Theorem (or IVT) need not be explicitly stated.

This point is earned for an affirmative conclusion and bounding 0 by two values of opposite signs pulled from the table.

A response could earn the first and third points without earning the second point if conditions are presented but not sufficiently justified and the conclusion is correct.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

Unit 1 AP Topics 16

- ☐ Connection to $x'(t)$
- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

$x'(t)$ is the velocity of the particle at time t .

x is twice differentiable. $\Rightarrow x'$ is differentiable.

$\Rightarrow x'$ is continuous for $0 \leq t \leq 24$.

$$x'(0) = 2.4 > 0 > -1.4 = x'(6)$$

Therefore, by the Intermediate Value Theorem, there is a time t , for $0 < t < 6$, such that $x'(t) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

This point is earned for a response that provides clear evidence of an understanding of Euler's method presented in a table or in the form shown in the model solution.

This point could still be earned in the presence of a single error.

If the method is displayed in a table with no column/row labels, this point is only earned for a completely correct final answer (and in that case, both points are earned).

If the method is displayed in a table with no column/row labels and there is an error in one or more of the values in the table, neither point is earned.

Second point:

This point is earned only by final answer of 7 or equivalent with accompanying work. That answer does not need to be simplified.

The second point cannot be earned without the first point.

Both points are earned for a response of $1 + 2.4 \cdot 6 - 1.4 \cdot 6$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

Unit 1 AP Topics 16

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$x(6) \approx x(0) + x'(0)(6 - 0) = 1 + 2.4 \cdot 6 = 15.4$$

$$x(12) \approx x(6) + x'(6)(12 - 6) \approx 15.4 + (-1.4) \cdot 6 = 7$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

All terms must be correct in order to earn the point.

Using another variable (such as x) is acceptable as long as that variable is used correctly in both non-constant terms.

Coefficients need not be simplified, i.e., $\frac{-6.8}{2!}$ is acceptable.

Additional terms or “+ . . .” added to the polynomial fail to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$1 + 2.4t + \frac{-6.8}{2!}t^2 = 1 + 2.4t - 3.4t^2$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

$s^{(6)}(-2)$ must be correct (and numerical) but need not be simplified.

Second point:

Unit 1 AP Topics 16

The second point will be earned for a correct structure for the sixth-degree term even if the derivative value is incorrect; that is, a form of $\frac{\text{derivative value}}{6!}(t+2)^6$ earns the point regardless of the derivative value's correctness.

The derivative value must be numerical; i.e., $(t+5)^7$ in the numerator is not acceptable.

Setting the derivative equal to the term will not earn the point, for example, $(t+5)^7 = (-2+5)^7 = \frac{3^7}{6!}(t+2)^6$.

Multiple terms of degree less than six that are presented (even if circled or boxed) will not be read.

A response such as $\frac{b}{6!}(t+2)^6$, where b is the previously incorrectly computed value of $s^{(6)}(-2)$, will earn the second point but not the first.

Acceptable forms of the correct term include:

$$\cdot \frac{3^7}{6!}(t+2)^6$$

$$\cdot \frac{2187}{6!}(t+2)^6$$

$$\cdot \frac{2187}{720}(t+2)^6$$

$$\cdot \frac{243}{80}(t+2)^6$$

$$\cdot 3.037(t+2)^6$$

$$\cdot 3.038(t+2)^6$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $s^{(6)}(-2)$

☐ Term

Solution:

$$s^{(6)}(t) = (t+5)^7$$

$$s^{(6)}(-2) = (-2+5)^7 = 3^7$$

$$\text{The sixth-degree term is } \frac{s^{(6)}(-2)}{6!}(t+2)^6 = \frac{3^7}{6!}(t+2)^6.$$

Unit 1 AP Topics 16

79. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	10	15	20
$x'(t)$ (meters per minute)	-2.4	3.8	6.0	7.6	7.0

The position of a particle moving along a straight line is modeled by a twice-differentiable function $x(t)$, where time t is measured in minutes and $x(t)$ is measured in meters. Selected values of $x'(t)$, the derivative of $x(t)$, are shown in the table above. The position of the particle at time $t = 0$ is $x(0) = 8$.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to

approximate $\int_0^{20} x'(t) \, dt$. Show the computations that lead to your answer.

(b) Using correct units, explain the meaning of $\int_0^{20} x'(t) \, dt$ in the context of the problem.

(c) Must there be a time t , for $0 < t < 20$, when the velocity of the particle is zero? Justify your answer.

(d) Use Euler's method, starting at $t = 0$ with two steps of equal size, to approximate the position of the particle at time $t = 10$. Show the computations that lead to your answer.

(e) It is known that $x''(0) = 3$. Write the second-degree Taylor polynomial for x about $t = 0$.

(f) Let s be a function such that the fifth derivative of s with respect to t is $(t - 3)^9$. Write the fifth-degree term of the fifth-degree Taylor polynomial for s about $t = 1$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The answer need not be simplified but must be equivalent to 114.

If errors are made in simplification, the point is not earned.

The answer must have accompanying work, specifically the sum of two products.

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\begin{aligned}\int_0^{20} x'(t) dt &\approx x'(5)(10 - 0) + x'(15)(20 - 10) \\ &= 3.8 \cdot 10 + 7.6 \cdot 10 = 114\end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

“Displacement” can be written as “change in position.”

“Distance traveled” or “total meters” is not acceptable.

The time interval must be specific, i.e., $t = 0$ to $t = 20$ minutes; at $t = 20$ minutes is not acceptable.

“Over 20 minutes” is not acceptable.

Second point:

“Meters” may be stated as the units in part (a) or in part (b). The point is earned in either case, unless some other unit is re(stated) in part (b).

“Minutes” need not be stated as the variable t was defined in the problem stem.

Units point may be earned without the interpretation point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Interpretation

☐ Units

Unit 1 AP Topics 16

Solution:

$\int_0^{20} x'(t) dt$ is the displacement of the particle, in meters, from $t = 0$ to $t = 20$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The connection of velocity to $x'(t)$ need not be explicit; any specific reference to $x'(t)$ or specific values in the table will indicate the connection and earn the point.

Second point:

Both conditions must be stated, i.e., the continuity of $x'(t)$ and the bounding of 0 by two values of $x'(t)$ of opposite signs.

The conditions point requires $x'(t)$ is differentiable and therefore continuous. The point is not earned for just $x'(t)$ is differentiable and continuous.

The interval upon which $x'(t)$ is continuous and/or differentiable does not need to be stated. If stated, it must be a correct interval with the possible exception of inclusion/exclusion of the endpoints.

The values of $x'(t)$ must explicitly bound 0, either verbally or symbolically.

Intervals need not be presented using mathematical notation; for example, “ $x'(0) < 0$ and $x'(5) > 0$ ” or “ $x'(t)$ changes signs between $t = 0$ and $t = 5$ ” are both acceptable.

A general statement such as “ $x'(t)$ changes signs” is not acceptable.

Third point:

Intermediate Value Theorem (or IVT) need not be explicitly stated.

This point is earned for an affirmative conclusion and bounding 0 by two values of opposite signs pulled from the table.

A response could earn the first and third points without earning the second point if conditions are presented but not sufficiently justified and the conclusion is correct.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

Unit 1 AP Topics 16

- ☐ Connection to $x'(t)$
- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

$x'(t)$ is the velocity of the particle at time t .

x is twice differentiable. $\Rightarrow x'$ is differentiable.

$\Rightarrow x'$ is continuous for $0 \leq t \leq 20$.

$$x'(0) = -2.4 < 0 < 3.8 = x'(5)$$

Therefore, by the Intermediate Value Theorem, there is a time t , for $0 < t < 5$, such that $x'(t) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

This point is earned for a response that provides clear evidence of an understanding of Euler's method presented in a table or in the form shown in the model solution.

This point could still be earned in the presence of a single error.

If the method is displayed in a table with no column/row labels, this point is only earned for a completely correct final answer (and in that case, both points are earned).

If the method is displayed in a table with no column/row labels and there is an error in one or more of the values in the table, neither point is earned.

Second point:

This point is earned only by final answer of 15 or equivalent with accompanying work. That answer does not need to be simplified.

The second point cannot be earned without the first point.

Both points are earned for a response of $8 - 2.4 \cdot 5 + 3.8 \cdot 5$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

Unit 1 AP Topics 16

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$x(5) \approx x(0) + x'(0)(5 - 0) = 8 + (-2.4) \cdot 5 = -4$$

$$x(10) \approx x(5) + x'(5)(10 - 5) \approx -4 + 3.8 \cdot 5 = 15$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

All terms must be correct in order to earn the point.

Using another variable (such as x) is acceptable as long as that variable is used correctly in both non-constant terms.

Coefficients need not be simplified, i.e., $\frac{3}{2!}$ is acceptable.

Additional terms or “+ $\dots$ ” added to the polynomial fail to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$8 - 2.4t + \frac{3}{2!}t^2 = 8 - 2.4t + 1.5t^2$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

$s^{(5)}(1)$ must be correct (and numerical) but need not be simplified.

Second point:

Unit 1 AP Topics 16

The second point will be earned for a correct structure for the fifth-degree term even if the derivative value is incorrect; that is, a form of $\frac{\text{derivative value}}{5!}(t-1)^5$ earns the point regardless of the derivative value's correctness.

The derivative value must be numerical; i.e., $(t-3)^9$ in the numerator is not acceptable.

Setting the derivative equal to the term will not earn the point, for example, $(t-3)^9 = (1-3)^9 = \frac{-2^9}{5!}(t-1)^5$.

Multiple terms of degree less than five that are presented (even if circled or boxed) will not be read.

A response such as $\frac{b}{5!}(t-1)^5$, where b is the previously incorrectly computed value of $s^{(5)}(1)$, will earn the second point but not the first.

Acceptable forms of the correct term include:

$$\cdot \frac{-2^9}{5!}(t-1)^5$$

$$\cdot \frac{-512}{5!}(t-1)^5$$

$$\cdot \frac{-512}{120}(t-1)^5$$

$$\cdot \frac{-64}{15}(t-1)^5$$

$$\cdot -4.266(t-1)^5$$

$$\cdot -4.267(t-1)^5$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $s^{(5)}(1)$

☐ Term

Solution:

$$s^{(5)}(t) = (t-3)^9$$

$$s^{(5)}(1) = (1-3)^9 = (-2)^9 = -2^9$$

$$\text{The fifth-degree term is } \frac{s^{(5)}(1)}{5!}(t-1)^5 = \frac{-2^9}{5!}(t-1)^5.$$

Unit 1 AP Topics 16

80. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	8	12	16
$x'(t)$ (meters per second)	1.2	-3.2	5.0	6.2	7.0

The position of a particle moving along a straight line is modeled by a twice-differentiable function $x(t)$, where time t is measured in seconds and $x(t)$ is measured in meters. Selected values of $x'(t)$, the derivative of $x(t)$, are shown in the table above. The position of the particle at time $t = 0$ is $x(0) = 11$.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to

approximate $\int_0^{16} x'(t) \, dt$. Show the computations that lead to your answer.

(b) Using correct units, explain the meaning of $\int_0^{16} x'(t) \, dt$ in the context of the problem.

(c) Must there be a time t , for $0 < t < 16$, when the velocity of the particle is zero? Justify your answer.

(d) Use Euler's method, starting at $t = 0$ with two steps of equal size, to approximate the position of the particle at time $t = 8$. Show the computations that lead to your answer.

(e) It is known that $x''(0) = -4.4$. Write the second-degree Taylor polynomial for x about $t = 0$.

(f) Let s be a function such that the seventh derivative of s with respect to t is $(t + 3)^6$. Write the seventh-degree term of the seventh-degree Taylor polynomial for s about $t = -1$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The answer need not be simplified but must be equivalent to 24.

If errors are made in simplification, the point is not earned.

The answer must have accompanying work, specifically the sum of two products.

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\int_0^{16} x'(t) dt \approx x'(4)(8 - 0) + x'(12)(16 - 8)$$

$$= (-3.2) \cdot 8 + 6.2 \cdot 8 = 24$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

“Displacement” can be written as “change in position.”

“Distance traveled” or “total meters” is not acceptable.

The time interval must be specific, i.e., $t = 0$ to $t = 16$ seconds; at $t = 16$ seconds is not acceptable.

“Over 16 seconds” is not acceptable.

Second point:

“Meters” may be stated as the units in part (a) or in part (b). The point is earned in either case, unless some other unit is re(stated) in part (b).

“Seconds” need not be stated as the variable t was defined in the problem stem.

Units point may be earned without the interpretation point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Interpretation

☐ Units

Unit 1 AP Topics 16

Solution:

$\int_0^{16} x'(t) dt$ is the displacement of the particle, in meters, from $t = 0$ to $t = 16$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The connection of velocity to $x'(t)$ need not be explicit; any specific reference to $x'(t)$ or specific values in the table will indicate the connection and earn the point.

Second point:

Both conditions must be stated, i.e., the continuity of $x'(t)$ and the bounding of 0 by two values of $x'(t)$ of opposite signs.

The conditions point requires $x'(t)$ is differentiable and therefore continuous. The point is not earned for just $x'(t)$ is differentiable and continuous.

The interval upon which $x'(t)$ is continuous and/or differentiable does not need to be stated. If stated, it must be a correct interval with the possible exception of inclusion/exclusion of the endpoints.

The values of $x'(t)$ must explicitly bound 0, either verbally or symbolically.

Intervals need not be presented using mathematical notation; for example, “ $x'(0) > 0$ and $x'(4) < 0$ ” or “ $x'(t)$ changes signs between $t = 0$ and $t = 4$ ” are both acceptable.

A general statement such as “ $x'(t)$ changes signs” is not acceptable.

Third point:

Intermediate Value Theorem (or IVT) need not be explicitly stated.

This point is earned for an affirmative conclusion and bounding 0 by two values of opposite signs pulled from the table.

A response could earn the first and third points without earning the second point if conditions are presented but not sufficiently justified and the conclusion is correct.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

Unit 1 AP Topics 16

- ☐ Connection to $x'(t)$
- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

$x'(t)$ is the velocity of the particle at time t .

x is twice differentiable. $\Rightarrow x'$ is differentiable.

$\Rightarrow x'$ is continuous for $0 \leq t \leq 16$.

$$x'(0) = 1.2 > 0 > -3.2 = x'(4)$$

Therefore, by the Intermediate Value Theorem, there is a time t , for $0 < t < 4$, such that $x'(t) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

This point is earned for a response that provides clear evidence of an understanding of Euler's method presented in a table or in the form shown in the model solution.

This point could still be earned in the presence of a single error.

If the method is displayed in a table with no column/row labels, this point is only earned for a completely correct final answer (and in that case, both points are earned).

If the method is displayed in a table with no column/row labels and there is an error in one or more of the values in the table, neither point is earned.

Second point:

This point is earned only by final answer of 3 or equivalent with accompanying work. That answer does not need to be simplified.

The second point cannot be earned without the first point.

Both points are earned for a response of $11 + 1.2 \cdot 4 - 3.2 \cdot 4$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

Unit 1 AP Topics 16

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$x(4) \approx x(0) + x'(0)(4 - 0) = 11 + 1.2 \cdot 4 = 15.8$$

$$x(8) \approx x(4) + x'(4)(8 - 4) \approx 15.8 + (-3.2) \cdot 4 = 3$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

All terms must be correct in order to earn the point.

Using another variable (such as x) is acceptable as long as that variable is used correctly in both non-constant terms.

Coefficients need not be simplified, i.e., $\frac{-4.4}{2!}$ is acceptable.

Additional terms or “+ ...” added to the polynomial fail to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$11 + 1.2t + \frac{-4.4}{2!}t^2 = 11 + 1.2t - 2.2t^2$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

$s^{(7)}(-1)$ must be correct (and numerical) but need not be simplified.

Second point:

Unit 1 AP Topics 16

The second point will be earned for a correct structure for the seventh-degree term even if the derivative value is incorrect; that is, a form of $\frac{\text{derivative value}}{7!}(t+1)^7$ earns the point regardless of the derivative value's correctness.

The derivative value must be numerical; i.e., $(t+3)^6$ in the numerator is not acceptable.

Setting the derivative equal to the term will not earn the point, for example, $(t+3)^6 = (-1+3)^6 = \frac{2^6}{7!}(t+1)^7$.

Multiple terms of degree less than seven that are presented (even if circled or boxed) will not be read.

A response such as $\frac{b}{7!}(t+1)^7$, where b is the previously incorrectly computed value of $s^{(7)}(-1)$, will earn the second point but not the first.

Acceptable forms of the correct term include:

$$\cdot \frac{2^6}{7!}(t+1)^7$$

$$\cdot \frac{64}{7!}(t+1)^7$$

$$\cdot \frac{64}{5040}(t+1)^7$$

$$\cdot \frac{4}{315}(t+1)^7$$

$$\cdot 0.012(t+1)^7$$

$$\cdot 0.013(t+1)^7$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $s^{(7)}(-1)$

☐ Term

Solution:

$$s^{(7)}(t) = (t+3)^6$$

$$s^{(7)}(-1) = (-1+3)^6 = 2^6$$

The seventh-degree term is $\frac{s^{(7)}(-1)}{7!}(t+1)^7 = \frac{2^6}{7!}(t+1)^7$.

Unit 1 AP Topics 16

81. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	3	6	9	12
$x'(t)$ (feet per minute)	-1.8	2.6	4.0	5.2	-3.6

The position of a particle moving along a straight line is modeled by a twice-differentiable function $x(t)$, where time t is measured in minutes and $x(t)$ is measured in feet. Selected values of $x'(t)$, the derivative of $x(t)$, are shown in the table above. The position of the particle at time $t = 0$ is $x(0) = 5$.

- (a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_0^{12} x'(t) \, dt$. Show the computations that lead to your answer.
- (b) Using correct units, explain the meaning of $\int_0^{12} x'(t) \, dt$ in the context of the problem.
- (c) Must there be a time t , for $0 < t < 12$, when the velocity of the particle is zero? Justify your answer.
- (d) Use Euler's method, starting at $t = 0$ with two steps of equal size, to approximate the position of the particle at time $t = 6$. Show the computations that lead to your answer.
- (e) It is known that $x''(0) = 3.4$. Write the second-degree Taylor polynomial for x about $t = 0$.
- (f) Let s be a function such that the fourth derivative of s with respect to t is $(t - 4)^7$. Write the fourth-degree term of the fourth-degree Taylor polynomial for s about $t = 2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The answer need not be simplified but must be equivalent to 46.8.

If errors are made in simplification, the point is not earned.

The answer must have accompanying work, specifically the sum of two products.

Unit 1 AP Topics 16



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\int_0^{12} x'(t) dt \approx x'(3)(6 - 0) + x'(9)(12 - 6)$$

$$= 2.6 \cdot 6 + 5.2 \cdot 6 = 46.8$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

“Displacement” can be written as “change in position.”

“Distance traveled” or “total feet” is not acceptable.

The time interval must be specific, i.e., $t = 0$ to $t = 12$ minutes; at $t = 12$ minutes is not acceptable.

“Over 12 minutes” is not acceptable.

Second point:

“Feet” may be stated as the units in part (a) or in part (b). The point is earned in either case, unless some other unit is re(stated) in part (b).

“Minutes” need not be stated as the variable t was defined in the problem stem.

Units point may be earned without the interpretation point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Interpretation

☐ Units

Unit 1 AP Topics 16

Solution:

$\int_0^{12} x'(t) dt$ is the displacement of the particle, in feet, from $t = 0$ to $t = 12$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The connection of velocity to $x'(t)$ need not be explicit; any specific reference to $x'(t)$ or specific values in the table will indicate the connection and earn the point.

Second point:

Both conditions must be stated, i.e., the continuity of $x'(t)$ and the bounding of 0 by two values of $x'(t)$ of opposite signs.

The conditions point requires $x'(t)$ is differentiable and therefore continuous. The point is not earned for just $x'(t)$ is differentiable and continuous.

The interval upon which $x'(t)$ is continuous and/or differentiable does not need to be stated. If stated, it must be a correct interval with the possible exception of inclusion/exclusion of the endpoints.

The values of $x'(t)$ must explicitly bound 0, either verbally or symbolically.

Intervals need not be presented using mathematical notation; for example, “ $x'(0) < 0$ and $x'(3) > 0$ ” or “ $x'(t)$ changes signs between $t = 0$ and $t = 3$ ” are both acceptable.

A general statement such as “ $x'(t)$ changes signs” is not acceptable.

Third point:

Intermediate Value Theorem (or IVT) need not be explicitly stated.

This point is earned for an affirmative conclusion and bounding 0 by two values of opposite signs pulled from the table.

A response could earn the first and third points without earning the second point if conditions are presented but not sufficiently justified and the conclusion is correct.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

Unit 1 AP Topics 16

- ☐ Connection to $x'(t)$
- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

$x'(t)$ is the velocity of the particle at time t .

x is twice differentiable. $\Rightarrow x'$ is differentiable.

$\Rightarrow x'$ is continuous for $0 \leq t \leq 12$.

$$x'(0) = -1.8 < 0 < 2.6 = x'(3)$$

Therefore, by the Intermediate Value Theorem, there is a time t , for $0 < t < 3$, such that $x'(t) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

This point is earned for a response that provides clear evidence of an understanding of Euler's method presented in a table or in the form shown in the model solution.

This point could still be earned in the presence of a single error.

If the method is displayed in a table with no column/row labels, this point is only earned for a completely correct final answer (and in that case, both points are earned).

If the method is displayed in a table with no column/row labels and there is an error in one or more of the values in the table, neither point is earned.

Second point:

This point is earned only by final answer of 7.4 or equivalent with accompanying work. That answer does not need to be simplified.

The second point cannot be earned without the first point.

Both points are earned for a response of $5 - 1.8 \cdot 3 + 2.6 \cdot 3$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

Unit 1 AP Topics 16

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$x(3) \approx x(0) + x'(0)(3 - 0) = 5 + (-1.8) \cdot 3 = -0.4$$

$$x(6) \approx x(3) + x'(3)(6 - 3) \approx -0.4 + 2.6 \cdot 3 = 7.4$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

All terms must be correct in order to earn the point.

Using another variable (such as x) is acceptable as long as that variable is used correctly in both non-constant terms.

Coefficients need not be simplified, i.e., $\frac{3.4}{2!}$ is acceptable.

Additional terms or “+ . . .” added to the polynomial fail to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$5 - 1.8t + \frac{3.4}{2!}t^2 = 5 - 1.8t + 1.7t^2$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

$s^{(4)}(2)$ must be correct (and numerical) but need not be simplified.

Second point:

Unit 1 AP Topics 16

The second point will be earned for a correct structure for the fourth-degree term even if the derivative value is incorrect; that is, a form of $\frac{\text{derivative value}}{4!}(t - 2)^4$ earns the point regardless of the derivative value's correctness.

The derivative value must be numerical; i.e., $(t - 4)^7$ in the numerator is not acceptable.

Setting the derivative equal to the term will not earn the point, for example, $(t - 4)^7 = (2 - 4)^7 = \frac{-2^7}{4!}(t - 2)^4$.

Multiple terms of degree less than four that are presented (even if circled or boxed) will not be read.

A response such as $\frac{b}{4!}(t - 2)^4$, where b is the previously incorrectly computed value of $s^{(4)}(2)$, will earn the second point but not the first.

Acceptable forms of the correct term include:

$$\cdot \frac{-2^7}{4!}(t - 2)^4$$

$$\cdot \frac{-128}{4!}(t - 2)^4$$

$$\cdot \frac{-128}{24}(t - 2)^4$$

$$\cdot \frac{-16}{3}(t - 2)^4$$

$$\cdot -5.333(t - 2)^4$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $s^{(4)}(2)$

☐ Term

Solution:

$$s^{(4)}(t) = (t - 4)^7$$

$$s^{(4)}(2) = (2 - 4)^7 = (-2)^7 = -2^7$$

$$\text{The fourth-degree term is } \frac{s^{(4)}(2)}{4!}(t - 2)^4 = \frac{-2^7}{4!}(t - 2)^4.$$

Unit 1 AP Topics 16

82.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	-4	4	2
4	-1	3	6	7

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x . The function h is given by $h(x) = f(g(x)) - 6$.

(a) Explain why there must be a value r for $1 < r < 3$ such that $h(r) = -5$.

(b) Explain why there must be a value c for $1 < c < 3$ such that $h'(c) = -5$.

(c) Let w be the function given by $w(x) = \int_1^{g(x)} f(t) dt$. Find the value of $w'(3)$.

(d) If g^{-1} is the inverse function of g , write an equation for the line tangent to the graph of $y = g^{-1}(x)$ at $x = 2$.

Part A

1 point is earned for $h(1)$ and $h(3)$

1 point is earned for the conclusion, using IVT

$$h(1) = f(g(1)) - 6 = f(2) - 6 = 9 - 6 = 3$$

$$h(3) = f(g(3)) - 6 = f(4) - 6 = -1 - 6 = -7$$

Since $h(3) < -5 < h(1)$ and h is continuous, by the Intermediate Value Theorem, there exists a value r , $1 < r < 3$, such that $h(r) = -5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $h(1)$ and $h(3)$

1 point is earned for the conclusion, using IVT

$$h(1) = f(g(1)) - 6 = f(2) - 6 = 9 - 6 = 3$$

$$h(3) = f(g(3)) - 6 = f(4) - 6 = -1 - 6 = -7$$

Unit 1 AP Topics 16

Since $h(3) < -5 < h(1)$ and h is continuous, by the Intermediate Value Theorem, there exists a value r , $1 < r < 3$, such that $h(r) = -5$.

Part B

1 point is earned for: $\frac{h(3) - h(1)}{3 - 1}$

1 point is earned for the conclusion, using MVT

$$\frac{h(3) - h(1)}{3 - 1} = \frac{-7 - 3}{3 - 1} = -5$$

Since h is continuous and differentiable, by the Mean Value Theorem, there exists a value c , $1 < c < 3$, such that $h'(c) = -5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{h(3) - h(1)}{3 - 1}$

1 point is earned for the conclusion, using MVT

$$\frac{h(3) - h(1)}{3 - 1} = \frac{-7 - 3}{3 - 1} = -5$$

Since h is continuous and differentiable, by the Mean Value Theorem, there exists a value c , $1 < c < 3$, such that $h'(c) = -5$.

Part C

1 point is earned for applying chain rule

1 point is earned for correct answer

$$w'(3) = f(g(3)) \cdot g'(3) = f(4) \cdot 2 = -2$$

Unit 1 AP Topics 16



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for applying chain rule

1 point is earned for correct answer

$$w'(3) = f(g(3)) \cdot g'(3) = f(4) \cdot 2 = -2$$

Part D

1 point is earned for $g^{-1}(2)$

1 point is earned for $(g^{-1})'(2)$

1 point is earned for the tangent line equation

$$g(1) = 2, \text{ so } g^{-1}(2) = 1.$$

$$(g^{-1})'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g^{-1}(2)$

1 point is earned for $(g^{-1})'(2)$

1 point is earned for the tangent line equation

Unit 1 AP Topics 16

$$g(1) = 2, \text{ so } g^{-1}(2) = 1.$$

$$(g^{-1})'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.

Unit 1 AP Topics 16

83.

t (minutes)	0	2	5	8	12
$v_A(t)$ (meters/minute)	0	100	40	-120	-150

Train A runs back and forth on an east-west section of railroad track. Train A 's velocity, measured in meters per minute, is given by a differentiable function $v_A(t)$, where time t is measured in minutes. Selected values for $v_A(t)$ are given in the table above.

- (a) Find the average acceleration of train A over the interval $2 \leq t \leq 8$.
- (b) Do the data in the table support the conclusion that train A 's velocity is -100 meters per minute at some time t with $5 < t < 8$? Give a reason for your answer.
- (c) At time $t = 2$, train A 's position is 300 meters east of the Origin Station, and the train is moving to the east. Write an expression involving an integral that gives the position of train A , in meters from the Origin Station, at time $t = 12$. Use a trapezoidal sum with three subintervals indicated by the table to approximate the position of the train at time $t = 12$.
- (d) A second train, train B , travels north from the Origin Station. At time t the velocity of train B is given by $v_B(t) = -5t^2 + 60t + 25$, and at time $t = 2$ the train is 400 meters north of the station. Find the rate, in meters per minute, at which the distance between train A and train B is changing at time $t = 2$.

Part A

1 point is earned for average acceleration

$$\text{average accel} = \frac{v_A(8) - v_A(2)}{8 - 2} = \frac{-120 - 100}{6} = -\frac{110}{3} \text{ m/min}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for average acceleration

$$\text{average accel} = \frac{v_A(8) - v_A(2)}{8 - 2} = \frac{-120 - 100}{6} = -\frac{110}{3} \text{ m/min}^2$$

Part B

1 point is earned for $v_A(8) < -100 < v_A(5)$

1 point is earned for the conclusion, using IVT

Unit 1 AP Topics 16

v_A is differentiable $\Rightarrow v_A$ is continuous

$$v_A(8) = -120 < -100 < 40 = v_A(5)$$

Therefore, by the Intermediate Value Theorem, there is a time t , $5 < t < 8$, such that $v_A(t) = -100$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $v_A(8) < -100 < v_A(5)$

1 point is earned for the conclusion, using IVT

v_A is differentiable $\Rightarrow v_A$ is continuous

$$v_A(8) = -120 < -100 < 40 = v_A(5)$$

Therefore, by the Intermediate Value Theorem, there is a time t , $5 < t < 8$, such that $v_A(t) = -100$.

Part C

1 point is earned for the position expression

1 point is earned for the trapezoidal sum

1 point is earned for position at time $t = 12$

$$\begin{aligned}
 s_A(12) &= s_A(2) + \int_2^{12} v_A(t) dt = 300 + \int_2^{12} v_A(t) dt \\
 \int_2^{12} v_A(t) dt &\approx 3 \cdot \frac{100+40}{2} + 3 \cdot \frac{40-120}{2} + 4 \cdot \frac{-120-150}{2} \\
 &= -450 \\
 s_A(12) &\approx 300 - 450 = -150
 \end{aligned}$$

Unit 1 AP Topics 16

The position of Train A at time $t = 12$ minutes is approximately 150 meters west of Origin Station.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the position expression

1 point is earned for the trapezoidal sum

1 point is earned for position at time $t = 12$

$$s_A(12) = s_A(2) + \int_2^{12} v_A(t) dt = 300 + \int_2^{12} v_A(t) dt$$

$$\int_2^{12} v_A(t) dt \approx 3 \cdot \frac{100+40}{2} + 3 \cdot \frac{40-120}{2} + 4 \cdot \frac{-120-150}{2}$$

$$= -450$$

$$s_A(12) \approx 300 - 450 = -150$$

The position of Train A at time $t = 12$ minutes is approximately 150 meters west of Origin Station.

Part D

2 points are earned for implicit differentiation of distance relationship

1 point is earned for the answer

Let x be train A 's position, y train B 's position, and z the distance between train A and train B .

$$z^2 = x^2 + y^2 \Rightarrow 2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$x = 300, y = 400 \Rightarrow z = 500$$

$$v_B(2) = -20 + 120 + 25 = 125$$

$$500 \frac{dz}{dt} = (300)(100) + (400)(125)$$

$$\frac{dz}{dt} = \frac{80000}{500} = 160 \text{ meters per minute}$$



0	1	2	3
---	---	---	---

Unit 1 AP Topics 16

The student response earns all of the following points:

2 points are earned for implicit differentiation of distance relationship

1 point is earned for the answer

Let x be train A 's position, y train B 's position, and z the distance between train A and train B .

$$z^2 = x^2 + y^2 \Rightarrow 2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$x = 300, y = 400 \Rightarrow z = 500$$

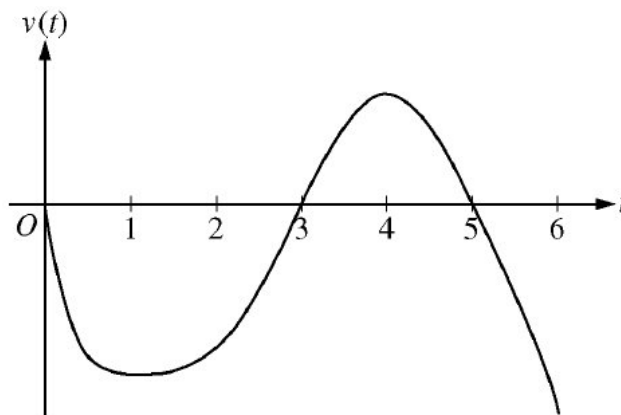
$$v_B(2) = -20 + 120 + 25 = 125$$

$$500 \frac{dz}{dt} = (300)(100) + (400)(125)$$

$$\frac{dz}{dt} = \frac{80000}{500} = 160 \text{ meters per minute}$$

Unit 1 AP Topics 16

84.

Graph of v

A particle moves along the x -axis so that its velocity at time t , for $0 \leq t \leq 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t -axis and the graph of v on the intervals $[0, 3]$, $[3, 5]$, and $[5, 6]$ are 8, 3, and 2, respectively. At time $t = 0$, the particle is at $x = -2$.

- (a) For $0 \leq t \leq 6$, find both the time and the position of the particle when the particle is farthest to the left. Justify your answer.
- (b) For how many values of t , where $0 \leq t \leq 6$, is the particle at $x = -8$? Explain your reasoning.
- (c) On the interval $2 < t < 3$, is the speed of the particle increasing or decreasing? Give a reason for your answer.
- (d) During what time intervals, if any, is the acceleration of the particle negative? Justify your answer.

Part A

The response can earn up to 3 points:

1 point is earned for identifying $t = 3$ as a candidate

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

1 point is earned for consideration of $\int_0^6 v(t) dt$

$$x(3) = -2 + \int_0^3 v(t) dt = -2 - 8 = -10 \quad x(6) = -2 + \int_0^6 v(t) dt = -2 - 8 + 3 - 2 = -9$$

1 point is earned for the correct conclusion

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.



0	1	2	3
---	---	---	---

Unit 1 AP Topics 16

The response earns all of the following points:

1 point is earned for identifying $t = 3$ as a candidate

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

1 point is earned for consideration of $\int_0^6 v(t) dt$

$$x(3) = -2 + \int_0^3 v(t) dt = -2 - 8 = -10 \quad x(6) = -2 + \int_0^6 v(t) dt = -2 - 8 + 3 - 2 = -9$$

1 point is earned for the correct conclusion

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

Part B

The response can earn up to 3 points:

1 point is earned for the positions at $t = 3$, $t = 5$, and $t = 6$

1 point is earned for the description of motion

1 point is earned for the correctly conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point is earned for the positions at $t = 3$, $t = 5$, and $t = 6$

1 point is earned for the description of motion

1 point is earned for the correctly conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.

Unit 1 AP Topics 16**Part C**

The response can earn up to 1 point:

1 point is earned for the correct answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.



0	1
---	---

The response earns all of the following point:

1 point is earned for the correct answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.

Part D

The response can earn up to 2 points:

1 point is earned for the correct answer

1 point is earned for correct justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for the correct answer

1 point is earned for correct justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.

Unit 1 AP Topics 16

85.

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in gigaleters (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t = 30$, the reservoir contains 125 gigaleters of water.

(a) Use the tangent line approximation to W at time $t = 30$ to predict the volume of water $W(t)$, in gigaleters, in the reservoir at time $t = 32$. Show the computations that lead to your answer.

(b) Use a left Riemann sum, with the three subintervals indicated by the data in the table, to approximate $\int_0^{30} W'(t) dt$. Use this approximation to estimate the volume of water $W(t)$, in gigaleters, in the reservoir at time $t = 0$. Show the computations that lead to your answer.

(c) Explain why there must be at least one time t , other than $t = 10$, such that $W'(t) = 0.7$ GL/day.

(d) The equation $A = 0.3W^{2/3}$ gives the relationship between the area A , in square kilometers, of the surface of the reservoir, and the volume of water $W(t)$, in gigaleters, in the reservoir. Find the instantaneous rate of change of A , in square kilometers per day, with respect to t when $t = 30$ days.

Part A

The response can earn up to 1 point:

1 point: For the correct answer

An equation of the tangent line is $y = 0.5(t - 30) + 125$.

$$W(32) \approx 0.5(32 - 30) + 125 = 126$$



0	1
---	---

The response earns all of the following point:

1 point: For the correct answer

An equation of the tangent line is $y = 0.5(t - 30) + 125$.

$$W(32) \approx 0.5(32 - 30) + 125 = 126$$

Unit 1 AP Topics 16

Part B

The response can earn up to 3 points:

1 point: For the left Riemann sum

1 point: For the approximation

1 point: For the correct answer

$$\int_0^{30} W'(t) dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

$$W(0) = W(30) - \int_0^{30} W'(t) dt = 125 - 22.4 = 102.6$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: For the left Riemann sum

1 point: For the approximation

1 point: For the correct answer

$$\int_0^{30} W'(t) dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

$$W(0) = W(30) - \int_0^{30} W'(t) dt = 125 - 22.4 = 102.6$$

Part C

The response can earn up to 2 points:

2 points: For the explanation

W' is differentiable $\Rightarrow W'$ is continuous.

$$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$$

By the Intermediate Value Theorem, there must be at least one time t , $22 \leq t \leq 30$, such that $W'(t) = 0.7$.

Unit 1 AP Topics 16



0	1	2
---	---	---

The response earns all of the following points:

2 points: For the explanation

W' is differentiable $\Rightarrow W'$ is continuous.

$$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$$

By the Intermediate Value Theorem, there must be at least one time t , $22 \leq t \leq 30$, such that $W'(t) = 0.7$.

Part D

The response can earn up to 3 points:

2 points: For stating $\frac{dA}{dt}$

1 point: For the correct answer

$$\frac{dA}{dt} = (0.3)^{\frac{2}{3}} W^{-1/3} \cdot \frac{dW}{dt} = \frac{0.2}{\sqrt[3]{W}} \cdot \frac{dW}{dt}$$

$$\left. \frac{dA}{dt} \right|_{t=30} = \frac{0.2}{\sqrt[3]{125}} \cdot 0.5 = 0.02$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

2 points: For stating $\frac{dA}{dt}$

1 point: For the correct answer

$$\frac{dA}{dt} = (0.3)^{\frac{2}{3}} W^{-1/3} \cdot \frac{dW}{dt} = \frac{0.2}{\sqrt[3]{W}} \cdot \frac{dW}{dt}$$

$$\left. \frac{dA}{dt} \right|_{t=30} = \frac{0.2}{\sqrt[3]{125}} \cdot 0.5 = 0.02$$